O17e - Diffraction

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May 11, 2026

1 Introduction

Diffraction is one of the fundamental manifestations of the wave nature of light. When a coherent light wave encounters an aperture or a periodic structure whose dimensions are comparable to the wavelength, the transmitted wave is redistributed in space due to interference between secondary wavelets originating from different parts of the aperture. In this experiment, diffraction patterns produced by a single slit, a double slit, and a diffraction grating are recorded and analyzed using a CCD line camera. The aim is to determine the geometric parameters of these diffracting objects and to compare the observed intensity distributions with theoretical predictions.

Two diffraction regimes are investigated: Fraunhofer diffraction and Fresnel diffraction. Fraunhofer diffraction corresponds to the far-field limit, where the observed intensity pattern is directly related to the Fourier transform of the aperture transmission function. In the experiment, this regime is realized by placing the CCD camera in the focal plane of a lens. For a single slit, the Fraunhofer pattern consists of a central maximum and weaker side maxima separated by minima whose positions depend on the slit width. For a double slit, the intensity distribution contains rapidly oscillating interference fringes modulated by the single-slit diffraction envelope. By analyzing the positions of the minima and by fitting the theoretical intensity functions to the measured data, the slit width and slit separation can be determined.

Fourier analysis provides an additional method for interpreting the measured Fraunhofer patterns. Since the camera measures intensity rather than the electric-field amplitude, the Fourier transform of the measured intensity profile is related to the autocorrelation of the aperture transmission function. Therefore, the fast Fourier transform (FFT) of the recorded intensity profiles can be used to extract the characteristic dimensions of the single slit and double slit. This also illustrates the close connection between diffraction optics and Fourier-transform methods.

In the Fresnel regime, the observation plane is at a finite distance from the aperture, so the curvature of the wavefronts can no longer be neglected. This produces diffraction profiles that depend explicitly on the propagation distance. In this experiment, Fresnel diffraction is studied by removing the lens and placing the CCD camera directly behind the single slit, double slit, or grating at selected distances. The measured profiles are compared with theoretical Fresnel diffraction curves calculated using the slit parameters obtained from the Fraunhofer analysis.

The final part of the experiment investigates diffraction from a grating and the Talbot effect. A periodic grating can reproduce self-images of its transmission structure at characteristic propagation distances known as Talbot distances. Instead of identifying these self-images directly in real space, the Fourier transforms of the measured grating profiles are analyzed. The grating constant is determined from the positions of the Fourier peaks, while the distance-dependent amplitude of the first Fourier peak is used to determine the Talbot length. The measured Talbot length is then compared with the theoretical value

$$t_B = \frac{2g^2}{\lambda},$$

where g is the grating constant and λ is the laser wavelength.

Overall, this experiment combines direct diffraction measurements, curve fitting, Fourier analysis, and uncertainty evaluation to determine the physical parameters of optical apertures and to test the theoretical descriptions of Fraunhofer and Fresnel diffraction.

2 Theoretical Analysis

2.1 Overview

The experiment is based on the diffraction of coherent laser light by apertures with different transmission functions. The three investigated objects are a single slit, a double slit, and a diffraction grating. The measured quantity is the light intensity recorded by a CCD line camera as a function of transverse position. The main theoretical idea is that, in the Fraunhofer regime, the diffraction pattern is determined by the Fourier transform of the aperture transmission function. In the Fresnel regime, the finite propagation distance must be included explicitly, leading to diffraction

profiles described by Fresnel integrals.

The laser wavelength used in the experiment is

$$\lambda = 636 \text{ nm}.$$

The wave number is

$$k = \frac{2\pi}{\lambda}.$$

The coordinate x' denotes the transverse coordinate in the aperture plane, while x denotes the transverse coordinate in the camera plane. The distance between the diffracting object and the observation plane is denoted by z. In the Fraunhofer setup, a lens of focal length f is used, and the CCD camera is placed in the focal plane of the lens.

The measured intensity profiles are used in three main ways. First, the positions of diffraction minima are used to determine slit parameters. Second, theoretical intensity functions are fitted directly to the measured profiles. Third, the fast Fourier transform (FFT) of the measured intensity profiles is used to obtain aperture dimensions from their real-space autocorrelation.

2.2 Diffraction Geometry and Huygens Principle

According to Huygens' principle, every point of a wavefront may be regarded as the source of a secondary spherical wave. The field contribution from a point source can be written as

$$\psi_s = A\frac{\exp(ikr - i\omega t)}{r},$$

where A is the amplitude, r is the distance between the source point and the observation point, ω is the angular frequency, and $k = 2\pi/\lambda$ is the wave number.

For an aperture described by a transmission function $f(x')$, the total field at a point x in the observation plane is obtained by superposing the contributions from all points of the aperture:

$$\psi(x,t) \propto \int f(x')\frac{\exp(ikr - i\omega t)}{r}\,dx',$$

where

$$r = \sqrt{(x - x')^2 + z^2}.$$

In this experiment the slits are assumed to be very long in the vertical direction, so the problem can be treated as one-dimensional in the transverse coordinate x.

2.3 Fraunhofer Diffraction

2.3.1 Fraunhofer Approximation

Fraunhofer diffraction corresponds to the far-field limit, in which the observation distance is sufficiently large that the diffracted waves arriving at the observation plane can be approximated as plane waves. For a diffraction angle α, the transverse component of the wave vector is

$$k_x = k\sin\alpha = \frac{2\pi}{\lambda}\sin\alpha.$$

The phase difference between light originating from the aperture point x' and light originating from the aperture center is

$$\Delta\phi = -k_x x'.$$

Thus, the field amplitude in the Fraunhofer limit is proportional to

$$\psi(k_x) \propto \int_{-\infty}^{\infty} f(x')\exp(-ik_x x')\,dx'.$$

Therefore, the Fraunhofer diffraction amplitude is the Fourier transform of the aperture transmission function:

$$\psi(k_x) \propto \mathcal{F}\{f(x')\}.$$

The measured intensity is proportional to the squared modulus of the field:

$$I(k_x) \propto |\psi(k_x)|^2.$$

In the focal plane of a lens, the position x on the CCD camera is related to the diffraction angle by

$$\tan\alpha = \frac{x}{f}.$$

Using

$$\sin\alpha = \frac{x}{\sqrt{x^2+f^2}},$$

the exact relation between camera position and transverse wave vector is

$$k_x = \frac{2\pi}{\lambda}\frac{x}{\sqrt{x^2+f^2}}.$$

For small diffraction angles, $x \ll f$, this becomes

$$k_x \approx \frac{2\pi}{\lambda}\frac{x}{f}.$$

This relation is used to convert measured intensity profiles from position space to k_x-space.

2.4 Single-Slit Fraunhofer Diffraction

2.4.1 Transmission Function

For a single slit of width b, the transmission function is

$$f(x') = \begin{cases} 1, & -\frac{b}{2} < x' < \frac{b}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

2.4.2 Derivation of the Intensity Distribution

The Fraunhofer amplitude is

$$\psi(k_x) \propto \int_{-b/2}^{b/2} \exp(-ik_x x')\,dx'.$$

Evaluating the integral gives

$$\psi(k_x) = \left[\frac{\exp(-ik_x x')}{-ik_x}\right]_{-b/2}^{b/2}.$$

Therefore,

$$\psi(k_x) = \frac{\exp(-ik_x b/2) - \exp(ik_x b/2)}{-ik_x}.$$

Using

$$\exp(-ia) - \exp(ia) = -2i\sin a,$$

with $a = k_x b/2$, we obtain

$$\psi(k_x) = \frac{2\sin(k_x b/2)}{k_x}.$$

This can be rewritten as

$$\psi(k_x) = b\frac{\sin(k_x b/2)}{k_x b/2}.$$

Using the definition

$$\operatorname{sinc}(u) = \frac{\sin u}{u},$$

the amplitude becomes

$$\psi(k_x) \propto b\operatorname{sinc}\left(\frac{bk_x}{2}\right).$$

The measured intensity is therefore

$$I(k_x) = I_0\operatorname{sinc}^2\left(\frac{bk_x}{2}\right) + I_{\text{bg}},$$

where I_0 is an amplitude factor and I_{bg} accounts for background intensity.

2.4.3 Minima of the Single-Slit Pattern

The minima occur when the numerator of the sinc function is zero:

$$\sin\left(\frac{bk_x}{2}\right) = 0.$$

Thus,

$$\frac{bk_x}{2} = n\pi, \qquad n = \pm1, \pm2, \pm3, \ldots$$

Using $k_x = (2\pi/\lambda)\sin\alpha$, this gives

$$\frac{b}{2}\frac{2\pi}{\lambda}\sin\alpha_n = n\pi.$$

Therefore,

$$b\sin\alpha_n = n\lambda.$$

In the focal plane of the lens,

$$x_n = f\tan\alpha_n.$$

For small angles,

$$\sin\alpha_n \approx \tan\alpha_n \approx \frac{x_n}{f}.$$

Therefore,

$$b\frac{x_n}{f} \approx n\lambda.$$

Solving for the minimum position gives

$$x_n \approx \frac{nf\lambda}{b}.$$

The experiment uses the distance between the minima of order $+n$ and $-n$:

$$2x_n \approx \frac{2nf\lambda}{b}.$$

This has the form of a linear relationship,

$$2x_n = mn + c,$$

where ideally $c = 0$, and the slope is

$$m = \frac{2f\lambda}{b}.$$

Thus, the single-slit width can be determined from the fitted slope by

$$b = \frac{2f\lambda}{m}.$$

2.5 Fourier Transform Analysis of the Single-Slit Pattern

In Fraunhofer diffraction, the field amplitude is proportional to the Fourier transform of the aperture function:

$$\psi(k_x) \propto \mathcal{F}\{f(x')\}.$$

However, the CCD camera measures intensity:

$$I(k_x) \propto |\psi(k_x)|^2.$$

Since

$$\psi(k_x) = \mathcal{F}\{f(x')\},$$

the measured intensity can be written as

$$I(k_x) \propto \mathcal{F}\{f(x')\}\mathcal{F}\{f(x')\}^*.$$

Using the convolution theorem, the inverse Fourier transform of the intensity is proportional to the autocorrelation of the aperture function:

$$\mathcal{F}^{-1}\{I(k_x)\} \propto f(x') * f(-x').$$

For a symmetric aperture such as a single slit, $f(x') = f(-x')$, so

$$\mathcal{F}^{-1}\{I(k_x)\} \propto f(x') * f(x').$$

The convolution of a rectangular slit function with itself is a triangular function. If the slit width is b, the resulting triangular function has a base width of $2b$. Thus, the FFT of the measured intensity profile can be used to determine the slit width from the width of the autocorrelation feature.

2.6 Double-Slit Fraunhofer Diffraction

2.6.1 Transmission Function

For a double slit with slit width b and center-to-center slit separation g, the transmission function can be written as the sum of two shifted single-slit functions:

$$f(x') = \operatorname{rect}\left(\frac{x' - g/2}{b}\right) + \operatorname{rect}\left(\frac{x' + g/2}{b}\right),$$

where

$$\operatorname{rect}\left(\frac{x'}{b}\right) = \begin{cases} 1, & |x'| < \frac{b}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

2.6.2 Derivation of the Double-Slit Intensity Distribution

The Fourier transform of a single slit centered at the origin is

$$b \operatorname{sinc}\left(\frac{bk_x}{2}\right).$$

Using the shift theorem of the Fourier transform,

$$\mathcal{F}\{f(x' - x_0)\} = \exp(-ik_x x_0) F(k_x),$$

the two shifted slits have amplitudes

$$b \operatorname{sinc}\left(\frac{bk_x}{2}\right) \exp\left(-i\frac{gk_x}{2}\right)$$

and

$$b \operatorname{sinc}\left(\frac{bk_x}{2}\right) \exp\left(i\frac{gk_x}{2}\right).$$

The total amplitude is the sum:

$$\psi(k_x) \propto b \operatorname{sinc}\left(\frac{bk_x}{2}\right) \left[\exp\left(-i\frac{gk_x}{2}\right) + \exp\left(i\frac{gk_x}{2}\right)\right].$$

Using

$$e^{-ia} + e^{ia} = 2\cos a,$$

we obtain

$$\psi(k_x) \propto 2b \operatorname{sinc}\left(\frac{bk_x}{2}\right) \cos\left(\frac{gk_x}{2}\right).$$

The measured intensity is therefore

$$I(k_x) = I_0 \operatorname{sinc}^2\left(\frac{bk_x}{2}\right) \cos^2\left(\frac{gk_x}{2}\right) + I_{\text{bg}}.$$

The factor

$$\operatorname{sinc}^2\left(\frac{bk_x}{2}\right)$$

is the single-slit diffraction envelope, while

$$\cos^2\left(\frac{gk_x}{2}\right)$$

describes the interference fringes due to the two slits.

2.6.3 Minima of the Double-Slit Interference Pattern

The destructive interference minima of the double-slit interference term occur when

$$\cos^2\left(\frac{gk_x}{2}\right) = 0.$$

Thus,

$$\cos\left(\frac{gk_x}{2}\right) = 0.$$

This happens for

$$\frac{gk_x}{2} = \left(n + \frac{1}{2}\right)\pi, \qquad n = 0, \pm 1, \pm 2, \ldots$$

Using

$$k_x = \frac{2\pi}{\lambda}\sin\alpha,$$

we obtain

$$\frac{g}{2}\frac{2\pi}{\lambda}\sin\alpha_n = \left(n + \frac{1}{2}\right)\pi.$$

Therefore,

$$g\sin\alpha_n = \left(n + \frac{1}{2}\right)\lambda.$$

In the small-angle approximation,

$$\sin\alpha_n \approx \frac{x_n}{f}.$$

Thus,

$$g\frac{x_n}{f} \approx \left(n + \frac{1}{2}\right)\lambda.$$

Solving for the minima positions gives

$$x_n \approx \frac{f\lambda}{g}\left(n + \frac{1}{2}\right).$$

The distance between symmetric minima is

$$2x_n \approx \frac{2f\lambda}{g}\left(n + \frac{1}{2}\right).$$

Thus, a linear fit of $2x_n$ against $n + 1/2$ gives

$$2x_n = m\left(n + \frac{1}{2}\right) + c,$$

where the slope is

$$m = \frac{2f\lambda}{g}.$$

Therefore, the slit separation is

$$g = \frac{2f\lambda}{m}.$$

The slit width b can be obtained from the envelope minima in the same way as for the single slit, but a more accurate method is to fit the full double-slit Fraunhofer intensity function.

2.7 Fourier Transform Analysis of the Double-Slit Pattern

As in the single-slit case, the CCD measures the intensity rather than the electric-field amplitude. Therefore,

$$I(k_x) \propto |\mathcal{F}\{f(x')\}|^2.$$

The inverse Fourier transform of the measured intensity is proportional to the autocorrelation of the double-slit transmission function:

$$\mathcal{F}^{-1}\{I(k_x)\} \propto f(x') * f(-x').$$

For a double slit, the autocorrelation contains three main features:

- a central triangular peak caused by the autocorrelation of each slit with itself,
- two side peaks caused by the cross-correlation between the two different slits,
- a characteristic separation between these side peaks and the central peak equal to the slit separation g.

The width of the triangular features is related to the slit width b, while the separation of the side peaks from the central peak is related to the slit separation g. Therefore, the FFT of the measured double-slit intensity profile can be used to determine both b and g.

2.8 Fresnel Diffraction

2.8.1 Fresnel Approximation

In the Fresnel regime, the observation distance is finite, so the curvature of the diffracted wavefronts must be retained. The distance between an aperture point x' and an observation point x is

$$r = \sqrt{(x - x')^2 + z^2}.$$

For z much larger than the transverse distances, this can be expanded as

$$r = z\sqrt{1 + \frac{(x - x')^2}{z^2}} \approx z + \frac{(x - x')^2}{2z}.$$

Substituting this into the phase factor gives

$$\exp(ikr) \approx \exp(ikz) \exp\left(i\frac{k}{2z}(x - x')^2\right).$$

Thus, the field in the Fresnel approximation is

$$\psi(x, z) \propto \int f(x') \exp\left(i\frac{k}{2z}(x - x')^2\right) dx'.$$

Using $k = 2\pi/\lambda$, this becomes

$$\psi(x, z) \propto \int f(x') \exp\left(i\frac{\pi}{\lambda z}(x - x')^2\right) dx'.$$

The measured intensity is

$$I(x, z) \propto |\psi(x, z)|^2.$$

2.8.2 Fresnel Integrals

The Fresnel integral is defined as

$$F(u) = C(u) + iS(u),$$

where

$$C(u) = \int_0^u \cos\left(\frac{\pi t^2}{2}\right) dt,$$

and

$$S(u) = \int_0^u \sin\left(\frac{\pi t^2}{2}\right) dt.$$

Equivalently,

$$F(u) = \int_0^u \exp\left(i\frac{\pi t^2}{2}\right) dt.$$

These integrals appear naturally when evaluating the Fresnel diffraction integral over finite aperture intervals.

2.9 Single-Slit Fresnel Diffraction

For a single slit,

$$f(x') = \begin{cases} 1, & -\frac{b}{2} < x' < \frac{b}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

The Fresnel field is

$$\psi(x,z) \propto \int_{-b/2}^{b/2} \exp\left(i\frac{\pi}{\lambda z}(x-x')^2\right)\, dx'.$$

To express the integral in terms of Fresnel integrals, define the substitution

$$u = \sqrt{\frac{2}{\lambda z}}(x - x').$$

Then

$$du = -\sqrt{\frac{2}{\lambda z}}\, dx',$$

or

$$dx' = -\sqrt{\frac{\lambda z}{2}}\, du.$$

The limits become

$$x' = -\frac{b}{2} \quad \Rightarrow \quad u_1 = \sqrt{\frac{2}{\lambda z}}\left(x + \frac{b}{2}\right) = \frac{b+2x}{\sqrt{2\lambda z}},$$

and

$$x' = \frac{b}{2} \quad \Rightarrow \quad u_2 = \sqrt{\frac{2}{\lambda z}}\left(x - \frac{b}{2}\right) = \frac{2x-b}{\sqrt{2\lambda z}}.$$

After reversing the integration limits, the field is proportional to the difference between two Fresnel integrals. The intensity can be written as

$$I(x,z) \propto \left| F\left(\frac{b+2x}{\sqrt{2\lambda z}}\right) + F\left(\frac{b-2x}{\sqrt{2\lambda z}}\right)\right|^2.$$

This formula is used to compare the measured single-slit Fresnel profiles with theory.

2.10 Double-Slit Fresnel Diffraction

For a double slit with slit width b and slit separation g, the two open intervals are

$$-\frac{g+b}{2} < x' < -\frac{g-b}{2}$$

and

$$\frac{g-b}{2} < x' < \frac{g+b}{2}.$$

The Fresnel field is the sum of the contributions from both slits:

$$\psi(x,z) \propto \int_{-(g+b)/2}^{-(g-b)/2} \exp\left(i\frac{\pi}{\lambda z}(x-x')^2\right)\, dx' + \int_{(g-b)/2}^{(g+b)/2} \exp\left(i\frac{\pi}{\lambda z}(x-x')^2\right)\, dx'.$$

Evaluating both integrals using the Fresnel substitution gives the theoretical double-slit Fresnel intensity

$$I(x,z) \propto \left| F\left(\frac{g+b+2x}{\sqrt{2\lambda z}}\right) + F\left(\frac{g+b-2x}{\sqrt{2\lambda z}}\right) - F\left(\frac{g-b+2x}{\sqrt{2\lambda z}}\right) - F\left(\frac{g-b-2x}{\sqrt{2\lambda z}}\right)\right|^2.$$

This expression is used to compare the measured double-slit Fresnel profiles with theory. The values of b and g used in the calculation are taken from the Fraunhofer analysis.

2.11 Fresnel Number

The Fresnel number is a dimensionless quantity used to characterize the diffraction regime. For a slit of width b, it is defined as

$$N_b = \frac{b^2}{4\lambda z}.$$

For a double slit, an additional Fresnel number based on the slit separation g can be defined:

$$N_g = \frac{g^2}{4\lambda z}.$$

When the Fresnel number is much smaller than one,

$$N \ll 1,$$

the Fraunhofer approximation is valid. When the Fresnel number is of order one or larger,

$$N \gtrsim 1,$$

the diffraction pattern belongs to the Fresnel regime and depends explicitly on the propagation distance z.

2.12 Diffraction Grating and the Talbot Effect

2.12.1 Periodic Grating Transmission Function

A diffraction grating is a periodic aperture with grating constant g. Its transmission function satisfies

$$f(x' + g) = f(x').$$

Therefore, it can be written as a Fourier series:

$$f(x') = \sum_{n=-\infty}^{\infty} c_n \exp\left(-i\frac{2\pi n}{g}x'\right),$$

where c_n are the Fourier coefficients of the grating.

2.12.2 Fresnel Propagation of a Periodic Structure

In Fresnel diffraction, each Fourier component of the grating acquires a propagation-dependent phase factor. After propagating a distance z, the field can be written in the form

$$\psi(x, z) \propto \sum_{n=-\infty}^{\infty} c_n \exp\left(-i\frac{2\pi n}{g}x\right) \exp\left(-i\pi\lambda z\frac{n^2}{g^2}\right).$$

The second exponential factor depends on z. A self-image occurs when this phase factor is equal to unity for all integer n. This requires

$$\pi\lambda z\frac{n^2}{g^2} = 2\pi m n^2,$$

where m is an integer. Therefore,

$$\lambda z = 2mg^2.$$

The self-imaging distances are

$$z_m = m\frac{2g^2}{\lambda}.$$

The fundamental Talbot length is therefore

$$t_B = \frac{2g^2}{\lambda}.$$

At integer multiples of t_B, the grating reproduces self-images of its transmission pattern.

2.12.3 Determination of the Grating Constant from Fourier Peaks

The FFT of a grating profile contains peaks corresponding to the spatial periodicity of the grating. If the FFT axis is expressed using spatial frequency κ, the grating peaks occur at

$$\kappa_n = \frac{n}{g}, \qquad n = 0, \pm 1, \pm 2, \ldots$$

The first-order peak is therefore located at

$$\kappa_1 = \frac{1}{g}.$$

Hence, the grating constant is

$$g = \frac{1}{\kappa_1}.$$

If the FFT axis is instead expressed using angular wave vector k, where

$$k = 2\pi\kappa,$$

then the first-order peak occurs at

$$k_1 = \frac{2\pi}{g}.$$

In that convention, the grating constant is

$$g = \frac{2\pi}{k_1}.$$

In the data analysis, the convention used for the FFT axis must therefore be stated clearly.

2.12.4 Determination of the Talbot Length from FFT Peak Amplitudes

In the Fresnel regime, the Fourier transform of the measured grating intensity contains peaks whose amplitudes depend on the propagation distance z. For the first-order Fourier peak, the amplitude oscillates periodically with z. The oscillation period Δz of the first Fourier peak amplitude is related to the Talbot length by

$$\Delta z = \frac{t_B}{2}.$$

Thus, the experimental Talbot length is

$$t_{B,\mathrm{exp}} = 2\Delta z.$$

The theoretical Talbot length is calculated from the grating constant as

$$t_{B,\mathrm{theory}} = \frac{2g^2}{\lambda}.$$

The experimental value obtained from the oscillation of the first Fourier peak amplitude is then compared with this theoretical value.

2.13 Uncertainty Propagation for Theoretical Quantities

The quantities obtained from the theoretical relations depend on measured variables such as slit positions, focal length, wavelength, FFT peak positions, and propagation distance. Their uncertainties are propagated using standard Gaussian uncertainty propagation.

For a single-slit width determined from the slope m of the linear fit,

$$b = \frac{2f\lambda}{m}.$$

The relative uncertainty is

$$\left(\frac{u(b)}{b}\right)^2 = \left(\frac{u(f)}{f}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2 + \left(\frac{u(m)}{m}\right)^2.$$

Therefore,

$$u(b) = b\sqrt{\left(\frac{u(f)}{f}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2 + \left(\frac{u(m)}{m}\right)^2}.$$

For the double-slit separation determined from the slope m,

$$g = \frac{2f\lambda}{m},$$

so

$$u(g) = g\sqrt{\left(\frac{u(f)}{f}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2 + \left(\frac{u(m)}{m}\right)^2}.$$

For a grating constant determined from the first FFT peak position κ_1,

$$g = \frac{1}{\kappa_1},$$

so

$$u(g) = \frac{u(\kappa_1)}{\kappa_1^2}.$$

If angular wave vector k_1 is used instead,

$$g = \frac{2\pi}{k_1},$$

and

$$u(g) = \frac{2\pi}{k_1^2}u(k_1).$$

For the theoretical Talbot length,

$$t_B = \frac{2g^2}{\lambda},$$

the relative uncertainty is

$$\left(\frac{u(t_B)}{t_B}\right)^2 = \left(2\frac{u(g)}{g}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2.$$

Hence,

$$u(t_B) = t_B\sqrt{4\left(\frac{u(g)}{g}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2}.$$

For the experimental Talbot length,

$$t_{B,\mathrm{exp}} = 2\Delta z,$$

so

$$u(t_{B,\mathrm{exp}}) = 2u(\Delta z).$$

These expressions are used to compare experimentally determined quantities with theoretical or manufacturer values within uncertainty.

3 Experimental Analysis

3.1 Experimental Setup

The experiment was performed on an optical bench using a diode laser, polarizer, slide holder, collimating lens, and CCD line camera. The laser wavelength was

$$\lambda = 636 \text{ nm}.$$

The general arrangement of the optical components was

$$\text{laser} \rightarrow \text{polarizer} \rightarrow \text{slide holder} \rightarrow \text{lens} \rightarrow \text{CCD camera}.$$

The polarizer was used to attenuate the laser intensity before the light reached the diffracting object. This was necessary to prevent saturation of the CCD camera. In some measurements, two polarizers may be used if the intensity is still too high after one polarizer. The diffracting objects were mounted in the slide holder. Depending on the task,

a single slit, double slit, or diffraction grating was inserted.

For the Fraunhofer diffraction measurements, the lens was placed behind the slit or double slit, and the CCD camera was positioned in the focal plane of the lens. In this arrangement, the transverse position on the camera corresponds to the diffraction angle, and therefore to the transverse wave-vector component k_x. The focal length of the lens was recorded as

$$\mathrm{f} = 300.8 \text{ mm}$$

The physical position on the CCD was obtained from the pixel number by using the pixel width of the camera. The new camera was used in the experiment with pixel width

$$w_{\mathrm{pix}} = 14 \ \mu\mathrm{m}.$$

The conversion from pixel number p to transverse position x was performed using

$$x = (p - p_0) w_{\mathrm{pix}},$$

where p_0 is the pixel position of the principal maximum. This shift places the central maximum at $x = 0$, which is necessary for the analysis of symmetric diffraction patterns and for the FFT.

For the Fresnel diffraction measurements, the lens was removed from the optical path. The CCD camera was then placed directly behind the single slit, double slit, or grating at a distance z. In this configuration, the measured diffraction pattern depends explicitly on the propagation distance. For the single slit and double slit, profiles were recorded at three different distances. For the grating, profiles were recorded at several distances in order to study the Talbot effect.

The optical components were carefully aligned in height and along the optical axis. Particular attention was paid to the height and horizontal position of the CCD camera, since the camera records only a single line of pixels. A small misalignment could therefore strongly reduce the measured intensity or distort the diffraction pattern. The laser beam was adjusted so that it passed through the center of the selected slit or grating and illuminated the active region of the CCD camera.

The main experimental configurations are summarized in Table 1.

Table 1: Summary of the experimental configurations used for the different measurements.

Task	Diffracting object	Lens	Camera position
1a–1b	Single slit	Present	Focal plane of lens
1c	Single slit	Removed	Directly behind slit at distance z
2a–2c	Double slit	Present	Focal plane of lens
2d	Double slit	Removed	Directly behind double slit at distance z
3	Grating	Removed	Directly behind grating at distance z

3.2 Data Acquisition and Measurement Procedure

The CCD line camera was controlled using the data acquisition software. Each recorded file contained two columns: the first column was the pixel number, and the second column was the measured light intensity in arbitrary units. The intensity values were not absolute optical powers, but were proportional to the light intensity incident on the CCD.

Before recording the diffraction patterns, the integration time and the number of averages were adjusted. The integration time was chosen so that the diffraction pattern was clearly visible while avoiding detector saturation. The maximum recorded intensity was kept below approximately 80% of the maximum camera signal. The number of averages was chosen to reduce random noise in the measured intensity profile.

The general data acquisition parameters were recorded as

$$\text{integration time} = 2.000 \text{ ms},$$

$$\text{number of averages} = 100.$$

For each measurement, the following quantities were recorded:

- diffracting object used,
- slit or grating label, if applicable,
- focal length f, for Fraunhofer measurements,

- distance z, for Fresnel measurements,
- camera type and pixel width,
- integration time,
- number of averages,
- filename of the recorded CSV data,
- maximum recorded intensity.

Single-Slit Measurements

For the single-slit Fraunhofer measurement, the single-slit slide was inserted into the slide holder and the CCD camera was placed in the focal plane of the lens. The optical components were adjusted until a symmetric diffraction pattern with a clear central maximum and visible side minima was obtained. The intensity profile was then saved as a CSV file.

The single-slit Fraunhofer data were used for two analyses. First, the positions of the minima on both sides of the central maximum were determined. The distances $2x_n$ between the minima of order $+n$ and $-n$ were calculated and used in a linear fit to determine the slit width b. Second, the measured profile was shifted so that the principal maximum was located at $x = 0$, converted to $I(k_x)$, and Fourier transformed. The FFT result was then used to obtain a second value for the slit width.

For the single-slit Fresnel measurements, the lens was removed. The CCD camera was placed directly behind the single slit at three different distances,

$$z_1 = 27\,\text{cm}, \qquad z_2 = 31\,\text{cm}, \qquad z_3 = 35\,\text{cm}.$$

At each distance, the diffraction profile was recorded. These profiles were later compared with the theoretical Fresnel diffraction curves calculated using the slit width obtained from the Fraunhofer analysis.

Double-Slit Measurements

For the double-slit Fraunhofer measurement, the double-slit slide was inserted into the slide holder and the CCD camera was again placed in the focal plane of the lens. The setup was aligned until a clear interference pattern was visible. The recorded profile consisted of narrow interference fringes modulated by a broader single-slit diffraction envelope.

The double-slit Fraunhofer profile was analyzed in three ways. First, the positions of the interference minima were used to determine the slit separation g from a linear fit. Second, the full theoretical double-slit Fraunhofer function was fitted to the measured intensity profile to determine both the slit width b and the slit separation g. Third, the profile was converted to $I(k_x)$ and Fourier transformed. The FFT was used to determine both b and g from the autocorrelation structure of the aperture.

For the double-slit Fresnel measurements, the lens was removed and the CCD camera was placed directly behind the double slit at three different distances,

$$z_1 = 40\,\text{cm}, \qquad z_2 = 50\,\text{cm}, \qquad z_3 = 60\,\text{cm}.$$

The measured Fresnel profiles were later compared with the theoretical double-slit Fresnel curves using the values of b and g obtained from the Fraunhofer analysis.

Grating Measurements

For the grating measurements, the diffraction grating was placed in the slide holder and the lens was removed. The CCD camera was positioned directly behind the grating. The grating-camera distance z was varied systematically in order to observe the distance-dependent Fresnel diffraction pattern and the Talbot effect.

For each distance, a CSV file was saved. The recorded intensity profiles were Fourier transformed. From the positions of the FFT maxima, the grating constant g was determined. The amplitude of the first Fourier peak was then extracted for each propagation distance z. Finally, the first peak amplitude was plotted as a function of z, and the oscillation period was used to determine the experimental Talbot length.

General Data Treatment

The same basic data treatment was applied to all recorded profiles. First, the CSV files were imported and the pixel number was converted into position using the appropriate pixel width. The position axis was shifted so that the main maximum or center of symmetry was located at $x = 0$. If necessary, a background intensity was subtracted from the data. The corrected intensity was then normalized to simplify comparison with theory:

$$I_{\mathrm{norm}} = \frac{I - I_{\mathrm{bg}}}{I_{\mathrm{max}} - I_{\mathrm{bg}}}.$$

For Fraunhofer measurements, the position axis was converted to transverse wave vector using

$$k_x = \frac{2\pi}{\lambda} \frac{x}{\sqrt{x^2 + f^2}},$$

or, in the small-angle approximation,

$$k_x \approx \frac{2\pi x}{\lambda f}.$$

The choice between the exact relation and the small-angle approximation is stated in the corresponding analysis section.

For FFT analysis, the intensity data were evaluated on an evenly spaced axis. The Fourier transform was calculated after subtracting the background and, where necessary, applying a suitable window or restricting the fit range to avoid edge effects. The FFT axis was labelled with the appropriate physical unit, and the convention used for spatial frequency or angular wave vector was stated explicitly.

For all fits, the model function, fitted parameters, parameter uncertainties, and goodness-of-fit quantities were reported. The quality of each fit was evaluated using the residuals and numerical indicators such as the root-mean-square error,

$$\mathrm{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (I_i - I_{\mathrm{fit},i})^2},$$

and, where appropriate, the coefficient of determination,

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (I_i - I_{\mathrm{fit},i})^2}{\sum_{i=1}^{N} (I_i - \bar{I})^2}.$$

The uncertainties of the final quantities were determined by propagating the uncertainties in pixel position, focal length, wavelength, fitted slopes, FFT peak positions, and propagation distances.

4 Determination of the Single-Slit Width from Fraunhofer Minima (Task 1a)

4.1 Goal of the Analysis

The aim of this part of the experiment was to determine the width b of a single slit from the positions of the minima in its Fraunhofer diffraction pattern. The measurement was performed using the new CCD line camera and a single slit with manufacturer-specified width

$$b_{\mathrm{man}} = 0.400 \text{ mm}.$$

The recorded Fraunhofer profile was analyzed by identifying pairs of minima of order $+n$ and $-n$. The distances $2x_n$ between corresponding minima were then plotted as a function of diffraction order n. A linear fit was used to determine the slit width.

4.2 Relevant Theory

For Fraunhofer diffraction at a single slit of width b, the minima of the diffraction pattern satisfy

$$b \sin \alpha_n = n\lambda, \qquad n = 1, 2, 3, \ldots,$$

where α_n is the diffraction angle of the n-th minimum and λ is the laser wavelength. In the focal plane of a lens with focal length f, the minimum position is related to the diffraction angle by

$$x_n = f \tan \alpha_n.$$

For small diffraction angles,

$$\sin\alpha_n \approx \tan\alpha_n \approx \frac{x_n}{f}.$$

Therefore,

$$b\frac{x_n}{f} \approx n\lambda.$$

Solving for x_n gives

$$x_n \approx \frac{nf\lambda}{b}.$$

The measured quantity in this task is the distance between the two symmetric minima of order $+n$ and $-n$:

$$2x_n \approx \frac{2nf\lambda}{b}.$$

This can be written as a linear relation,

$$2x_n = mn + c,$$

where the slope is

$$m = \frac{2f\lambda}{b}.$$

Thus, the slit width is obtained from the fitted slope by

$$b = \frac{2f\lambda}{m}.$$

4.3 Experimental Parameters

The Fraunhofer diffraction pattern was recorded using the new CCD camera. Since the new camera has a pixel width of

$$w_{\text{pix}} = 14\ \mu\text{m},$$

the pixel coordinate p was converted into a physical position x using

$$x = (p - p_0)w_{\text{pix}},$$

where p_0 is the pixel position of the central maximum. From the measured profile, the central maximum was found at

$$p_0 = 1127.$$

The laser wavelength and lens focal length used in the analysis were

$$\lambda = 636\ \text{nm},$$

and

$$f = 300.8\ \text{mm}.$$

The uncertainty of the focal length was taken as

$$u(f) = 10\ \text{mm}.$$

The uncertainty in each manually identified minimum position was estimated as

$$u(p) = 2\ \text{pixels}.$$

Therefore, the position uncertainty of a single minimum was

$$u(x) = u(p)w_{\text{pix}} = 2 \cdot 14\ \mu\text{m} = 28\ \mu\text{m}.$$

Since each value $2x_n$ is obtained from the difference between a right and left minimum position, the uncertainty is

$$u(2x_n) = \sqrt{u(x_{+n})^2 + u(x_{-n})^2}.$$

Assuming equal uncertainties for the left and right minima,

$$u(2x_n) = \sqrt{2}\,u(x) = \sqrt{2} \cdot 28\ \mu\text{m} = 39.6\ \mu\text{m}.$$

Thus,

$$u(2x_n) = 0.0396\ \text{mm}.$$

4.4 Method

The intensity profile was first shifted so that the central maximum was located at $x = 0$. The expected position of the first minimum was estimated using the manufacturer slit width,

$$x_1 \approx \frac{f\lambda}{b_{\text{man}}}.$$

With $f = 300.8$ mm, $\lambda = 636$ nm, and $b_{\text{man}} = 0.400$ mm, this gives

$$x_1 \approx 0.478\ \text{mm},$$

corresponding to approximately 34.2 pixels. This estimate was used only to guide the automatic search for the minima. The actual minimum positions were determined from the measured intensity profile.

The minima of orders $n = 1$ to $n = 5$ were identified on both sides of the central maximum. The distance

$$2x_n = x_{+n} - x_{-n}$$

was calculated for each order. A weighted linear fit of the form

$$2x_n = mn + c$$

was then performed using the uncertainty $u(2x_n) = 0.0396$ mm for each measured distance. The fitted slope was used to calculate the slit width.

4.5 Measured Minima and Linear Fit

The measured minima positions and calculated distances are shown in Table 2. All positions are given relative to the central maximum.

Table 2: Measured positions of the single-slit Fraunhofer minima and calculated distances $2x_n$. The uncertainty of each distance is $u(2x_n) = 0.0396$ mm.

Order n	Left pixel	Right pixel	x_{-n} [mm]	x_{+n} [mm]	$2x_n$ [mm]
1	1092	1164	-0.490	0.518	1.008
2	1056	1199	-0.994	1.008	2.002
3	1022	1238	-1.470	1.554	3.024
4	988	1266	-1.946	1.946	3.892
5	951	1301	-2.464	2.436	4.900

The corresponding linear fit was

$$2x_n = mn + c,$$

with the fitted parameters

$$m = (0.9674 \pm 0.0125)\ \text{mm},$$

and

$$c = (0.0630 \pm 0.0415)\ \text{mm}.$$

The fit quality was evaluated using the sum of squared errors, mean squared error, root-mean-square error, and coefficient of determination:

$$\text{SSE} = 5.6252 \times 10^{-9}\ \text{m}^2,$$

$$\text{MSE} = 1.1250 \times 10^{-9}\ \text{m}^2,$$

$$\text{RMSE} = 0.0335\ \text{mm},$$

and

$$R^2 = 0.99999.$$

The high value of R^2 indicates that the measured minima positions follow the expected linear dependence on diffraction order very well.

4.6 Plots

Figure 1 shows the measured Fraunhofer diffraction profile of the single slit. The central maximum was shifted to $x = 0$, and the detected minima used for the analysis are marked.

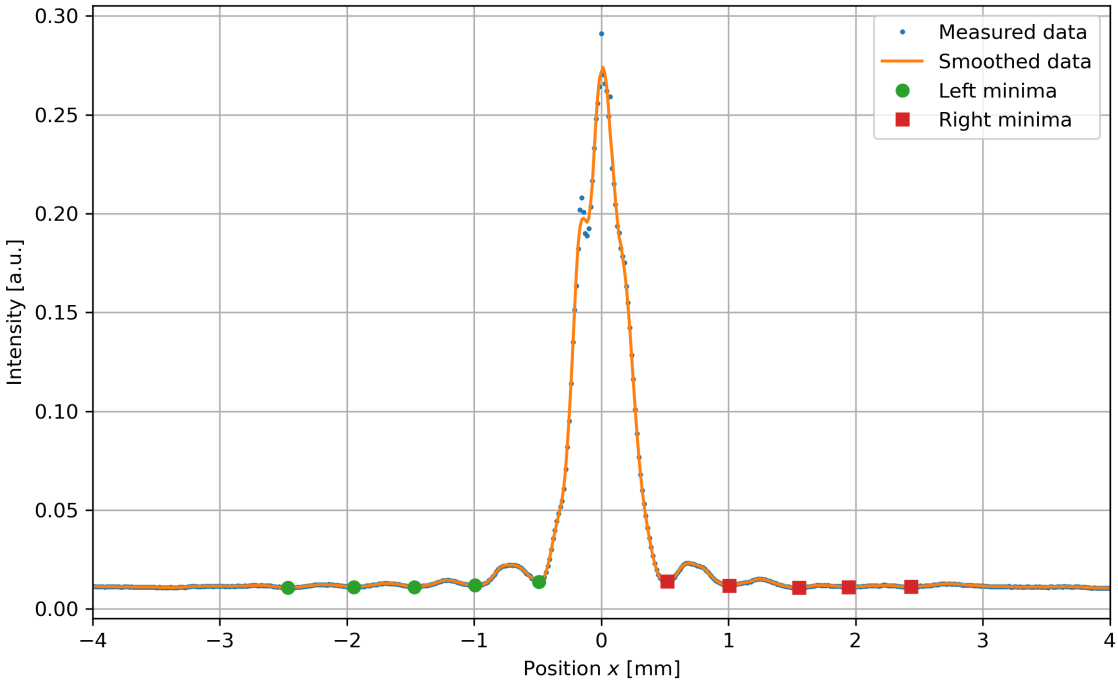


Figure 1: Measured Fraunhofer diffraction profile of the single slit with the detected left and right minima.

Figure 2 shows the measured distances $2x_n$ as a function of diffraction order n, together with the weighted linear fit.

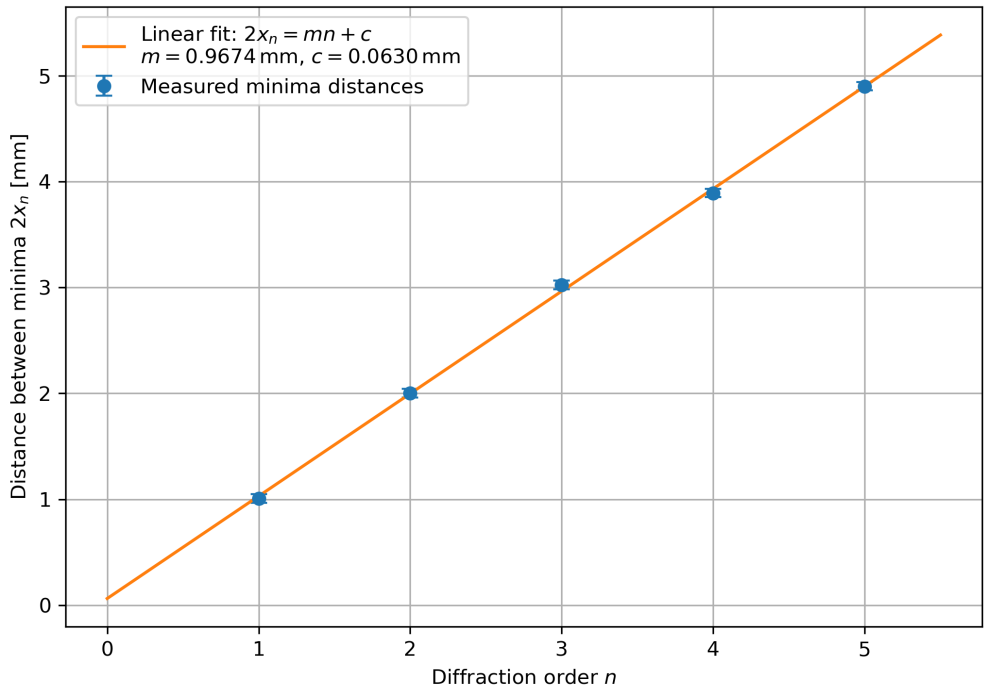


Figure 2: Linear fit of the distance $2x_n$ between the $\pm n$-th order minima as a function of diffraction order n.

Figure 3 shows the residuals of the linear fit. The residuals are small compared with the measured distances and are of the same order as the estimated position uncertainty.

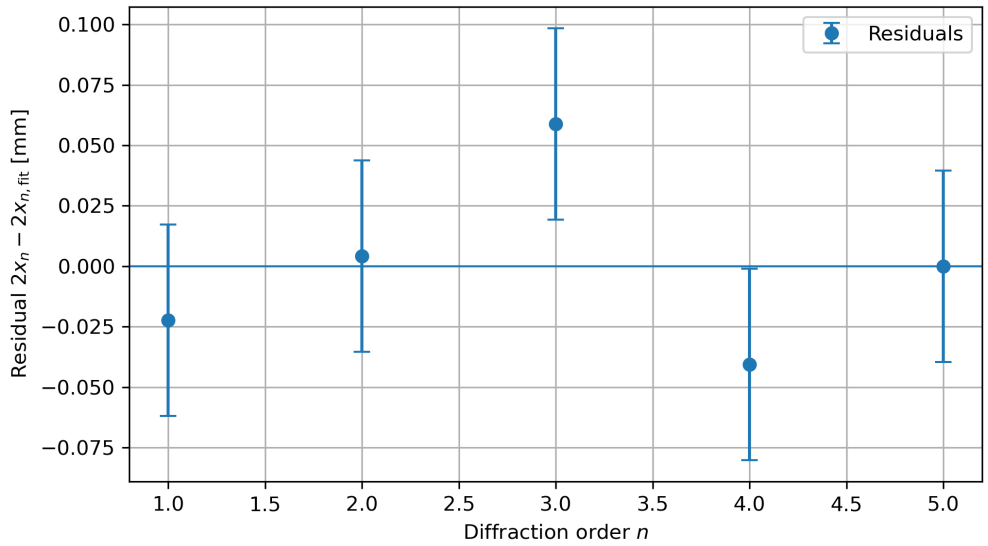


Figure 3: Residuals of the linear fit for the single-slit minima analysis.

4.7 Calculation of the Slit Width

Using

$$b = \frac{2f\lambda}{m},$$

with

$$f = 300.8 \text{ mm}, \qquad \lambda = 636 \text{ nm}, \qquad m = 0.9674 \text{ mm},$$

the experimental slit width was found to be

$$b_{\text{exp}} = 0.3955 \text{ mm}.$$

The uncertainty was calculated using Gaussian uncertainty propagation. Since

$$b = \frac{2f\lambda}{m},$$

the relative uncertainty is

$$\left(\frac{u(b)}{b}\right)^2 = \left(\frac{u(f)}{f}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2 + \left(\frac{u(m)}{m}\right)^2.$$

The wavelength uncertainty was not specified and was therefore neglected compared with the uncertainty in the focal length and fitted slope. Thus,

$$u(b) = b\sqrt{\left(\frac{u(f)}{f}\right)^2 + \left(\frac{u(m)}{m}\right)^2}.$$

Substituting

$$u(f) = 10 \text{ mm}, \qquad u(m) = 0.0125 \text{ mm},$$

gives

$$u(b) = 0.0141 \text{ mm}.$$

Therefore, the final result from the minima analysis is

$$\boxed{b_{\text{exp}} = (0.3955 \pm 0.0141) \text{ mm}}$$

or equivalently,

$$\boxed{b_{\text{exp}} = (395.5 \pm 14.1)\ \mu\text{m}.}$$

4.8 Comparison with Manufacturer Value

The manufacturer value of the slit width was

$$b_{\text{man}} = 0.400 \text{ mm}.$$

The absolute deviation is

$$\Delta b = |b_{\text{exp}} - b_{\text{man}}| = |0.3955 \text{ mm} - 0.400 \text{ mm}| = 0.0045 \text{ mm}.$$

The relative deviation is

$$\frac{\Delta b}{b_{\text{man}}} \times 100\% = \frac{0.0045 \text{ mm}}{0.400 \text{ mm}} \times 100\% = 1.12\%.$$

The manufacturer value lies within the experimental uncertainty interval,

$$0.3955 \text{ mm} \pm 0.0141 \text{ mm}.$$

Therefore, the measured slit width agrees with the manufacturer value within uncertainty.

4.9 Discussion

The measured distances $2x_n$ increase approximately linearly with the diffraction order n, as expected for Fraunhofer diffraction at a single slit. The fitted intercept,

$$c = (0.0630 \pm 0.0415) \text{ mm},$$

is small compared with the measured values of $2x_n$, although it is not exactly zero. A nonzero intercept may result from small asymmetries in the profile, imperfect centering of the main maximum, uncertainty in determining the minima, or slight deviations from the small-angle approximation.

The value obtained for the slit width,

$$b_{\text{exp}} = (0.3955 \pm 0.0141) \text{ mm},$$

is close to the manufacturer value of 0.400 mm. The relative deviation of 1.12% is small, and the two values are consistent within the experimental uncertainty. The dominant contribution to the uncertainty comes from the focal length uncertainty $u(f) = 10$ mm, which corresponds to a relative uncertainty of about 3.3%. The uncertainty from the fitted slope is smaller, about 1.3%.

The fit quality is very good, with

$$R^2 = 0.99999$$

and

$$\text{RMSE} = 0.0335 \text{ mm}.$$

The RMSE is comparable to the assigned uncertainty in the measured minima distances,

$$u(2x_n) = 0.0396 \text{ mm}.$$

This indicates that the assigned position uncertainty of 2 pixels per minimum is reasonable for this measurement.

Overall, the results confirm the expected Fraunhofer relation between the minima positions and diffraction order. The analysis satisfies the requirements of Task 1a by determining the distances $2x_n$ between the minima of order $\pm n$, fitting them linearly, reporting the fit parameters and fit quality, and using the fitted slope to obtain the slit width with uncertainty.

5 Determination of the Single-Slit Width from the Fourier Transform (Task 1b)

5.1 Goal of the Analysis

The aim of Task 1b was to determine the single-slit width b from the fast Fourier transform (FFT) of the measured Fraunhofer intensity profile. In Task 1a, the slit width was determined from the positions of the diffraction minima. In this task, the same measured Fraunhofer profile was analyzed in Fourier space.

According to the checklist, this analysis requires three main steps:

1. plotting the measured intensity profile as a function of the transverse wave vector k_x,
2. calculating and plotting the FFT of this profile with a physically meaningful unit,
3. determining the slit width from the FFT result and comparing it with the result from Task 1a.

The FFT result was compared with both the manufacturer value

$$b_{\text{man}} = 0.400 \text{ mm}$$

and the result from Task 1a,

$$b_{1a} = (0.3955 \pm 0.0141) \text{ mm}.$$

5.2 Relevant Theory

In the Fraunhofer regime, the electric-field amplitude behind a single slit is proportional to the Fourier transform of the aperture transmission function. For a slit of width b, the transmission function is

$$f(x') = \begin{cases} 1, & -\frac{b}{2} < x' < \frac{b}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

The Fraunhofer field amplitude is

$$\psi(k_x) \propto \mathcal{F}\{f(x')\}.$$

For a rectangular aperture, this gives

$$\psi(k_x) \propto b \,\text{sinc}\left(\frac{bk_x}{2}\right),$$

and therefore the measured intensity is

$$I(k_x) \propto \text{sinc}^2\left(\frac{bk_x}{2}\right).$$

The CCD camera measures intensity rather than field amplitude. Therefore,

$$I(k_x) \propto |\mathcal{F}\{f(x')\}|^2.$$

Using the convolution theorem, the inverse Fourier transform of the intensity profile is proportional to the autocorrelation of the aperture transmission function:

$$\mathcal{F}^{-1}\{I(k_x)\} \propto f(x') * f(-x').$$

Since the single-slit transmission function is symmetric,

$$f(x') = f(-x'),$$

this becomes

$$\mathcal{F}^{-1}\{I(k_x)\} \propto f(x') * f(x').$$

The autocorrelation of a rectangular slit is a triangular function. Its central triangular feature has zeros at

$$x' = \pm b.$$

Therefore, the slit width can be determined from the first zero, or minimum, of the FFT magnitude in the aperture-coordinate representation:

$$b_{\text{FFT}} = |x'_{\text{zero}}|.$$

5.3 Conversion from Camera Position to k_x

The recorded CSV file contained the pixel number and the corresponding intensity in arbitrary units. The new CCD camera was used, so the pixel width was

$$w_{\mathrm{pix}} = 14\ \mu\mathrm{m}.$$

The central maximum was found at

$$p_0 = 1127.$$

The camera position was therefore calculated as

$$x = (p - p_0)w_{\mathrm{pix}}.$$

For the Fraunhofer measurement, the transverse wave vector was calculated using the small-angle relation

$$k_x \approx \frac{2\pi}{\lambda}\frac{x}{f}.$$

Here,

$$\lambda = 636\ \mathrm{nm},$$

and

$$f = 300.8\ \mathrm{mm}.$$

The resulting k_x-axis spacing was

$$\Delta k_x = 459.804\ \mathrm{rad\,m^{-1}}.$$

The measured background intensity was estimated from the edges of the profile as

$$I_{\mathrm{bg}} = 0.00977\ \mathrm{a.u.}$$

The corrected intensity was then normalized using

$$I_{\mathrm{norm}} = \frac{I - I_{\mathrm{bg}}}{I_{\mathrm{max}} - I_{\mathrm{bg}}}.$$

5.4 FFT Method

The FFT was calculated from the normalized intensity profile $I(k_x)$. Since the Fourier transform was performed with respect to the variable k_x, the conjugate FFT coordinate has the dimension of length. This coordinate corresponds to the aperture coordinate x'.

For a discrete FFT, the aperture-coordinate spacing was obtained from the k_x-axis spacing and the number of data points. The dataset contained

$$N = 2048$$

points, and the FFT coordinate spacing was

$$\Delta x' = 0.00672\ \mathrm{mm}.$$

The FFT magnitude was normalized to its maximum value. The first minimum of the central triangular autocorrelation feature was then identified on the positive x'-side. This position was interpreted as the slit width:

$$b_{\mathrm{FFT}} = x'_{\mathrm{min}}.$$

5.5 Results

The main numerical results of the FFT analysis are summarized in Table 3.

Table 3: Results of the single-slit FFT analysis.

Quantity	Value	Unit
Number of data points N	2048	–
Central maximum pixel p_0	1127	pixel
Background intensity I_{bg}	0.00977	a.u.
k_x-axis spacing Δk_x	459.804	$\mathrm{rad\,m^{-1}}$
FFT coordinate spacing $\Delta x'$	0.00672	mm
FFT resolution uncertainty	0.00336	mm
Slit width from FFT b_{FFT}	0.4738	mm
Uncertainty $u(b_{\mathrm{FFT}})$	0.0161	mm
Slit width from Task 1a b_{1a}	0.3955	mm
Uncertainty $u(b_{1a})$	0.0141	mm
Manufacturer slit width b_{man}	0.4000	mm
Relative deviation from manufacturer	18.43	%
Relative difference from Task 1a	19.78	%
Normalized difference from Task 1a	3.65	–

The slit width obtained from the FFT was

$$\boxed{b_{\mathrm{FFT}} = (0.4738 \pm 0.0161)\ \mathrm{mm}.}$$

5.6 Plots

Figure 4 shows the measured single-slit Fraunhofer intensity profile as a function of the transverse wave vector k_x. This plot is required because the FFT was performed on the intensity profile in k_x-space.

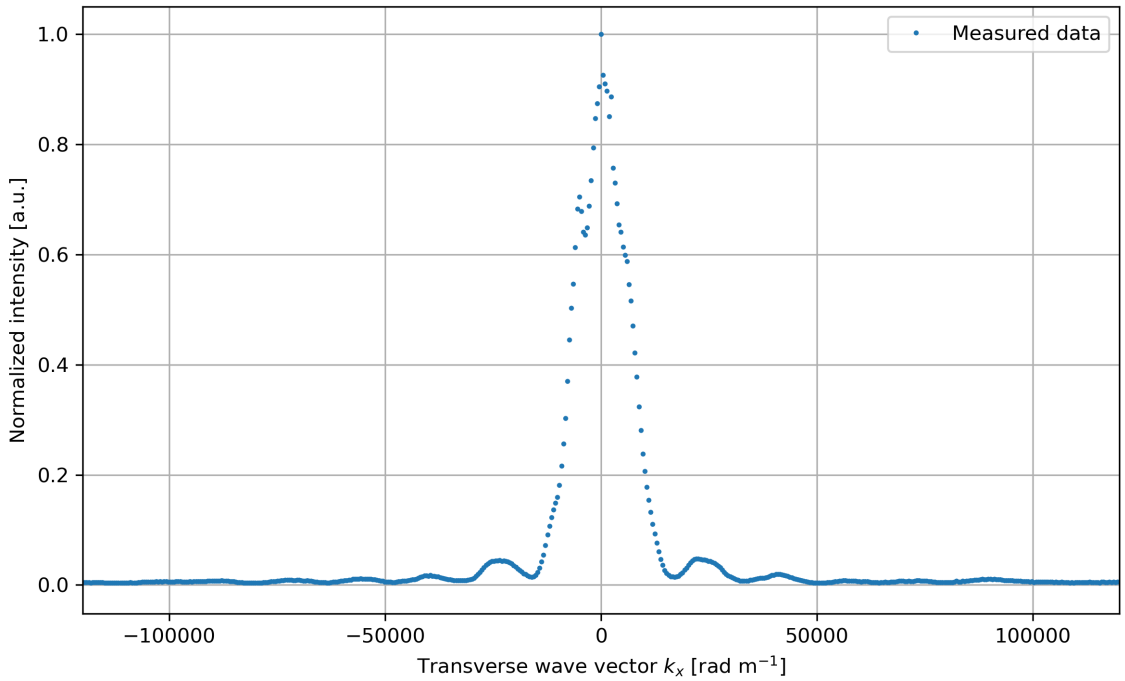


Figure 4: Measured single-slit Fraunhofer intensity profile plotted as a function of the transverse wave vector k_x.

Figure 5 shows the FFT magnitude of the measured intensity profile as a function of the aperture coordinate x'. The central feature corresponds to the autocorrelation of the slit transmission function. For an ideal single slit, the first zero of this autocorrelation feature occurs at $|x'| = b$.

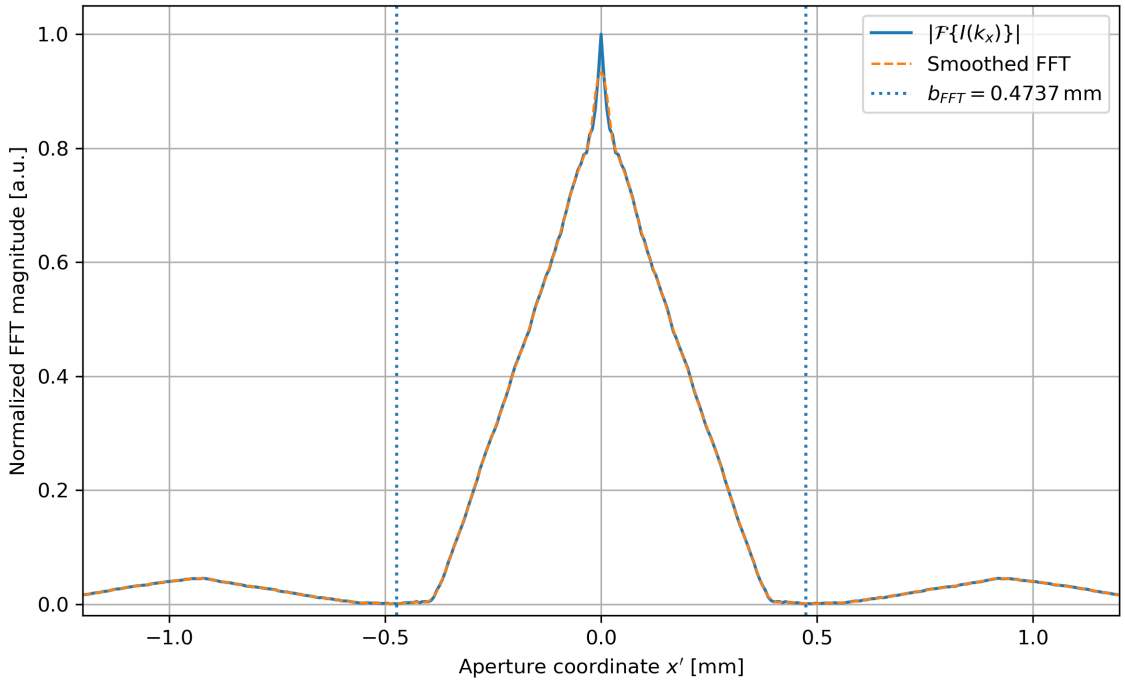


Figure 5: Normalized FFT magnitude of the measured single-slit intensity profile as a function of aperture coordinate x'.

Figure 6 shows a zoomed view of the positive first zero region used to determine b_{FFT}.

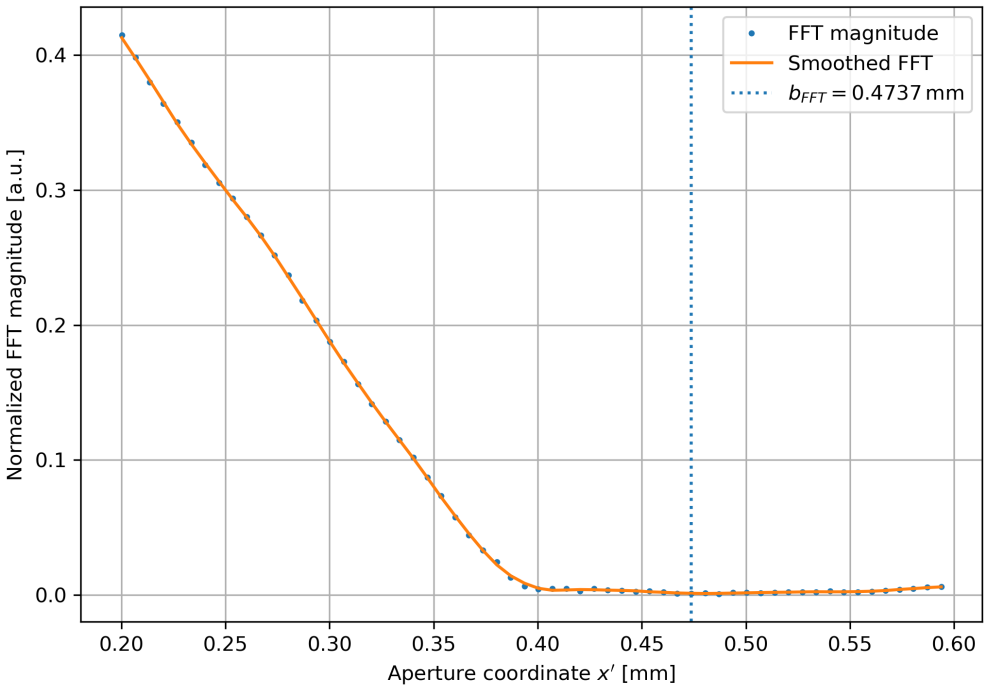


Figure 6: Zoomed view of the FFT magnitude near the positive first zero of the autocorrelation feature.

5.7 Uncertainty Analysis

The uncertainty of the FFT result has two main contributions. The first is the finite resolution of the FFT coordinate axis. Since the spacing of the aperture coordinate was

$$\Delta x' = 0.00672 \text{ mm},$$

the resolution uncertainty was estimated as half of this spacing:

$$u_{\mathrm{res}}(b) = \frac{\Delta x'}{2} = 0.00336 \text{ mm}.$$

The second contribution comes from the scaling of the FFT coordinate. Since the k_x-axis was calculated using

$$k_x \approx \frac{2\pi x}{\lambda f},$$

the conjugate coordinate x' depends on the focal length, wavelength, and pixel width. The focal length uncertainty was

$$u(f) = 10 \text{ mm}.$$

The wavelength and pixel-width uncertainties were not specified and were therefore neglected in this analysis. The total uncertainty was calculated as

$$u(b_{\text{FFT}}) = b_{\text{FFT}} \sqrt{\left(\frac{u(f)}{f}\right)^2 + \left(\frac{u_{\text{res}}(b)}{b_{\text{FFT}}}\right)^2}.$$

Substituting

$$b_{\text{FFT}} = 0.4738 \text{ mm}, \qquad f = 300.8 \text{ mm}, \qquad u(f) = 10 \text{ mm}, \qquad u_{\text{res}}(b) = 0.00336 \text{ mm},$$

gives

$$u(b_{\text{FFT}}) = 0.0161 \text{ mm}.$$

Therefore,

$$b_{\text{FFT}} = (0.4738 \pm 0.0161) \text{ mm}.$$

5.8 Comparison with Task 1a and Manufacturer Value

The value obtained from the minima analysis in Task 1a was

$$b_{1a} = (0.3955 \pm 0.0141) \text{ mm}.$$

The absolute difference between the FFT result and the Task 1a result is

$$\Delta b = |b_{\text{FFT}} - b_{1a}| = |0.4738 \text{ mm} - 0.3955 \text{ mm}| = 0.0782 \text{ mm}.$$

The relative difference is

$$\frac{\Delta b}{b_{1a}} \times 100\% = 19.78\%.$$

The combined uncertainty is

$$u_{\text{comb}} = \sqrt{u(b_{\text{FFT}})^2 + u(b_{1a})^2}.$$

Using the measured uncertainties,

$$u_{\text{comb}} = \sqrt{(0.0161 \text{ mm})^2 + (0.0141 \text{ mm})^2} = 0.0214 \text{ mm}.$$

The normalized difference is

$$\frac{\Delta b}{u_{\text{comb}}} = 3.65.$$

Thus, the FFT result differs from the Task 1a result by approximately 3.65 combined standard uncertainties.

The deviation from the manufacturer value is

$$|b_{\text{FFT}} - b_{\text{man}}| = |0.4738 \text{ mm} - 0.4000 \text{ mm}| = 0.0738 \text{ mm}.$$

The relative deviation from the manufacturer value is

$$\frac{0.0738 \text{ mm}}{0.4000 \text{ mm}} \times 100\% = 18.43\%.$$

5.9 Discussion

The FFT analysis successfully produced an autocorrelation-like feature from the measured single-slit Fraunhofer intensity profile. The FFT coordinate has the correct physical unit of length, and the first minimum of the central feature was used to estimate the slit width. However, the resulting value,

$$b_{\mathrm{FFT}} = (0.4738 \pm 0.0161)\ \mathrm{mm},$$

is significantly larger than both the Task 1a result and the manufacturer value. The difference from Task 1a corresponds to 3.65 combined standard uncertainties, indicating that the discrepancy is unlikely to be explained by random uncertainty alone.

Several systematic effects may explain this difference. First, the FFT method is sensitive to background subtraction and finite data-window effects. Since the measured intensity profile is only available over a finite CCD range, the FFT can be broadened or distorted by truncation at the edges. Second, any residual background, asymmetry, or noise in the outer parts of the diffraction pattern can influence the position of the first minimum in the FFT magnitude. Third, smoothing was used only to locate the FFT minimum; although this helps reduce noise, it can also shift shallow minima slightly. Fourth, the real experimental diffraction profile is not an ideal infinite-range sinc-squared function, so the autocorrelation feature may not have a perfectly sharp zero at $x' = \pm b$.

The Task 1a method is based directly on the minima positions in the measured diffraction pattern and gave

$$b_{1a} = (0.3955 \pm 0.0141)\ \mathrm{mm},$$

which agrees well with the manufacturer value of 0.400 mm. Therefore, for this dataset, the minima method appears to provide a more reliable estimate of the slit width than the FFT zero-position method.

Nevertheless, Task 1b demonstrates the expected Fourier-transform relationship between the Fraunhofer intensity profile and the real-space autocorrelation of the slit. The required $I(k_x)$ plot and FFT plot were produced, the FFT result was given with units, and the extracted slit width was quantitatively compared with the result from Task 1a.

6 Fresnel Diffraction of a Single Slit (Task 1c)

6.1 Goal of the Analysis

The aim of Task 1c was to measure the Fresnel diffraction pattern of the single slit at three different propagation distances and compare the measured intensity profiles with the theoretical Fresnel diffraction curves. In contrast to Tasks 1a and 1b, the lens was removed for this measurement, so the CCD camera was placed directly behind the slit. Therefore, the finite propagation distance z had to be included explicitly in the diffraction model.

The three measured distances were

$$z_1 = 27\,\mathrm{cm}, \qquad z_2 = 31\,\mathrm{cm}, \qquad z_3 = 35\,\mathrm{cm}.$$

Following the instruction for this task, the slit width determined from the Fraunhofer analysis was used in the Fresnel theory calculation:

$$b = (0.3955 \pm 0.0141)\,\mathrm{mm}.$$

The goal was not to determine a new slit width, but to test whether the measured near-field diffraction profiles agree with the Fresnel diffraction model.

6.2 Relevant Theory

In the Fresnel regime, the observation distance is finite, so the curvature of the wavefronts must be retained. The field at the observation point x behind an aperture is proportional to

$$\psi(x, z) \propto \int f(x') \exp\left(i \frac{\pi}{\lambda z}(x - x')^2\right) dx',$$

where x' is the coordinate in the aperture plane, x is the coordinate in the CCD plane, z is the propagation distance, and λ is the laser wavelength.

For a single slit of width b, the transmission function is

$$f(x') = \begin{cases} 1, & -\frac{b}{2} < x' < \frac{b}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, the Fresnel field is

$$\psi(x,z) \propto \int_{-b/2}^{b/2} \exp\left(i\frac{\pi}{\lambda z}(x-x')^2\right)\, dx'.$$

This integral can be expressed using the complex Fresnel integral

$$F(u) = C(u) + iS(u),$$

where

$$C(u) = \int_0^u \cos\left(\frac{\pi t^2}{2}\right)\, dt,$$

and

$$S(u) = \int_0^u \sin\left(\frac{\pi t^2}{2}\right)\, dt.$$

The theoretical single-slit Fresnel intensity is then

$$I(x,z) \propto \left| F\left(\frac{b+2x}{\sqrt{2\lambda z}}\right) + F\left(\frac{b-2x}{\sqrt{2\lambda z}}\right)\right|^2.$$

Since the measured CCD intensity is given in arbitrary units, the theoretical curve was scaled and shifted before comparison with the data. The fit model used for the comparison was

$$I_{\mathrm{fit}}(x) = A\, I_{\mathrm{th}}(x - x_0, z; b, \lambda) + C,$$

where A is an amplitude scaling factor, C is a constant background offset, and x_0 is a small horizontal offset accounting for imperfect centering of the measured profile. The slit width b, wavelength λ, and propagation distance z were kept fixed to their measured values.

6.3 Fresnel Number

The Fresnel number for a single slit is

$$N_b = \frac{b^2}{4\lambda z}.$$

It indicates whether the diffraction pattern is in the near-field or far-field regime. Using

$$b = (0.3955 \pm 0.0141)\,\mathrm{mm}, \qquad \lambda = 636\,\mathrm{nm},$$

the Fresnel numbers were calculated for each measured distance. The uncertainty was propagated using

$$u(N_b) = N_b\sqrt{\left(2\frac{u(b)}{b}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2 + \left(\frac{u(z)}{z}\right)^2}.$$

The wavelength uncertainty was not specified and was neglected. The distance uncertainty was taken as

$$u(z) = 0.5\,\mathrm{cm}.$$

The calculated Fresnel numbers were all below one, but not negligibly small, which means that finite-distance Fresnel effects are still relevant.

6.4 Experimental Parameters and Data Treatment

The new CCD camera was used, so the pixel width was

$$w_{\mathrm{pix}} = 14\,\mu\mathrm{m}.$$

Each CSV file contained the pixel number and intensity in arbitrary units. For each profile, the pixel position of the maximum intensity was used as the center p_0. The transverse position was calculated using

$$x = (p - p_0)w_{\mathrm{pix}}.$$

The central pixels were found to be

$$p_0 = 1021 \quad \text{for } z = 27\,\mathrm{cm},$$

$$p_0 = 1029 \quad \text{for } z = 31\,\text{cm},$$

and

$$p_0 = 1035 \quad \text{for } z = 35\,\text{cm}.$$

The background intensity was estimated from the outer regions of each profile and subtracted. The background-corrected intensity was normalized according to

$$I_{\text{norm}} = \frac{I - I_{\text{bg}}}{I_{\text{max}} - I_{\text{bg}}}.$$

The noise level was estimated from the standard deviation of the normalized background regions. The theoretical Fresnel curves were then fitted to the normalized data using only the amplitude A, offset C, and horizontal shift x_0 as free parameters.

6.5 Results

The results of the Fresnel-profile comparison are summarized in Table 4. The fit quality was evaluated using the root-mean-square error and the coefficient of determination R^2.

Table 4: Results of the single-slit Fresnel diffraction analysis.

z [cm]	p_0	I_{bg} [a.u.]	N_b	$u(N_b)$	x_0 [mm]	RMSE
27	1021	0.00663	0.22774	0.01679	0.02203	0.003264
31	1029	0.00602	0.19835	0.01451	-0.00997	0.003332
35	1035	0.00758	0.17568	0.01278	-0.00913	0.002665

The fitted amplitude and offset parameters are given in Table 5.

Table 5: Fit parameters for the Fresnel theory comparison.

z [cm]	A	$u(A)$	C	$u(C)$	R^2
27	0.99488	0.00072	0.001525	0.000073	0.998925
31	0.97456	0.00069	0.001220	0.000075	0.998975
35	0.97507	0.00052	0.001783	0.000060	0.999417

The small RMSE values and high R^2 values indicate very good agreement between the measured Fresnel profiles and the theoretical model.

6.6 Plots

Figure 7 shows the measured Fresnel profile at $z = 27\,\text{cm}$ together with the theoretical Fresnel curve. The uncertainty band was calculated by varying the slit width and propagation distance within their uncertainties.

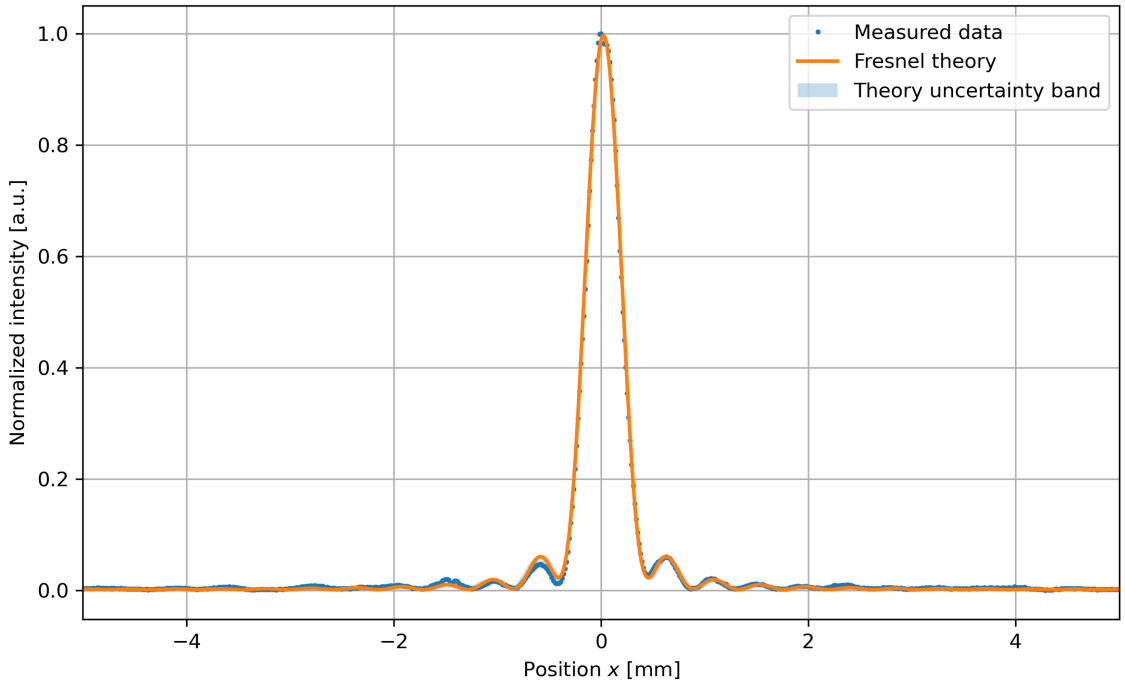


Figure 7: Measured and theoretical single-slit Fresnel diffraction profile at $z = 27\,\mathrm{cm}$.

Figure 8 shows the corresponding comparison for $z = 31\,\mathrm{cm}$.

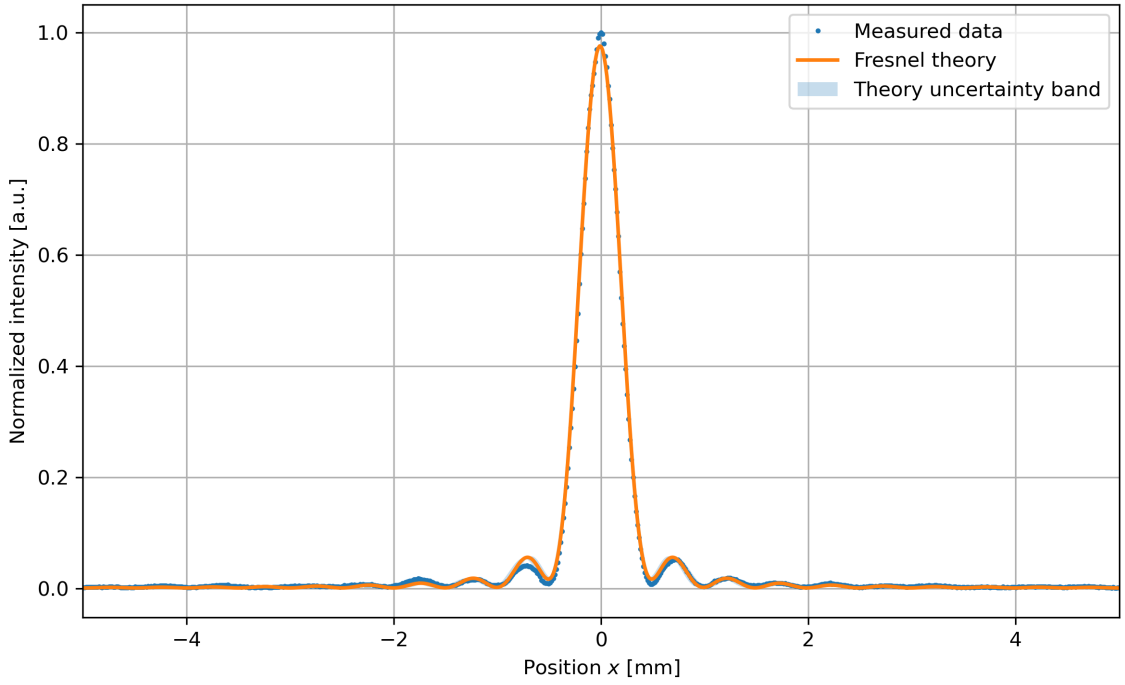


Figure 8: Measured and theoretical single-slit Fresnel diffraction profile at $z = 31\,\mathrm{cm}$.

Figure 9 shows the comparison for $z = 35\,\mathrm{cm}$.

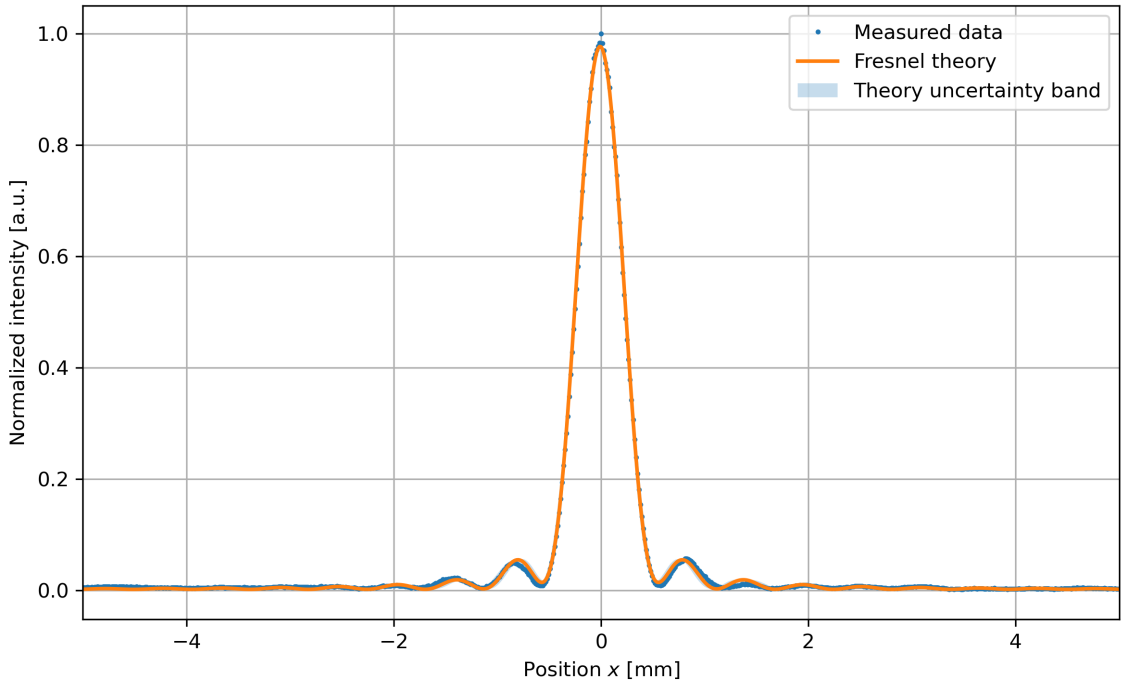


Figure 9: Measured and theoretical single-slit Fresnel diffraction profile at $z = 35\,\text{cm}$.

For direct comparison, all three measured Fresnel profiles and their corresponding theory curves are shown together in Figure 10. This plot illustrates the gradual change in the diffraction profile as the propagation distance increases.

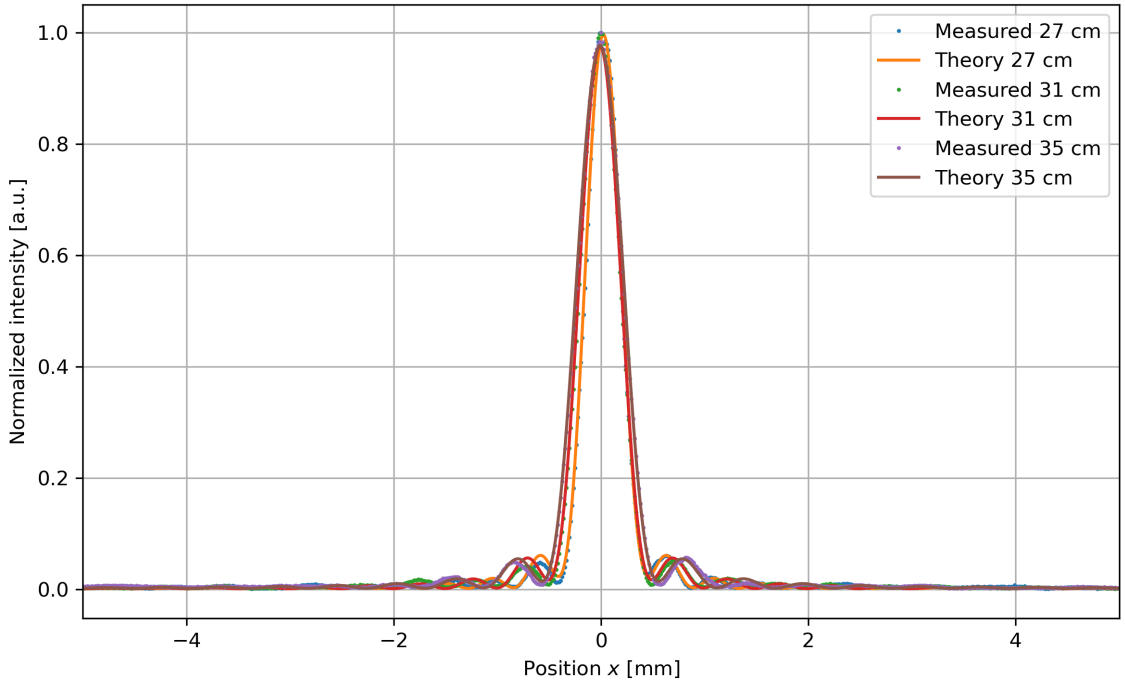


Figure 10: Comparison of all three measured single-slit Fresnel diffraction profiles with their corresponding theoretical curves.

The residuals for the three fits are shown in Figures 11, 12, and 13. These plots show the difference between the normalized measured intensity and the fitted Fresnel theory curve.

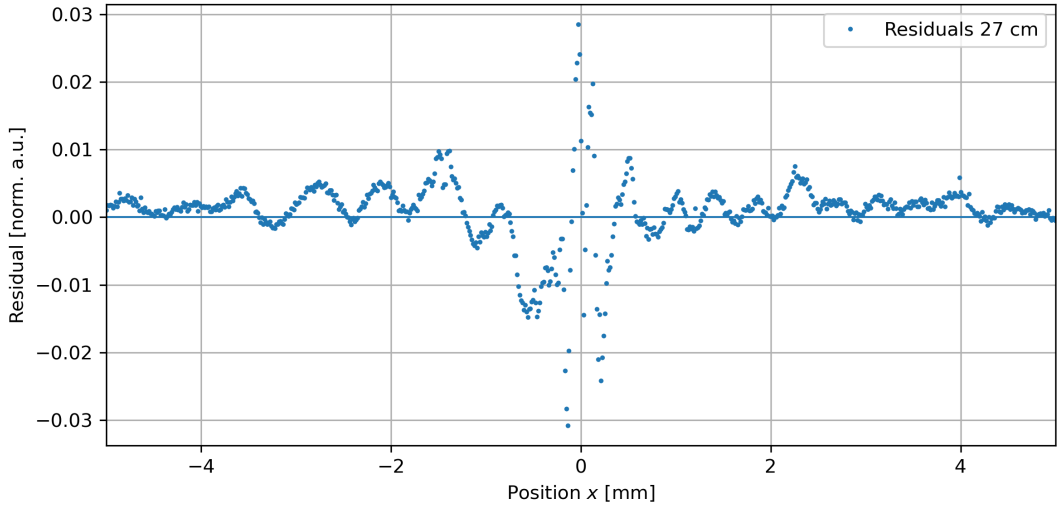


Figure 11: Residuals of the Fresnel theory comparison at $z = 27\,\mathrm{cm}$.

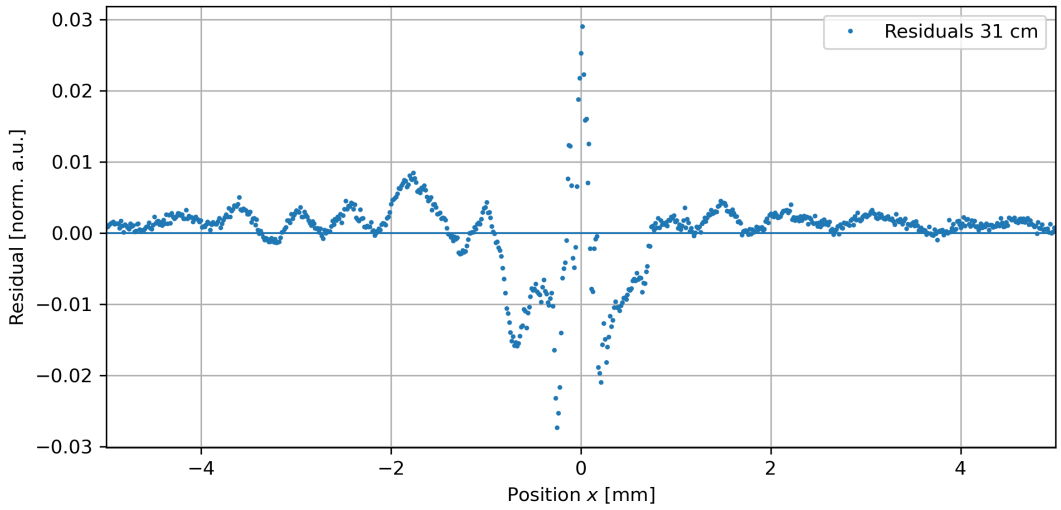


Figure 12: Residuals of the Fresnel theory comparison at $z = 31\,\mathrm{cm}$.

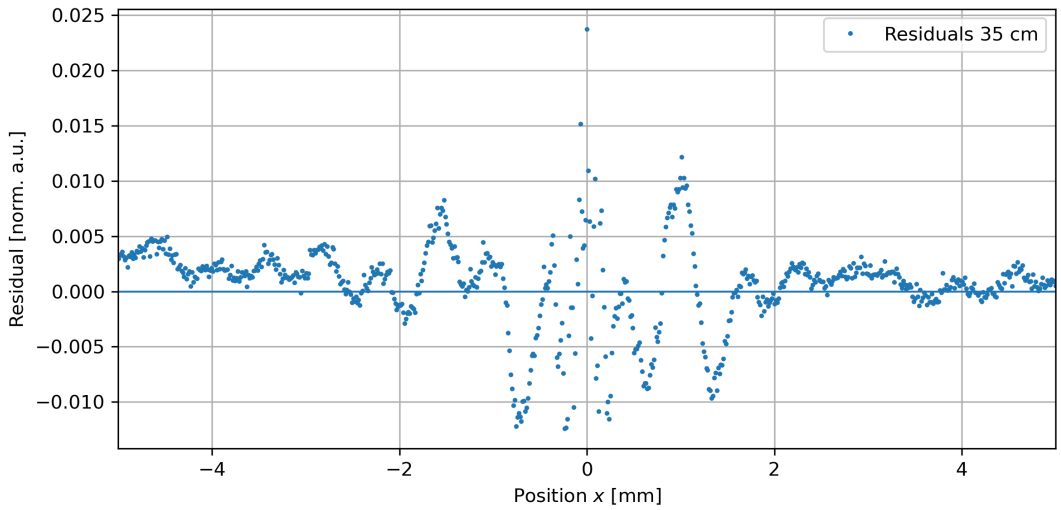


Figure 13: Residuals of the Fresnel theory comparison at $z = 35\,\mathrm{cm}$.

The Fresnel number as a function of propagation distance is shown in Figure 14. As expected from

$$N_b \propto \frac{1}{z},$$

the Fresnel number decreases as the camera is moved farther away from the slit.

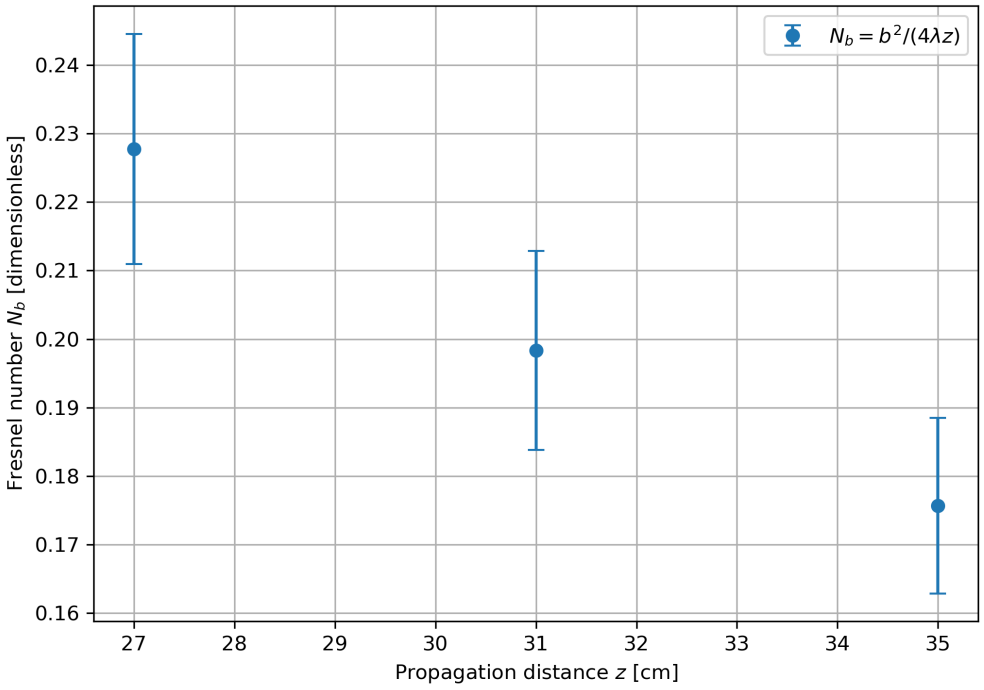


Figure 14: Fresnel number $N_b = b^2/(4\lambda z)$ as a function of propagation distance.

6.7 Uncertainty Analysis

The uncertainty analysis for this task includes the uncertainties in the slit width, propagation distance, centering, and intensity noise.

The slit width used in the theory was taken from Task 1a:

$$b = (0.3955 \pm 0.0141)\,\text{mm}.$$

The propagation distance uncertainty was estimated as

$$u(z) = 0.5\,\text{cm}.$$

The uncertainty in locating the profile center was taken as

$$u(p_0) = 2 \text{ pixels}.$$

Since the pixel width was

$$w_{\text{pix}} = 14\,\mu\text{m},$$

this corresponds to a centering uncertainty of

$$u(x_0) = 2 \cdot 14\,\mu\text{m} = 28\,\mu\text{m}.$$

The uncertainty in the Fresnel number was calculated from

$$N_b = \frac{b^2}{4\lambda z}.$$

Using Gaussian error propagation,

$$u(N_b) = N_b \sqrt{\left(2\frac{u(b)}{b}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2 + \left(\frac{u(z)}{z}\right)^2}.$$

Because no wavelength uncertainty was specified, $u(\lambda)$ was neglected. The resulting uncertainties are included in Table 4.

The theory uncertainty bands shown in the profile plots were calculated by varying b and z within their uncertainties. This gives a visual estimate of how much the theoretical Fresnel curve changes due to the main experimental uncertainties.

The intensity noise was estimated from the background regions of the normalized profile. The estimated normalized noise levels were

$$u(I) = 0.000759 \quad \text{for } z = 27\,\text{cm},$$

$$u(I) = 0.001094 \quad \text{for } z = 31\,\text{cm},$$

and

$$u(I) = 0.001151 \quad \text{for } z = 35\,\text{cm}.$$

These values are smaller than the RMSE values of the theory comparison, indicating that the residual deviations are not caused only by random detector noise but also include small systematic differences between the ideal Fresnel model and the measured profile.

6.8 Interpretation and Discussion

The measured Fresnel profiles at all three distances agree well with the theoretical single-slit Fresnel diffraction model. The fitted amplitude factors were close to one, and the fitted offsets were small. The horizontal shifts were also small, with the largest shift being

$$x_0 = 0.0220\,\text{mm}$$

at $z = 27\,\text{cm}$. This is comparable to the expected uncertainty from centering and alignment.

The fit quality was high for all three distances. The coefficient of determination was

$$R^2 = 0.998925$$

for $z = 27\,\text{cm}$,

$$R^2 = 0.998975$$

for $z = 31\,\text{cm}$, and

$$R^2 = 0.999417$$

for $z = 35\,\text{cm}$. The best agreement was obtained at $z = 35\,\text{cm}$, where the RMSE was lowest:

$$\text{RMSE} = 0.002665.$$

The Fresnel number decreased from

$$N_b = 0.2277$$

at $z = 27\,\text{cm}$ to

$$N_b = 0.1757$$

at $z = 35\,\text{cm}$. This trend is expected because the Fresnel number is inversely proportional to the propagation distance. Although all values are below one, they are still sufficiently large that the finite-distance Fresnel structure is observable.

The main possible sources of systematic deviation are imperfect alignment of the slit and CCD camera, uncertainty in the exact slit-camera distance, uncertainty in the position of the CCD sensor plane, background subtraction, and the use of a one-dimensional ideal slit model. The finite height of the CCD pixels and the finite vertical extent of the laser beam may also affect the measured intensity profile. In addition, the theory assumes uniform illumination of the slit, whereas the real laser beam may have a nonuniform transverse intensity distribution.

Overall, Task 1c satisfies the required comparison between measured Fresnel profiles and theory. Three profiles at different propagation distances were recorded, the corresponding theoretical Fresnel curves were calculated using the slit width from the Fraunhofer analysis, uncertainties were included through the Fresnel number and theory bands, and the agreement was quantified using RMSE and R^2.

7 Determination of the Double-Slit Separation from Fraunhofer Minima (Task 2a)

7.1 Goal of the Analysis

The aim of Task 2a was to determine the geometrical parameter of the double slit from the positions of the minima in the Fraunhofer diffraction pattern. The double slit used in the experiment had manufacturer-specified values

$$g_{\text{man}} = 0.750 \text{ mm}, \qquad b_{\text{man}} = 0.200 \text{ mm},$$

where g is the center-to-center slit separation and b is the width of each slit.

7.2 Relevant Theory

For a double slit, the Fraunhofer intensity distribution is given by

$$I(k_x) = I_0 \text{sinc}^2\left(\frac{bk_x}{2}\right)\cos^2\left(\frac{gk_x}{2}\right) + I_{\text{bg}}.$$

The factor

$$\text{sinc}^2\left(\frac{bk_x}{2}\right)$$

is the single-slit diffraction envelope, while

$$\cos^2\left(\frac{gk_x}{2}\right)$$

describes the interference fringes due to the separation of the two slits.

The minima of the interference term occur when

$$\cos^2\left(\frac{gk_x}{2}\right) = 0.$$

This gives

$$\frac{gk_x}{2} = \left(n + \frac{1}{2}\right)\pi, \qquad n = 0, 1, 2, \ldots$$

Using

$$k_x = \frac{2\pi}{\lambda}\sin\alpha,$$

the condition for destructive interference becomes

$$g\sin\alpha_n = \left(n + \frac{1}{2}\right)\lambda.$$

In the focal plane of a lens,

$$x_n = f\tan\alpha_n.$$

For small diffraction angles,

$$\sin\alpha_n \approx \tan\alpha_n \approx \frac{x_n}{f}.$$

Therefore,

$$g\frac{x_n}{f} \approx \left(n + \frac{1}{2}\right)\lambda.$$

Solving for x_n gives

$$x_n \approx \frac{f\lambda}{g}\left(n + \frac{1}{2}\right).$$

The measured distance between symmetric minima of order $+n$ and $-n$ is

$$2x_n \approx \frac{2f\lambda}{g}\left(n + \frac{1}{2}\right).$$

This is a linear relation of the form

$$2x_n = m\left(n + \frac{1}{2}\right) + c,$$

where the slope is

$$m = \frac{2f\lambda}{g}.$$

Thus, the slit separation is

$$g = \frac{2f\lambda}{m}.$$

7.3 Experimental Parameters

The Fraunhofer diffraction profile was recorded using the new CCD line camera, for which the pixel width is

$$w_{\mathrm{pix}} = 14\ \mu\mathrm{m}.$$

The central maximum of the recorded profile was found at

$$p_0 = 1125.$$

The position axis was obtained from the pixel coordinate using

$$x = (p - p_0)w_{\mathrm{pix}}.$$

The laser wavelength and lens focal length were

$$\lambda = 636\ \mathrm{nm},$$

and

$$f = 300.8\ \mathrm{mm}.$$

The uncertainty in the focal length was taken as

$$u(f) = 10\ \mathrm{mm}.$$

The uncertainty in locating each minimum was estimated as

$$u(p) = 2\ \mathrm{pixels}.$$

This corresponds to a position uncertainty of

$$u(x) = u(p)w_{\mathrm{pix}} = 2 \cdot 14\ \mu\mathrm{m} = 28\ \mu\mathrm{m}.$$

Since the measured quantity $2x_n$ is the distance between a left and right minimum, its uncertainty is

$$u(2x_n) = \sqrt{u(x_{+n})^2 + u(x_{-n})^2}.$$

Assuming equal uncertainties for both minima,

$$u(2x_n) = \sqrt{2} \cdot 28\ \mu\mathrm{m} = 39.6\ \mu\mathrm{m} = 0.0396\ \mathrm{mm}.$$

7.4 Method

The double-slit Fraunhofer profile was first centered at the principal maximum. The expected positions of the interference minima were estimated using the manufacturer slit separation

$$g_{\mathrm{man}} = 0.750\ \mathrm{mm}.$$

For the n-th interference minimum, the expected position is

$$x_n \approx \frac{f\lambda}{g_{\mathrm{man}}}\left(n + \frac{1}{2}\right).$$

This estimate was used only to guide the search for the minima in the measured profile. The actual minimum positions were then identified from the measured intensity profile.

The minima from $n = 0$ to $n = 8$ were identified on both sides of the central maximum. The distance between each pair of symmetric minima was calculated as

$$2x_n = x_{+n} - x_{-n}.$$

A weighted linear fit of the form

$$2x_n = m\left(n + \frac{1}{2}\right) + c$$

was then performed. The fitted slope was used to calculate the experimental slit separation g.

7.5 Measured Minima and Linear Fit

The measured minima positions and corresponding distances are shown in Table 6. All positions are given relative to the central maximum.

Table 6: Measured positions of the double-slit interference minima and calculated distances $2x_n$.

Order n	$n + \frac{1}{2}$	Left pixel	Right pixel	x_{-n} [mm]	x_{+n} [mm]	$2x_n$ [mm]
0	0.5	1116	1135	-0.126	0.140	0.266
1	1.5	1098	1154	-0.378	0.406	0.784
2	2.5	1079	1173	-0.644	0.672	1.316
3	3.5	1058	1193	-0.938	0.952	1.890
4	4.5	1043	1204	-1.148	1.106	2.254
5	5.5	1024	1225	-1.414	1.400	2.814
6	6.5	1006	1244	-1.666	1.666	3.332
7	7.5	988	1266	-1.918	1.974	3.892
8	8.5	971	1275	-2.156	2.100	4.256

The fit model was

$$2x_n = m\left(n + \frac{1}{2}\right) + c.$$

The fitted parameters were

$$m = (0.5040 \pm 0.0051)\ \text{mm},$$

and

$$c = (0.0436 \pm 0.0265)\ \text{mm}.$$

The fit quality parameters were

$$\text{SSE} = 2.1342 \times 10^{-8}\ \text{m}^2,$$

$$\text{MSE} = 2.3716 \times 10^{-9}\ \text{m}^2,$$

$$\text{RMSE} = 0.0487\ \text{mm},$$

and

$$R^2 = 0.99860.$$

7.6 Plots

Figure 15 shows the measured double-slit Fraunhofer profile. The detected left and right interference minima used for the analysis are marked. The measured data are shown as markers, while the smoothed curve was used only to assist the identification of minima.

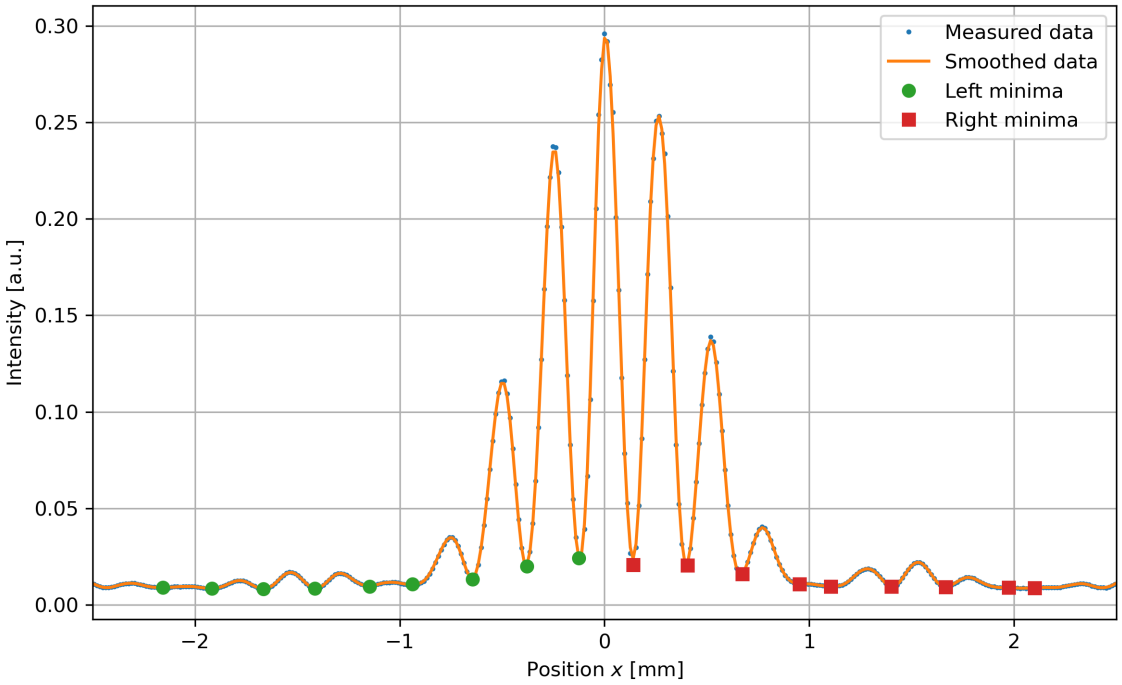


Figure 15: Measured double-slit Fraunhofer diffraction profile with the detected interference minima.

Figure 16 shows the measured distances $2x_n$ as a function of $n + \frac{1}{2}$, together with the weighted linear fit.

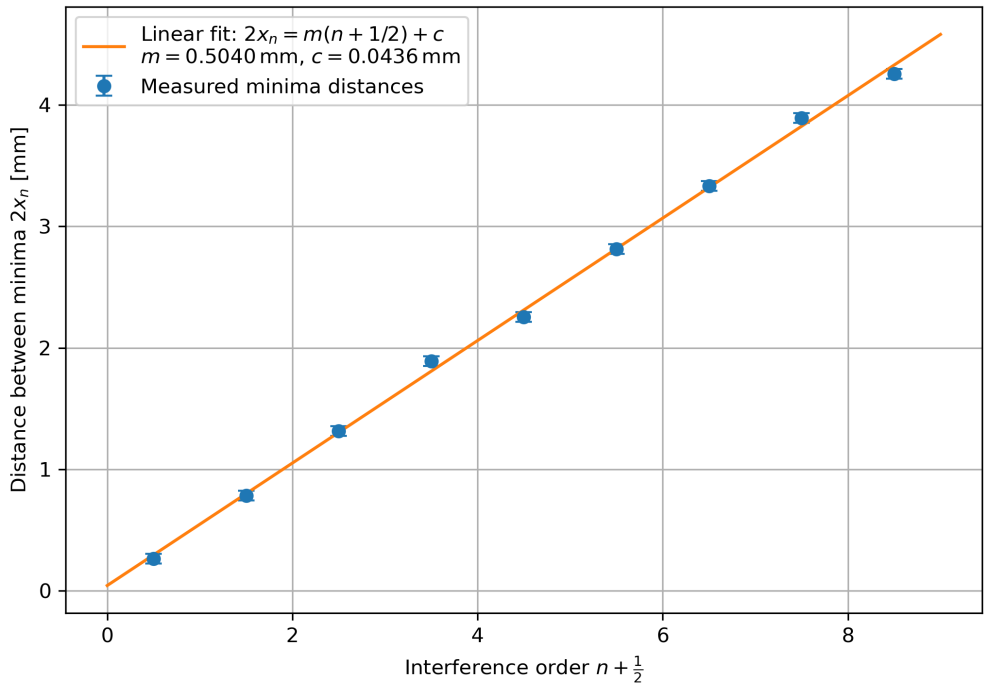


Figure 16: Linear fit of the distance $2x_n$ between symmetric double-slit interference minima as a function of $n + \frac{1}{2}$.

Figure 17 shows the residuals of the linear fit.

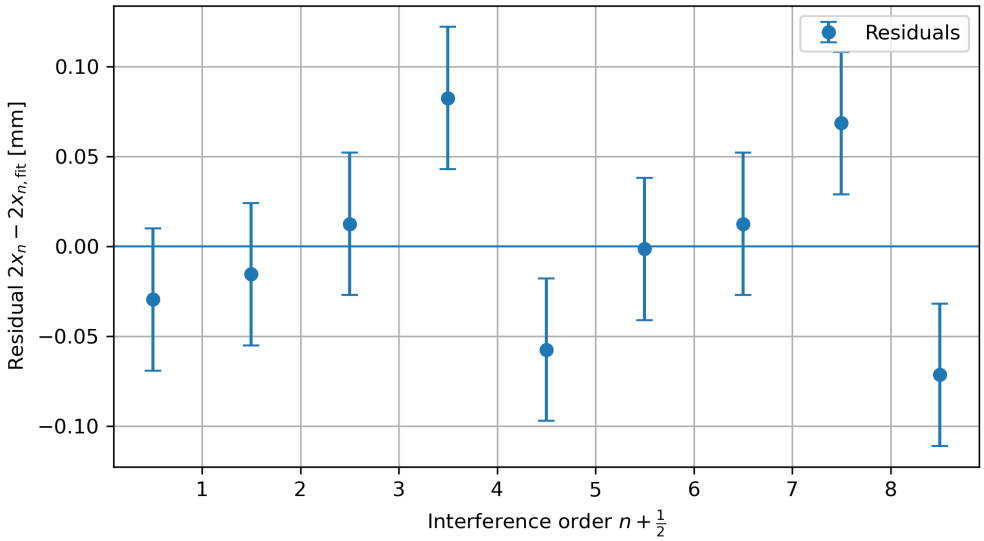


Figure 17: Residuals of the linear fit used to determine the double-slit separation.

7.7 Calculation of the Slit Separation

The slit separation was calculated from the fitted slope using

$$g = \frac{2f\lambda}{m}.$$

Substituting

$$f = 300.8 \text{ mm}, \qquad \lambda = 636 \text{ nm}, \qquad m = 0.5040 \text{ mm},$$

gives

$$g_{\text{exp}} = 0.7592 \text{ mm}.$$

The uncertainty was calculated by Gaussian uncertainty propagation. Since

$$g = \frac{2f\lambda}{m},$$

the relative uncertainty is

$$\left(\frac{u(g)}{g}\right)^2 = \left(\frac{u(f)}{f}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2 + \left(\frac{u(m)}{m}\right)^2.$$

The wavelength uncertainty was not specified and was neglected. Therefore,

$$u(g) = g\sqrt{\left(\frac{u(f)}{f}\right)^2 + \left(\frac{u(m)}{m}\right)^2}.$$

Using

$$u(f) = 10 \text{ mm}, \qquad u(m) = 0.0051 \text{ mm},$$

gives

$$u(g) = 0.0264 \text{ mm}.$$

Thus, the final result from the minima analysis is

$$\boxed{g_{\text{exp}} = (0.7592 \pm 0.0264) \text{ mm}}$$

or equivalently,

$$\boxed{g_{\text{exp}} = (759.2 \pm 26.4)\ \mu\text{m}.}$$

7.8 Comparison with Manufacturer Value

The manufacturer value of the slit separation was

$$g_{\text{man}} = 0.750 \text{ mm}.$$

The absolute deviation is

$$\Delta g = |g_{\text{exp}} - g_{\text{man}}| = |0.7592 \text{ mm} - 0.7500 \text{ mm}| = 0.0092 \text{ mm}.$$

The relative deviation is

$$\frac{\Delta g}{g_{\text{man}}} \times 100\% = \frac{0.0092 \text{ mm}}{0.7500 \text{ mm}} \times 100\% = 1.22\%.$$

The manufacturer value lies within the uncertainty interval of the measured value,

$$0.7592 \text{ mm} \pm 0.0264 \text{ mm}.$$

Therefore, the measured slit separation agrees with the manufacturer value within uncertainty.

7.9 Discussion

The measured minima distances $2x_n$ increase approximately linearly with $n+\frac{1}{2}$, as expected for destructive interference in a double-slit Fraunhofer pattern. The high coefficient of determination,

$$R^2 = 0.99860,$$

shows that the linear model describes the measured data well. The RMSE of the fit was

$$\text{RMSE} = 0.0487 \text{ mm},$$

which is slightly larger than the assigned uncertainty

$$u(2x_n) = 0.0396 \text{ mm}.$$

This suggests that, in addition to the estimated pixel uncertainty, small systematic effects may be present.

The fitted intercept was

$$c = (0.0436 \pm 0.0265) \text{ mm}.$$

Ideally, the intercept should be zero because the minima distances should vanish when $n + \frac{1}{2} = 0$. The nonzero intercept may be caused by imperfect centering of the diffraction pattern, slight asymmetry between the left and right side of the profile, uncertainty in the automatic identification of minima, or deviations from the small-angle approximation. The first few minima are close to the central maximum, and their exact positions may be affected by the finite pixel size and the shape of the central fringe.

The final slit separation obtained from the minima analysis was

$$g_{\text{exp}} = (0.7592 \pm 0.0264) \text{ mm}.$$

This agrees well with the manufacturer value

$$g_{\text{man}} = 0.750 \text{ mm},$$

with a relative deviation of only 1.22%. The result is therefore consistent with the expected double-slit geometry.

This analysis satisfies the requirements of Task 2a by determining the distances $2x_n$ between corresponding interference minima, fitting these distances linearly, reporting the model function, fit parameters, and fit quality, and using the fitted slope to determine the slit separation with uncertainty. The slit width b is not extracted here because the interference minima primarily determine g; b is determined more reliably in Task 2b from the full Fraunhofer fit.

8 Determination of Double-Slit Parameters by Fitting the Fraunhofer Pattern (Task 2b)

8.1 Goal of the Analysis

The aim of Task 2b was to determine both the slit width b and the slit separation g by fitting the full theoretical Fraunhofer diffraction function of a double slit to the measured intensity profile. This method uses the complete shape of the diffraction pattern, including both the interference fringes and the single-slit diffraction envelope.

The manufacturer values of the double slit were

$$b_{\text{man}} = 0.200 \text{ mm}, \qquad g_{\text{man}} = 0.750 \text{ mm}.$$

The result for the slit separation from Task 2a was

$$g_{2a} = (0.7592 \pm 0.0264) \text{ mm}.$$

The result of the fit in this task is compared with both the manufacturer values and the value of g from Task 2a.

8.2 Relevant Theory

For a double slit with slit width b and center-to-center slit separation g, the Fraunhofer diffraction intensity is

$$I(k_x) = I_0 \text{sinc}^2\left(\frac{bk_x}{2}\right)\cos^2\left(\frac{gk_x}{2}\right).$$

Here,

$$\text{sinc}(u) = \frac{\sin u}{u}.$$

The factor

$$\text{sinc}^2\left(\frac{bk_x}{2}\right)$$

describes the broad single-slit diffraction envelope, and the factor

$$\cos^2\left(\frac{gk_x}{2}\right)$$

describes the narrower interference fringes caused by the separation between the two slits.

In the fit, an amplitude factor, background offset, and small horizontal wave-vector offset were included. The model function used was

$$I_{\text{fit}}(k_x) = A\text{sinc}^2\left(\frac{b(k_x - k_{x,0})}{2}\right)\cos^2\left(\frac{g(k_x - k_{x,0})}{2}\right) + C.$$

The fit parameters were therefore

$$A, \quad b, \quad g, \quad C, \quad k_{x,0}.$$

The transverse wave vector was calculated from the CCD position using

$$k_x = \frac{2\pi}{\lambda}\frac{x}{\sqrt{x^2 + f^2}}.$$

For small angles this reduces to

$$k_x \approx \frac{2\pi x}{\lambda f}.$$

In the analysis, the exact expression was used for the fit.

8.3 Experimental Parameters and Data Treatment

The same double-slit Fraunhofer profile used in Task 2a was analyzed. The dataset contained

$$N = 2048$$

data points. The central maximum was located at

$$p_0 = 1125.$$

The new CCD camera was used, so the pixel width was

$$w_{\mathrm{pix}} = 14\ \mu\mathrm{m}.$$

The position coordinate was calculated as

$$x = (p - p_0)w_{\mathrm{pix}}.$$

The laser wavelength and focal length were

$$\lambda = 636\ \mathrm{nm},$$

and

$$f = 300.8\ \mathrm{mm}.$$

The focal-length uncertainty was taken as

$$u(f) = 10\ \mathrm{mm}.$$

The maximum measured intensity was

$$I_{\max} = 0.295956\ \mathrm{a.u.}$$

and the background intensity estimated from the outer regions of the profile was

$$I_{\mathrm{bg}} = 0.00752\ \mathrm{a.u.}$$

After background subtraction, the measured intensity was normalized according to

$$I_{\mathrm{norm}} = \frac{I - I_{\mathrm{bg}}}{I_{\max} - I_{\mathrm{bg}}}.$$

The estimated normalized intensity noise was

$$u(I) = 0.000905.$$

The fit was performed over the central region of the diffraction pattern,

$$-2.5\ \mathrm{mm} \le x \le 2.5\ \mathrm{mm},$$

to include the important interference fringes and the main single-slit envelope while avoiding unnecessary influence from noisy outer regions.

8.4 Fit Results

The fit model was

$$I_{\mathrm{fit}}(k_x) = A\mathrm{sinc}^2\left(\frac{b(k_x - k_{x,0})}{2}\right)\cos^2\left(\frac{g(k_x - k_{x,0})}{2}\right) + C.$$

The fitted parameters are summarized in Table 7.

Table 7: Fit results for the double-slit Fraunhofer diffraction pattern.

Quantity	Value	Unit
Amplitude A	1.00631 ± 0.00498	norm. a.u.
Slit width b	0.18431 ± 0.00620	mm
Slit separation g	0.73490 ± 0.02444	mm
Offset C	0.00952 ± 0.00105	norm. a.u.
Wave-vector offset $k_{x,0}$	289.57 ± 9.47	$\mathrm{rad\,m^{-1}}$

The goodness-of-fit values were

$$\mathrm{SSE} = 1.08605 \times 10^{-1},$$

$$\mathrm{MSE} = 3.04215 \times 10^{-4},$$

$$\mathrm{RMSE} = 0.01744,$$

and

$$R^2 = 0.99291.$$

The resulting slit parameters were therefore

$$\boxed{b_{\mathrm{fit}} = (0.1843 \pm 0.0062)\ \mathrm{mm}}$$

and

$$\boxed{g_{\mathrm{fit}} = (0.7349 \pm 0.0244)\ \mathrm{mm}.}$$

8.5 Uncertainty Analysis

The fit returned covariance uncertainties for the fitted parameters. For the slit width, the covariance contribution was

$$u_{\mathrm{cov}}(b) = 0.00093\ \mathrm{mm}.$$

For the slit separation, the covariance contribution was

$$u_{\mathrm{cov}}(g) = 0.00074\ \mathrm{mm}.$$

However, both b and g are obtained through the conversion from CCD position to wave vector k_x. Therefore, the uncertainty in the k_x-scale also contributes to the uncertainties in b and g. Since the dominant specified scale uncertainty was the focal length uncertainty, the relative scale uncertainty was taken as

$$\frac{u_{\mathrm{scale}}}{\mathrm{scale}} = \frac{u(f)}{f}.$$

With

$$u(f) = 10\ \mathrm{mm}, \qquad f = 300.8\ \mathrm{mm},$$

this gives

$$\frac{u(f)}{f} = 0.03324.$$

Therefore,

$$u_{\mathrm{scale}}(b) = b\frac{u(f)}{f} = 0.18431\ \mathrm{mm} \cdot \frac{10}{300.8} = 0.00613\ \mathrm{mm},$$

and

$$u_{\mathrm{scale}}(g) = g\frac{u(f)}{f} = 0.73490\ \mathrm{mm} \cdot \frac{10}{300.8} = 0.02443\ \mathrm{mm}.$$

The total uncertainties were calculated by adding the covariance and scale uncertainties in quadrature:

$$u(b) = \sqrt{u_{\mathrm{cov}}(b)^2 + u_{\mathrm{scale}}(b)^2},$$

and

$$u(g) = \sqrt{u_{\mathrm{cov}}(g)^2 + u_{\mathrm{scale}}(g)^2}.$$

This gives

$$u(b) = 0.00620\ \mathrm{mm},$$

and

$$u(g) = 0.02444\ \mathrm{mm}.$$

The scale uncertainty dominates the final uncertainty of both parameters.

8.6 Plots

Figure 18 shows the measured double-slit Fraunhofer profile and the fitted theoretical curve as a function of position. The vertical dashed lines indicate the fit region used in the analysis.

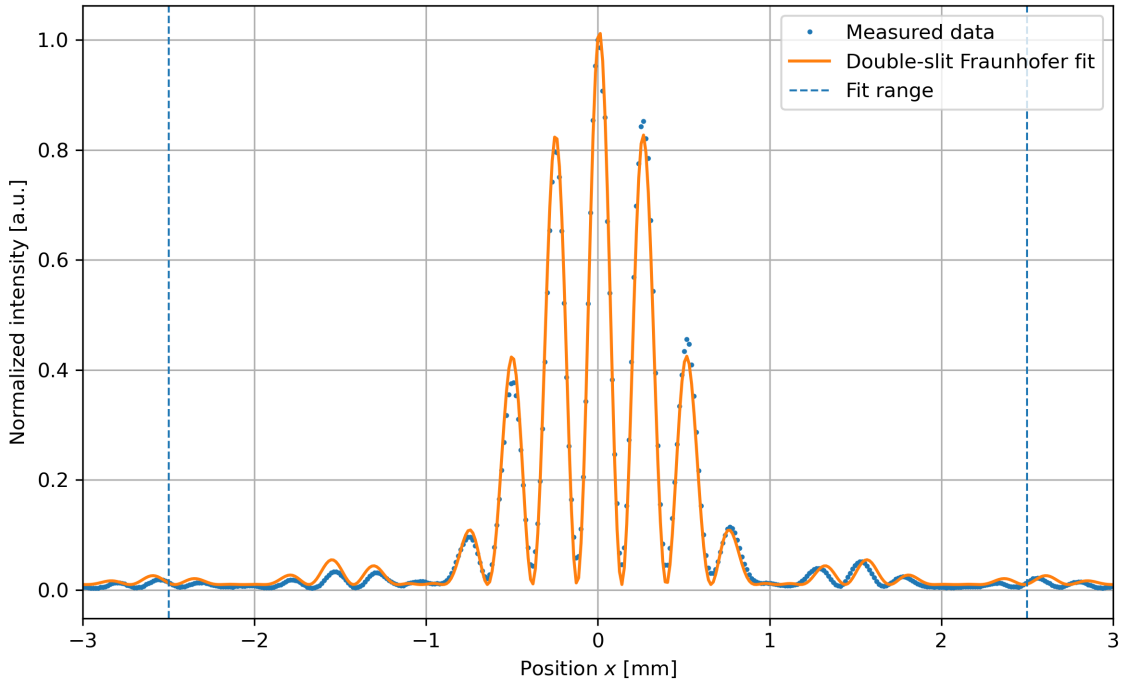


Figure 18: Measured double-slit Fraunhofer diffraction profile with the fitted theoretical double-slit model.

Figure 19 shows the same comparison as a function of transverse wave vector k_x. This representation is directly related to the theoretical Fraunhofer model.

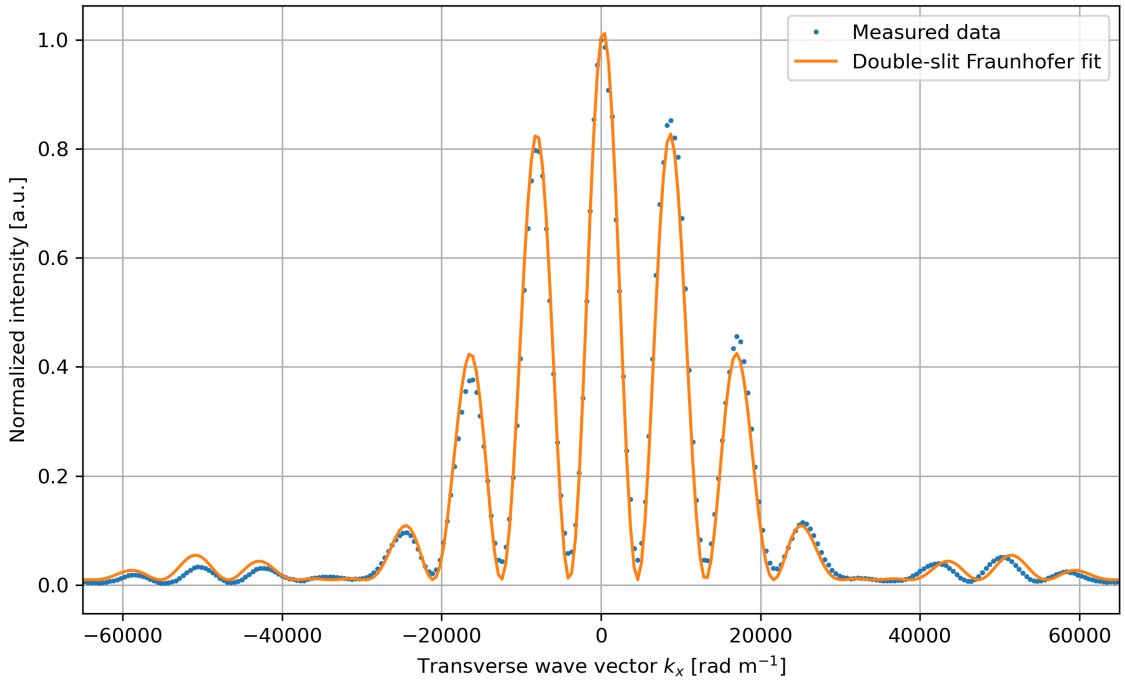


Figure 19: Measured double-slit Fraunhofer diffraction profile and fitted model as a function of transverse wave vector k_x.

Figure 20 shows a zoomed view of the central region of the diffraction pattern. This makes the agreement between the fitted interference fringes and measured data more visible.

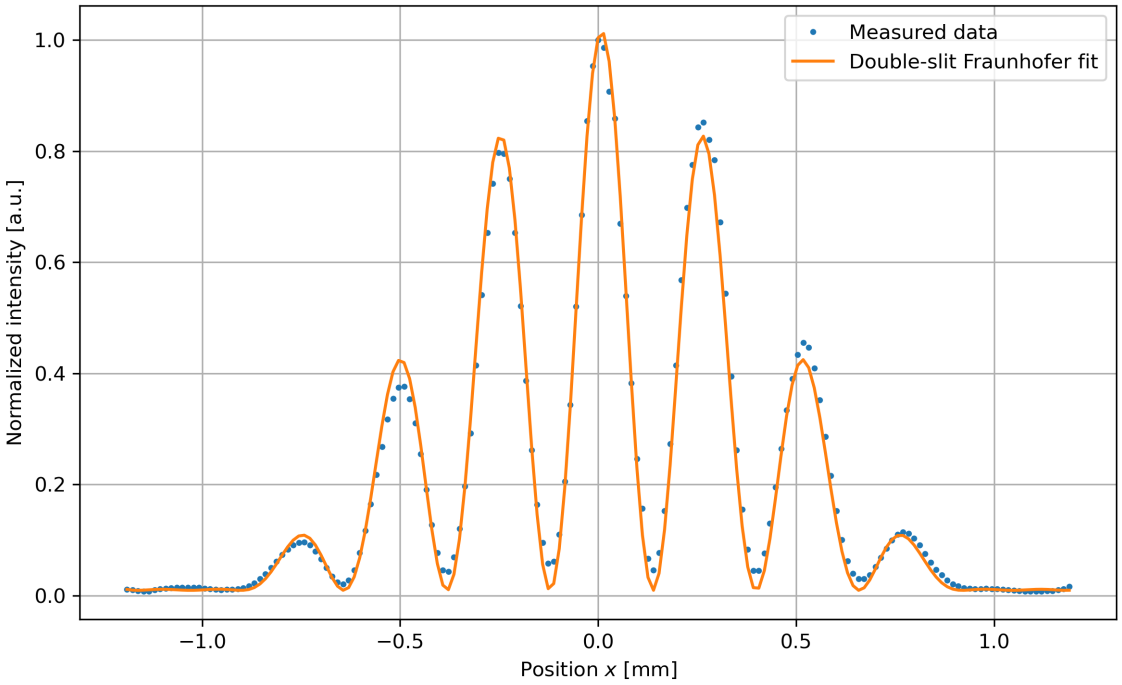


Figure 20: Zoomed view of the central region of the double-slit Fraunhofer profile and fit.

Figure 21 shows the residuals of the fit over the fitted range.

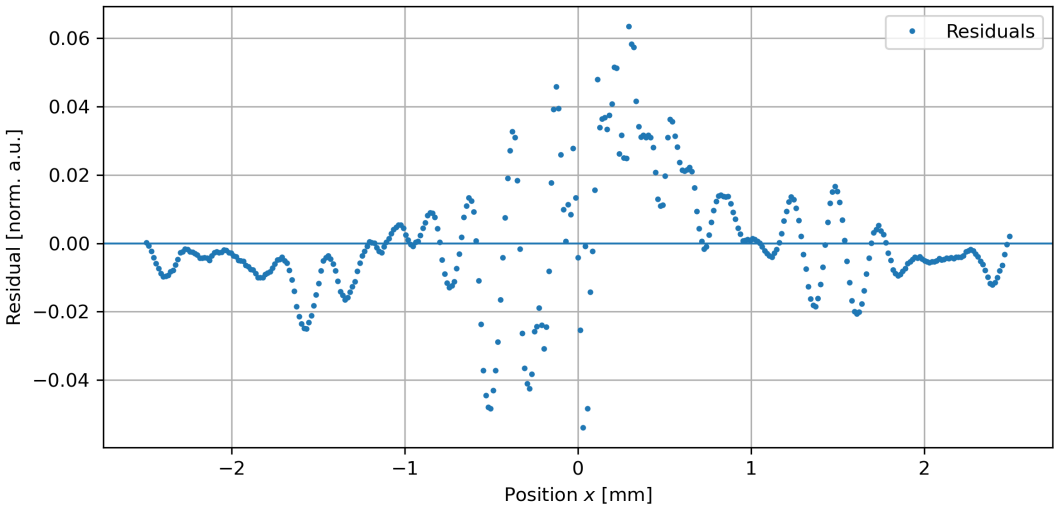


Figure 21: Residuals of the double-slit Fraunhofer fit.

Figure 22 shows the measured data and fitted curve on a semilogarithmic scale. This representation is useful for checking the agreement of the weaker side fringes and the envelope.

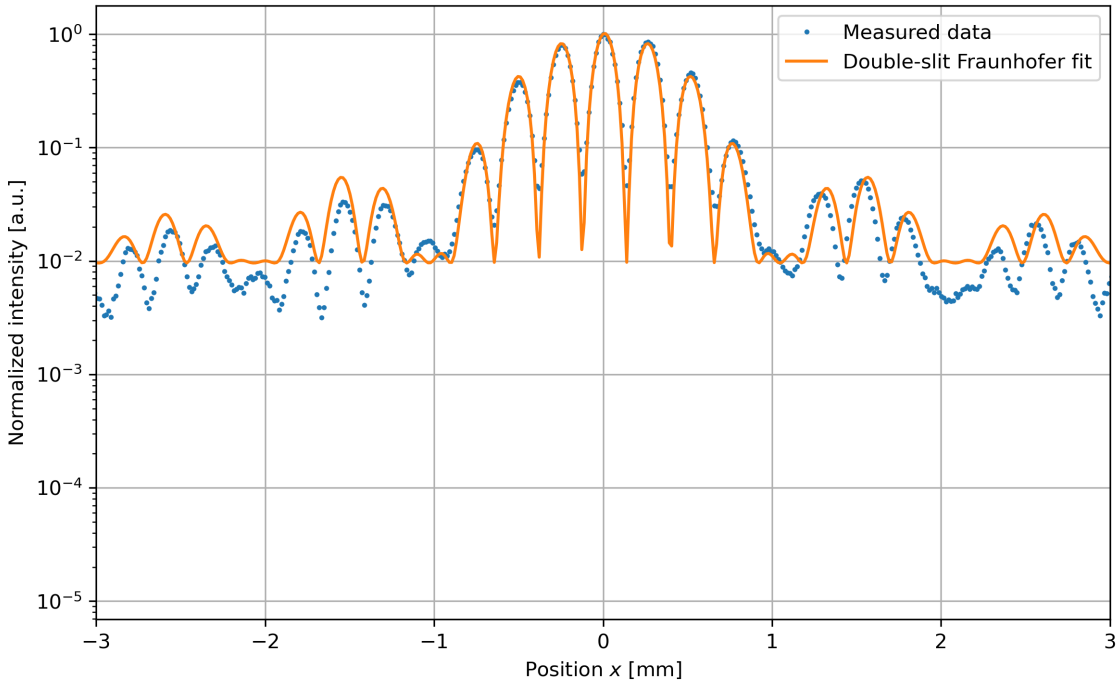


Figure 22: Double-slit Fraunhofer profile and fit shown on a semilogarithmic intensity scale.

8.7 Comparison with Manufacturer Values and Task 2a

The fitted slit width was

$$b_{\text{fit}} = (0.1843 \pm 0.0062)\ \text{mm}.$$

The manufacturer value was

$$b_{\text{man}} = 0.200\ \text{mm}.$$

The absolute deviation is

$$\Delta b = |b_{\text{fit}} - b_{\text{man}}| = |0.1843\ \text{mm} - 0.2000\ \text{mm}| = 0.0157\ \text{mm}.$$

The relative deviation is

$$\frac{\Delta b}{b_{\text{man}}} \times 100\% = 7.84\%.$$

The fitted slit separation was

$$g_{\text{fit}} = (0.7349 \pm 0.0244)\ \text{mm}.$$

The manufacturer value was

$$g_{\text{man}} = 0.750\ \text{mm}.$$

The absolute deviation is

$$\Delta g = |g_{\text{fit}} - g_{\text{man}}| = |0.7349\ \text{mm} - 0.7500\ \text{mm}| = 0.0151\ \text{mm}.$$

The relative deviation is

$$\frac{\Delta g}{g_{\text{man}}} \times 100\% = 2.01\%.$$

The value from Task 2a was

$$g_{2a} = (0.7592 \pm 0.0264)\ \text{mm}.$$

The relative difference between the Task 2b fit value and the Task 2a minima value is

$$\frac{|g_{\text{fit}} - g_{2a}|}{g_{2a}} \times 100\% = 3.20\%.$$

The normalized difference is

$$\frac{|g_{\text{fit}} - g_{2a}|}{\sqrt{u(g_{\text{fit}})^2 + u(g_{2a})^2}} = 0.674.$$

Since this value is below one, the two experimentally determined values of g agree within one combined standard uncertainty.

8.8 Discussion

The fitted double-slit Fraunhofer model describes the measured profile well. The coefficient of determination was

$$R^2 = 0.99291,$$

which indicates that the main structure of the diffraction pattern is reproduced by the theoretical model. The fit successfully captures both the rapid interference fringes and the broader single-slit diffraction envelope.

The fitted slit separation,

$$g_{\text{fit}} = (0.7349 \pm 0.0244)\ \text{mm},$$

agrees with the manufacturer value of 0.750 mm within uncertainty. It also agrees with the Task 2a result,

$$g_{2a} = (0.7592 \pm 0.0264)\ \text{mm},$$

within one combined standard uncertainty. This confirms that the interference fringe spacing is consistent across the minima-based and full-fit methods.

The fitted slit width,

$$b_{\text{fit}} = (0.1843 \pm 0.0062)\ \text{mm},$$

is smaller than the manufacturer value of 0.200 mm by 7.84%. The manufacturer value is not contained within the one-standard-uncertainty interval of the fitted value. This discrepancy may be caused by systematic effects such as imperfect background subtraction, nonuniform slit illumination, finite detector size, slight misalignment, or deviations of the real slit from the ideal rectangular aperture assumed in the model. The slit width is mainly determined by the broad envelope, which is more sensitive to background, normalization, and the chosen fit range than the slit separation.

The fitted background offset,

$$C = 0.00952,$$

is small but nonzero, showing that residual background or detector offset remained after the initial background correction. The fitted wave-vector offset,

$$k_{x,0} = 289.57\ \text{rad}\,\text{m}^{-1},$$

is also small compared with the full k_x-range used in the fit and accounts for a slight horizontal shift of the measured pattern.

Overall, Task 2b satisfies the checklist requirements by fitting the measured Fraunhofer double-slit profile with the correct theoretical model, determining both the slit width and slit separation, reporting the fitted parameters and fit quality, including uncertainties, and comparing the results with the manufacturer values and the Task 2a result.

9 Determination of Double-Slit Parameters from the Fourier Transform (Task 2c)

9.1 Goal of the Analysis

The aim of Task 2c was to determine the double-slit width b and slit separation g from the fast Fourier transform (FFT) of the measured Fraunhofer intensity profile. This provides an additional method for extracting the real-space aperture dimensions from the diffraction pattern.

The checklist requires that the FFT of the double-slit profile is calculated, represented with a proper physical unit, and used to determine b and g. The results must then be compared with the values obtained in Task 2a and Task 2b.

The reference values used for comparison were

$$b_{\text{man}} = 0.200 \text{ mm}, \qquad g_{\text{man}} = 0.750 \text{ mm}.$$

The previous experimental results were

$$g_{2a} = (0.7592 \pm 0.0264) \text{ mm},$$

$$b_{2b} = (0.1843 \pm 0.0062) \text{ mm},$$

and

$$g_{2b} = (0.7349 \pm 0.0244) \text{ mm}.$$

9.2 Relevant Theory

In Fraunhofer diffraction, the electric-field amplitude is proportional to the Fourier transform of the aperture transmission function. For a double slit, the intensity distribution is

$$I(k_x) = I_0 \text{sinc}^2\left(\frac{bk_x}{2}\right)\cos^2\left(\frac{gk_x}{2}\right).$$

The CCD camera measures intensity, not the electric field. Therefore,

$$I(k_x) \propto |\mathcal{F}\{f(x')\}|^2,$$

where $f(x')$ is the real-space double-slit transmission function.

Using the convolution theorem, the inverse Fourier transform of the measured intensity is proportional to the autocorrelation of the aperture function:

$$\mathcal{F}^{-1}\{I(k_x)\} \propto f(x') * f(-x').$$

For a symmetric double slit,

$$f(x') = f(-x'),$$

so this becomes

$$\mathcal{F}^{-1}\{I(k_x)\} \propto f(x') * f(x').$$

The autocorrelation of a double slit contains characteristic features related to both aperture parameters:

- The width of the central autocorrelation feature is related to the slit width b.
- The side autocorrelation peaks occur at positions corresponding to the slit separation g.

Thus, the FFT of the measured intensity profile can be used to determine both b and g. The first minimum or edge of the central autocorrelation feature gives an estimate of b, while the position of the side autocorrelation peak gives g.

9.3 Conversion to k_x-Space

The same Fraunhofer double-slit profile used in Tasks 2a and 2b was analyzed. The dataset contained

$$N = 2048$$

data points. The central maximum was located at

$$p_0 = 1125.$$

The new CCD camera was used, with pixel width

$$w_{\text{pix}} = 14 \ \mu\text{m}.$$

The transverse position was calculated as

$$x = (p - p_0)w_{\text{pix}}.$$

For the FFT, the intensity profile was represented as a function of the transverse wave vector k_x. In order to obtain an evenly spaced k_x-axis for the FFT, the small-angle approximation was used:

$$k_x \approx \frac{2\pi x}{\lambda f}.$$

With

$$\lambda = 636 \text{ nm}$$

and

$$f = 300.8 \text{ mm},$$

the resulting wave-vector spacing was

$$\Delta k_x = 459.804 \text{ rad m}^{-1}.$$

The estimated background intensity was

$$I_{\text{bg}} = 0.007352 \text{ a.u.}$$

The intensity was background-corrected and normalized before calculating the FFT:

$$I_{\text{norm}} = \frac{I - I_{\text{bg}}}{I_{\text{max}} - I_{\text{bg}}}.$$

9.4 FFT Method

The FFT was calculated from the normalized intensity profile $I(k_x)$. Since the Fourier transform was taken with respect to k_x, the conjugate coordinate has units of length. This coordinate corresponds to the real-space aperture coordinate x'.

For a discrete Fourier transform, the aperture-coordinate axis was calculated as

$$x' = 2\pi \,\text{fftfreq}(N, \Delta k_x).$$

The FFT coordinate spacing was

$$\Delta x' = 0.00672 \text{ mm}.$$

The corresponding resolution uncertainty was estimated as half of this spacing:

$$u_{\text{res}} = \frac{\Delta x'}{2} = 0.00336 \text{ mm}.$$

The slit width was extracted from the central autocorrelation feature:

$$b_{\text{FFT}} = |x'_{\text{edge}}|.$$

The slit separation was extracted from the side autocorrelation peak:

$$g_{\text{FFT}} = |x'_{\text{side peak}}|.$$

A smoothed FFT curve was used only to identify these features more robustly. The plotted FFT magnitude was normalized to its maximum value.

9.5 Results

The numerical results from the FFT analysis are summarized in Table 8.

Table 8: Results of the double-slit FFT analysis. The FFT was calculated from the Fraunhofer intensity profile represented as a function of k_x.

Quantity	Value	Unit
Number of data points N	2048	–
Central maximum pixel p_0	1125	pixel
Background intensity I_{bg}	0.007352	a.u.
k_x-axis spacing Δk_x	459.804	rad m^{-1}
FFT coordinate spacing $\Delta x'$	0.00672	mm
FFT resolution uncertainty	0.00336	mm
Slit width from FFT b_{FFT}	0.2313	mm
Uncertainty $u(b_{\text{FFT}})$	0.00838	mm
Slit separation from FFT g_{FFT}	0.7398	mm
Uncertainty $u(g_{\text{FFT}})$	0.0248	mm

Thus, the final FFT results were

$$\boxed{b_{\mathrm{FFT}} = (0.2313 \pm 0.0084)\ \mathrm{mm}}$$

and

$$\boxed{g_{\mathrm{FFT}} = (0.7398 \pm 0.0248)\ \mathrm{mm}.}$$

9.6 Plots

Figure 23 shows the measured double-slit Fraunhofer intensity profile as a function of transverse wave vector k_x. This plot verifies that the FFT was applied to the intensity profile in k_x-space.

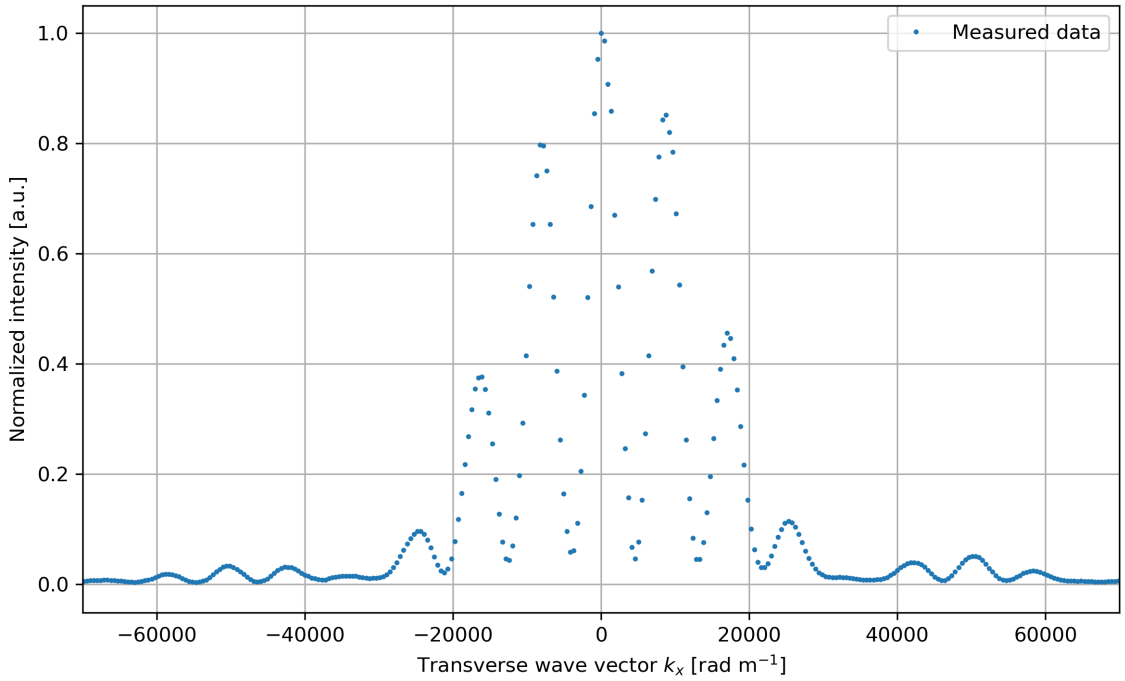


Figure 23: Measured double-slit Fraunhofer intensity profile plotted as a function of transverse wave vector k_x.

Figure 24 shows the normalized FFT magnitude as a function of aperture coordinate x'. The marked central feature was used to determine b_{FFT}, and the side peak was used to determine g_{FFT}.

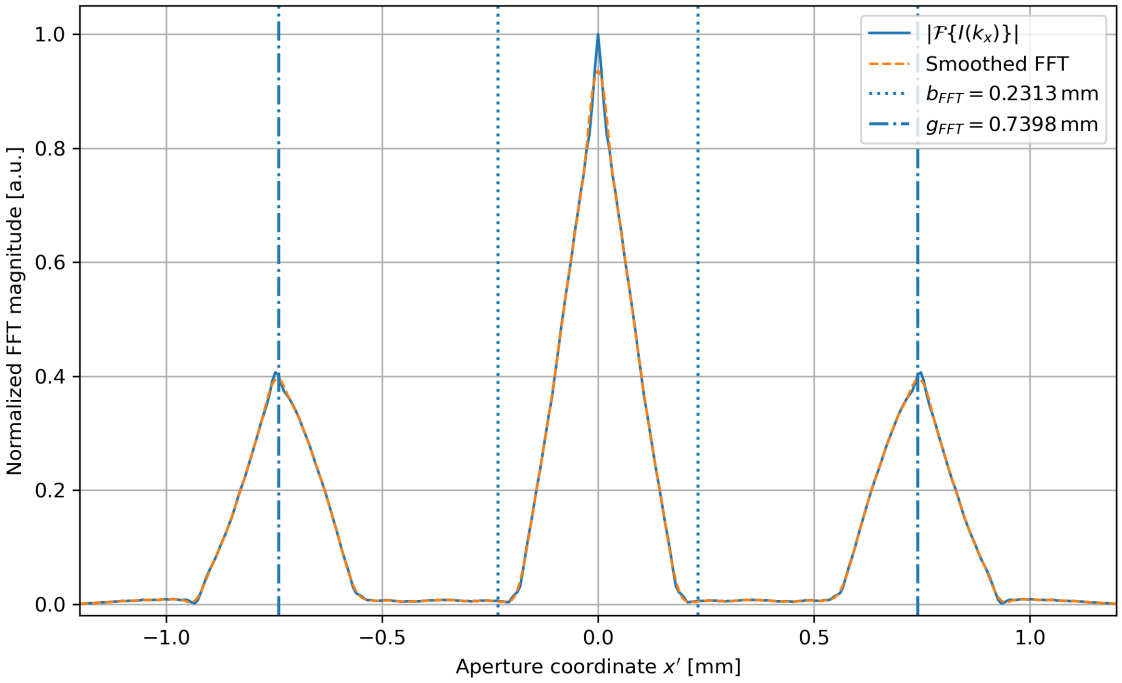


Figure 24: Normalized FFT magnitude of the double-slit Fraunhofer intensity profile as a function of aperture coordinate x'.

Figure 25 shows a zoomed view of the central autocorrelation feature used to determine the slit width b.

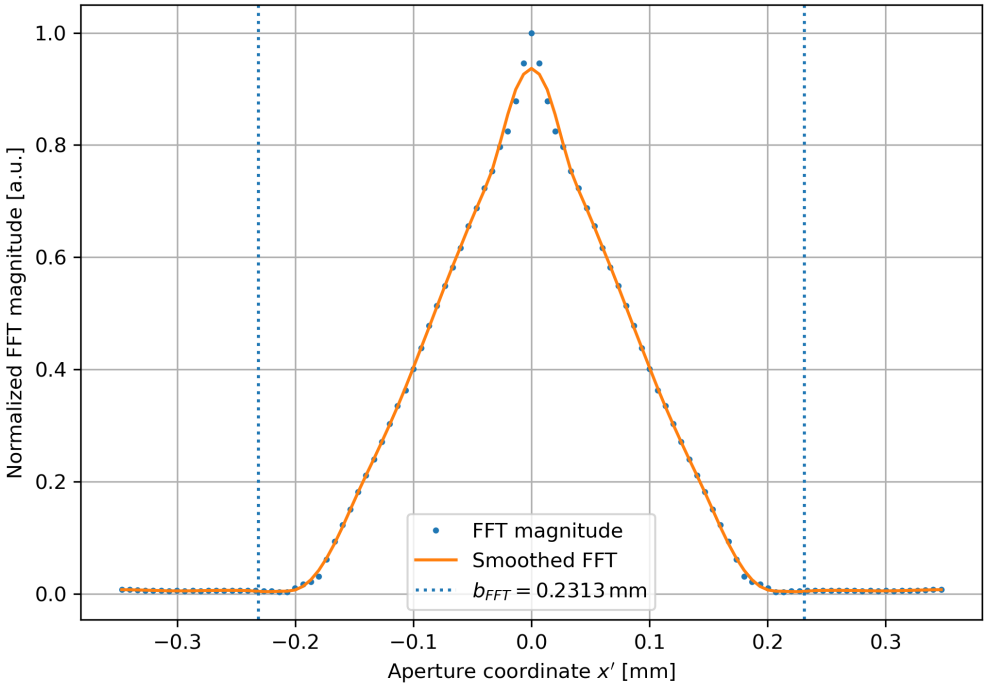


Figure 25: Zoomed view of the central autocorrelation feature in the FFT magnitude.

Figure 26 shows a zoomed view of the positive side autocorrelation peak used to determine the slit separation g.

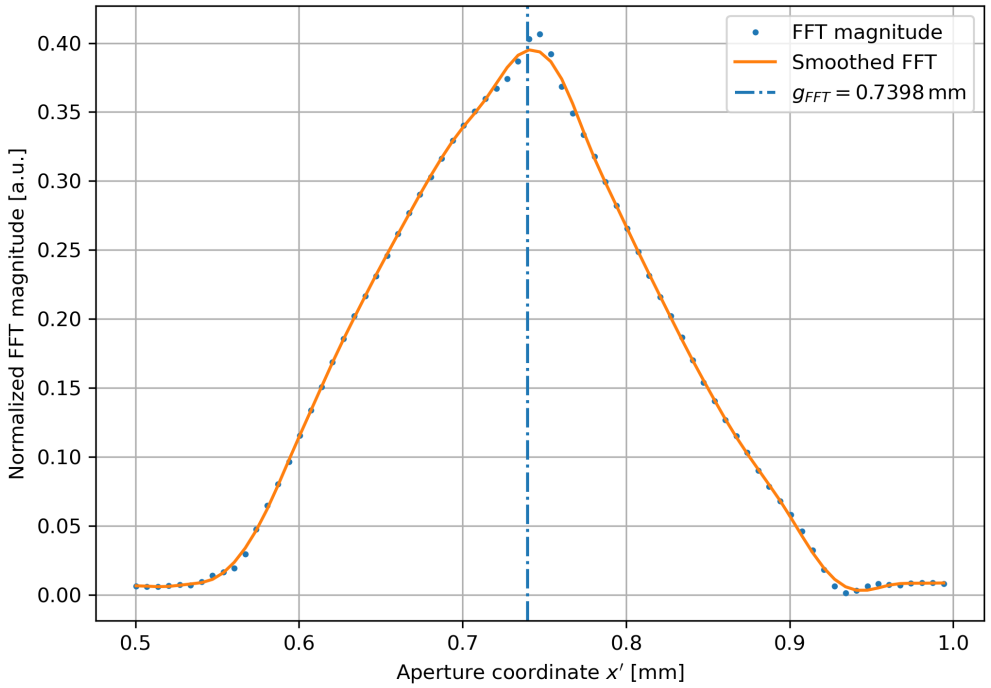


Figure 26: Zoomed view of the positive side autocorrelation peak.

Figure 27 shows the positive side of the FFT magnitude, including both the central feature and the side peak.

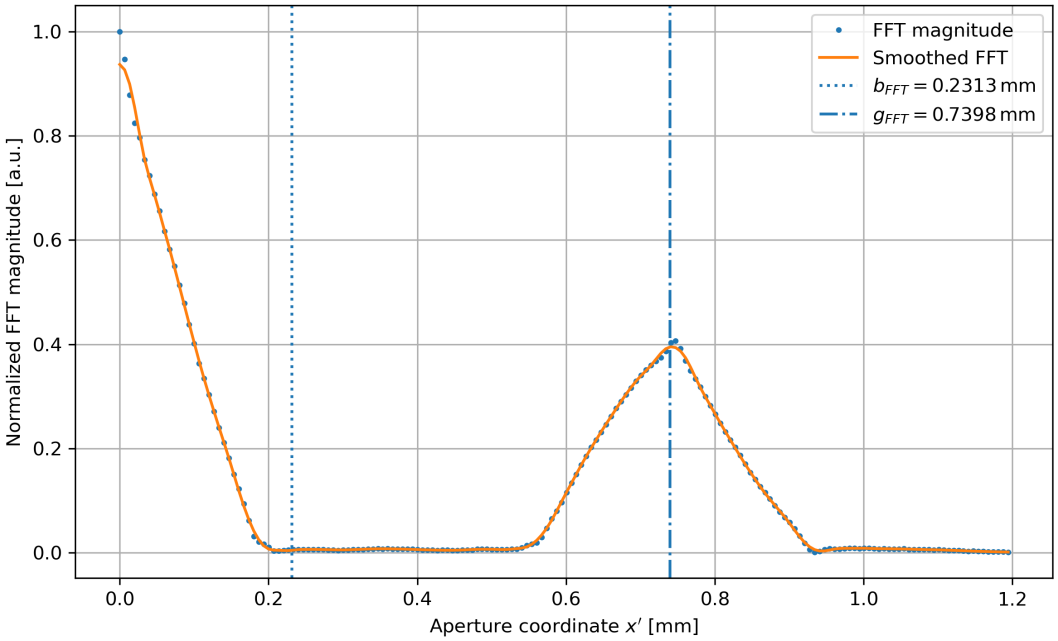


Figure 27: Positive side of the double-slit FFT magnitude showing the features used to extract b_{FFT} and g_{FFT}.

9.7 Uncertainty Analysis

The uncertainty in the FFT-derived quantities has two main contributions. The first is the finite resolution of the FFT coordinate axis,

$$u_{\mathrm{res}} = \frac{\Delta x'}{2} = 0.00336 \text{ mm}.$$

The second contribution comes from the uncertainty in the scale of the k_x-axis. Since

$$k_x \approx \frac{2\pi x}{\lambda f},$$

the conjugate FFT coordinate x' depends on the focal length, wavelength, and pixel width. The wavelength and pixel-width uncertainties were not specified and were neglected. The dominant scale uncertainty was therefore the focal-length uncertainty,

$$u(f) = 10 \text{ mm}.$$

The relative scale uncertainty is

$$\frac{u_{\text{scale}}}{\text{scale}} = \frac{u(f)}{f} = \frac{10 \text{ mm}}{300.8 \text{ mm}} = 0.03324.$$

Thus,

$$u_{\text{scale}}(b) = b_{\text{FFT}} \frac{u(f)}{f},$$

and

$$u_{\text{scale}}(g) = g_{\text{FFT}} \frac{u(f)}{f}.$$

The total uncertainties were calculated by adding the resolution and scale uncertainties in quadrature:

$$u(b_{\text{FFT}}) = \sqrt{u_{\text{res}}^2 + u_{\text{scale}}(b)^2},$$

and

$$u(g_{\text{FFT}}) = \sqrt{u_{\text{res}}^2 + u_{\text{scale}}(g)^2}.$$

This gives

$$u(b_{\text{FFT}}) = 0.00838 \text{ mm},$$

and

$$u(g_{\text{FFT}}) = 0.02482 \text{ mm}.$$

The uncertainty in g_{FFT} is dominated by the focal-length scale uncertainty, while the uncertainty in b_{FFT} contains comparable contributions from FFT resolution and scale uncertainty.

9.8 Comparison with Manufacturer Values

The manufacturer values were

$$b_{\text{man}} = 0.200 \text{ mm}, \qquad g_{\text{man}} = 0.750 \text{ mm}.$$

For the slit width,

$$\Delta b = |b_{\text{FFT}} - b_{\text{man}}| = |0.2313 \text{ mm} - 0.2000 \text{ mm}| = 0.0313 \text{ mm}.$$

The relative deviation is

$$\frac{\Delta b}{b_{\text{man}}} \times 100\% = 15.66\%.$$

For the slit separation,

$$\Delta g = |g_{\text{FFT}} - g_{\text{man}}| = |0.7398 \text{ mm} - 0.7500 \text{ mm}| = 0.0102 \text{ mm}.$$

The relative deviation is

$$\frac{\Delta g}{g_{\text{man}}} \times 100\% = 1.36\%.$$

Thus, the FFT-derived slit separation agrees well with the manufacturer value, while the FFT-derived slit width is noticeably larger.

9.9 Comparison with Tasks 2a and 2b

The slit width from Task 2b was

$$b_{2b} = (0.1843 \pm 0.0062) \text{ mm}.$$

The relative difference between the FFT and Task 2b slit widths is

$$\frac{|b_{\text{FFT}} - b_{2b}|}{b_{2b}} \times 100\% = 25.50\%.$$

The normalized difference is

$$\frac{|b_{\text{FFT}} - b_{2b}|}{\sqrt{u(b_{\text{FFT}})^2 + u(b_{2b})^2}} = 4.51.$$

This indicates that the two values of b do not agree within their combined uncertainties.

For the slit separation, the Task 2a value was

$$g_{2a} = (0.7592 \pm 0.0264) \text{ mm}.$$

The relative difference between g_{FFT} and g_{2a} is

$$\frac{|g_{\text{FFT}} - g_{2a}|}{g_{2a}} \times 100\% = 2.55\%.$$

The normalized difference is

$$\frac{|g_{\text{FFT}} - g_{2a}|}{\sqrt{u(g_{\text{FFT}})^2 + u(g_{2a})^2}} = 0.534.$$

The Task 2b value was

$$g_{2b} = (0.7349 \pm 0.0244) \text{ mm}.$$

The relative difference between g_{FFT} and g_{2b} is

$$\frac{|g_{\text{FFT}} - g_{2b}|}{g_{2b}} \times 100\% = 0.669\%.$$

The normalized difference is

$$\frac{|g_{\text{FFT}} - g_{2b}|}{\sqrt{u(g_{\text{FFT}})^2 + u(g_{2b})^2}} = 0.141.$$

Therefore, the FFT-derived slit separation agrees very well with both Task 2a and Task 2b.

9.10 Discussion

The FFT analysis clearly shows the autocorrelation structure expected from a double slit. The side autocorrelation peak gives a reliable value for the slit separation,

$$g_{\text{FFT}} = (0.7398 \pm 0.0248) \text{ mm}.$$

This value agrees with the manufacturer value and with the values obtained in Tasks 2a and 2b within uncertainty. This confirms that the separation of the two slits is robustly encoded in the periodic interference fringes and can be extracted accurately by the Fourier-transform method.

The slit width obtained from the FFT was

$$b_{\text{FFT}} = (0.2313 \pm 0.0084) \text{ mm}.$$

This is larger than both the manufacturer value and the Task 2b fit result. The discrepancy is significant, with a normalized difference of 4.51 relative to Task 2b. This suggests that the FFT method was less reliable for determining the slit width in this dataset.

A likely reason is that the slit width is determined from the width or edge of the central autocorrelation feature, which is sensitive to finite-window effects, background subtraction, smoothing, and noise. The central FFT feature may be broadened by the finite CCD range and by deviations from the ideal double-slit model. In contrast, the side peak position used to determine g is easier to identify and less sensitive to the exact shape of the central feature.

The use of the small-angle k_x relation for the FFT was necessary to obtain an evenly spaced k_x-axis. The resulting scale uncertainty was included through the focal-length uncertainty. Since $u(f) = 10$ mm is relatively large, the scale uncertainty is the dominant contribution to the final uncertainty of g_{FFT}.

Overall, Task 2c satisfies the checklist requirements by plotting the double-slit intensity profile as a function of k_x, calculating and plotting the FFT with a physical length unit, extracting both b and g, and comparing the FFT results with the results from Tasks 2a and 2b. The analysis shows that the FFT method gives a reliable slit separation but a less reliable slit width for the present measurement.

10 Fresnel Diffraction of a Double Slit (Task 2d)

10.1 Goal of the Analysis

The aim of Task 2d was to measure the Fresnel diffraction pattern of the double slit at three different propagation distances and compare the measured profiles with the theoretical Fresnel diffraction model. For this part of the experiment, the lens was removed and the CCD camera was placed directly behind the double slit.

The measured distances were

$$z_1 = 40\,\mathrm{cm}, \qquad z_2 = 50\,\mathrm{cm}, \qquad z_3 = 60\,\mathrm{cm}.$$

The slit parameters used in the theoretical calculation were taken from the Fraunhofer fit in Task 2b:

$$b = (0.1843 \pm 0.0062)\,\mathrm{mm},$$

and

$$g = (0.7349 \pm 0.0244)\,\mathrm{mm}.$$

Here, b is the width of each slit and g is the center-to-center slit separation. The goal of this task was not to refit b and g, but to check whether the Fresnel diffraction profiles measured at finite propagation distances are described by the theoretical model.

10.2 Relevant Theory

In the Fresnel regime, the finite propagation distance between the double slit and the CCD camera must be included. The field at position x in the observation plane is proportional to

$$\psi(x,z) \propto \int f(x') \exp\left(i\frac{\pi}{\lambda z}(x-x')^2\right)\,dx',$$

where x' is the transverse coordinate in the slit plane, x is the transverse coordinate in the camera plane, and z is the propagation distance.

For a double slit with slit width b and slit separation g, the two open intervals are

$$-\frac{g+b}{2} < x' < -\frac{g-b}{2},$$

and

$$\frac{g-b}{2} < x' < \frac{g+b}{2}.$$

The Fresnel field is therefore the sum of the contributions from both slits:

$$\psi(x,z) \propto \int_{-(g+b)/2}^{-(g-b)/2} \exp\left(i\frac{\pi}{\lambda z}(x-x')^2\right)\,dx' + \int_{(g-b)/2}^{(g+b)/2} \exp\left(i\frac{\pi}{\lambda z}(x-x')^2\right)\,dx'.$$

Using the complex Fresnel integral

$$F(u) = C(u) + iS(u),$$

with

$$C(u) = \int_0^u \cos\left(\frac{\pi t^2}{2}\right)\,dt,$$

and

$$S(u) = \int_0^u \sin\left(\frac{\pi t^2}{2}\right)\, dt,$$

the theoretical double-slit Fresnel intensity is

$$I(x,z) \propto \left| F\left(\frac{g+b+2x}{\sqrt{2\lambda z}}\right) + F\left(\frac{g+b-2x}{\sqrt{2\lambda z}}\right) - F\left(\frac{g-b+2x}{\sqrt{2\lambda z}}\right) - F\left(\frac{g-b-2x}{\sqrt{2\lambda z}}\right) \right|^2 .$$

Since the measured intensity is in arbitrary units, the theoretical curve was compared to the normalized measured data using the model

$$I_{\mathrm{fit}}(x) = A\, I_{\mathrm{th}}(x - x_0, z; b, g, \lambda) + C,$$

where A is an amplitude factor, C is a vertical offset, and x_0 is a small horizontal offset accounting for imperfect centering.

10.3 Fresnel Numbers

For the double slit, two Fresnel numbers were calculated. The slit-width Fresnel number is

$$N_b = \frac{b^2}{4\lambda z},$$

and the slit-separation Fresnel number is

$$N_g = \frac{g^2}{4\lambda z}.$$

The value N_b characterizes diffraction due to each individual slit, while N_g characterizes the finite-distance effects associated with the slit separation. Since $g > b$, the values of N_g are larger than those of N_b.

10.4 Experimental Parameters and Data Treatment

The new CCD camera was used, so the pixel width was

$$w_{\mathrm{pix}} = 14\,\mu\mathrm{m}.$$

Each CSV file contained the pixel number and the measured intensity in arbitrary units. For each distance, the position of the strongest maximum was used to center the profile. The position axis was calculated from

$$x = (p - p_0) w_{\mathrm{pix}}.$$

The central pixels were

$$p_0 = 890 \quad \text{for } z = 40\,\mathrm{cm},$$

$$p_0 = 904 \quad \text{for } z = 50\,\mathrm{cm},$$

and

$$p_0 = 924 \quad \text{for } z = 60\,\mathrm{cm}.$$

The background intensity was estimated from the outer regions of each measured profile and subtracted. The corrected intensity was normalized according to

$$I_{\mathrm{norm}} = \frac{I - I_{\mathrm{bg}}}{I_{\mathrm{max}} - I_{\mathrm{bg}}}.$$

The theoretical Fresnel curve was then fitted to each normalized profile using A, C, and x_0 as free parameters, while b, g, z, and λ were fixed to their measured or previously determined values.

The uncertainty in the propagation distance was taken as

$$u(z) = 0.5\,\mathrm{cm}.$$

The uncertainty in centering the profile was estimated as

$$u(p_0) = 2 \text{ pixels},$$

which corresponds to

$$u(x_0) = 2 \cdot 14\,\mu\mathrm{m} = 28\,\mu\mathrm{m}.$$

10.5 Results

The main results of the Fresnel comparison are summarized in Table 9. The fit quality was evaluated using the root-mean-square error and the coefficient of determination R^2.

Table 9: Results of the double-slit Fresnel diffraction analysis.

z [cm]	p_0	I_{bg} [a.u.]	N_b	N_g	x_0 [mm]	RMSE	R^2
40	890	0.003952	0.03338	0.53074	0.00678	0.04610	0.88251
50	904	0.005709	0.02671	0.42459	0.00151	0.01920	0.97664
60	924	0.004423	0.02226	0.35383	0.01027	0.03585	0.93756

The fitted amplitude and offset parameters are shown in Table 10.

Table 10: Fit parameters for the comparison between measured double-slit Fresnel profiles and theory.

z [cm]	A	$u(A)$	C	$u(C)$
40	1.11970	0.00910	0.008274	0.001067
50	1.04938	0.00359	0.007466	0.000458
60	1.05212	0.00601	0.005402	0.000828

The calculated uncertainties of the Fresnel numbers are summarized in Table 11.

Table 11: Fresnel numbers and uncertainties for the double-slit Fresnel measurements.

z [cm]	N_b	$u(N_b)$	N_g	$u(N_g)$
40	0.03338	0.00228	0.53074	0.03592
50	0.02671	0.00182	0.42459	0.02856
60	0.02226	0.00151	0.35383	0.02372

10.6 Plots

Figure 28 shows the measured double-slit Fresnel profile at $z = 40\,\mathrm{cm}$ with the corresponding theoretical curve. The measured data are shown as markers, while the theoretical curve is shown as a continuous line.

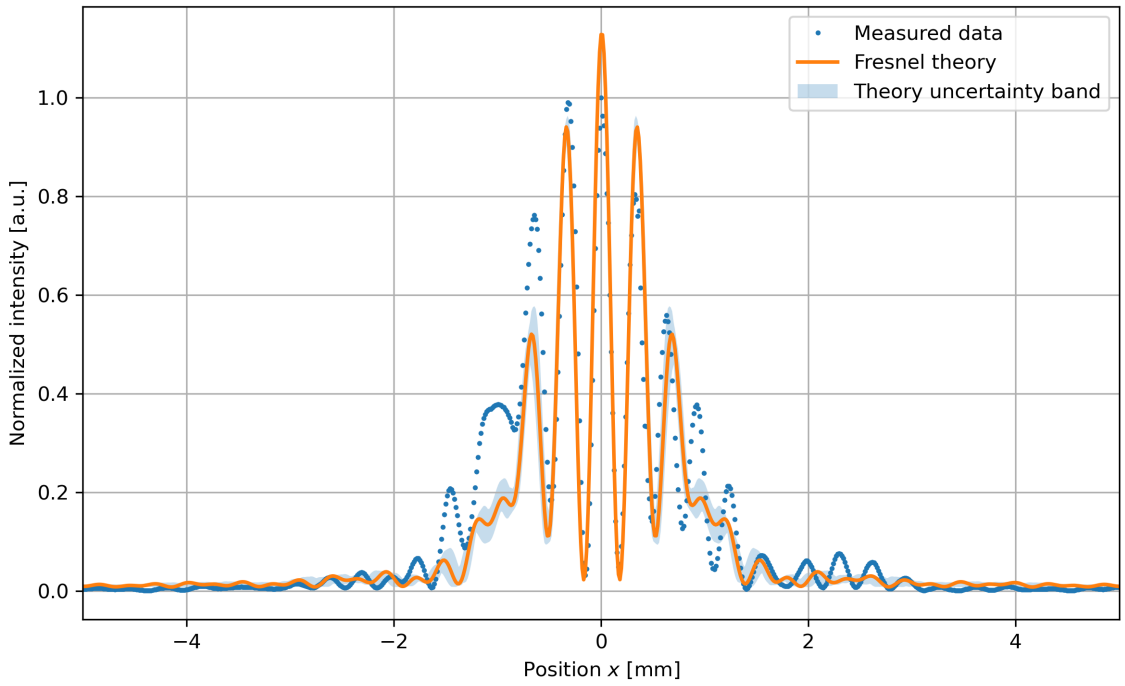


Figure 28: Measured and theoretical double-slit Fresnel diffraction profile at $z = 40\,\mathrm{cm}$.

Figure 29 shows the comparison for $z = 50\,\mathrm{cm}$.

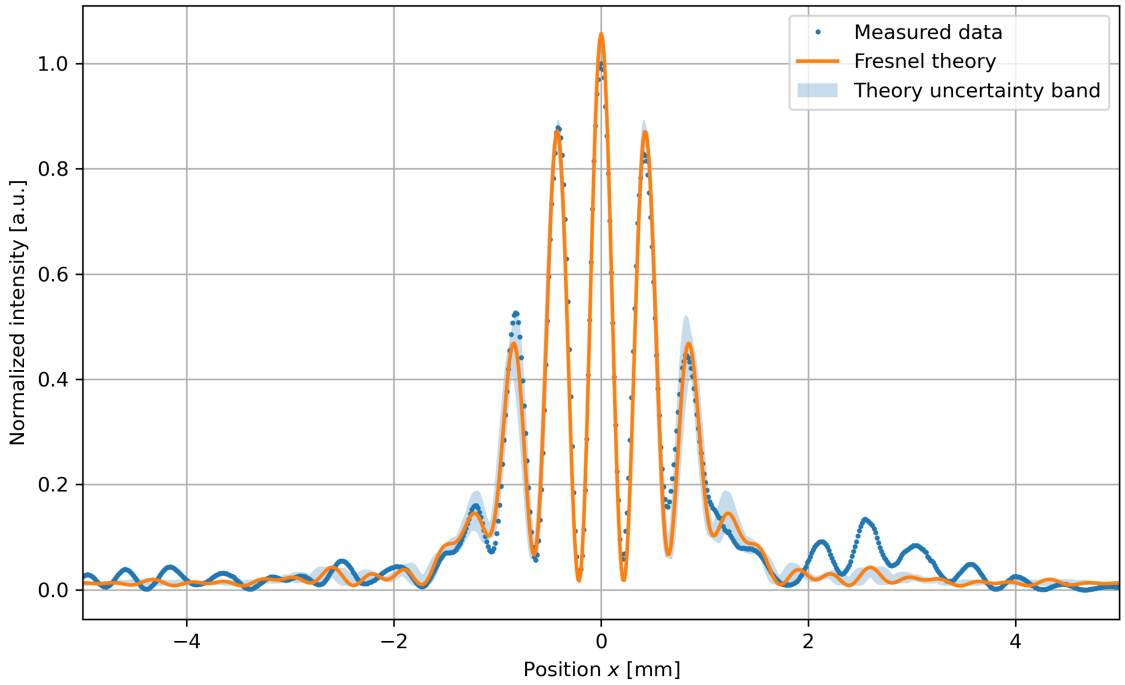


Figure 29: Measured and theoretical double-slit Fresnel diffraction profile at $z = 50\,\mathrm{cm}$.

Figure 30 shows the comparison for $z = 60\,\mathrm{cm}$.

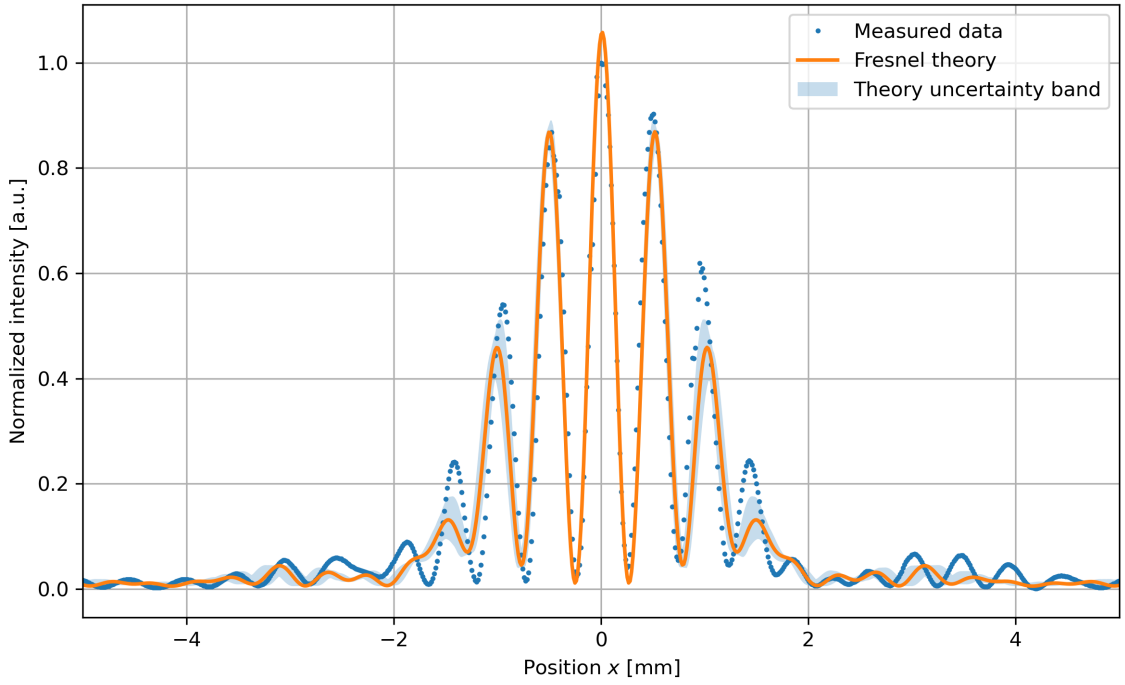


Figure 30: Measured and theoretical double-slit Fresnel diffraction profile at $z = 60\,\mathrm{cm}$.

Figure 31 compares all three measured Fresnel profiles and their corresponding theory curves in one plot.

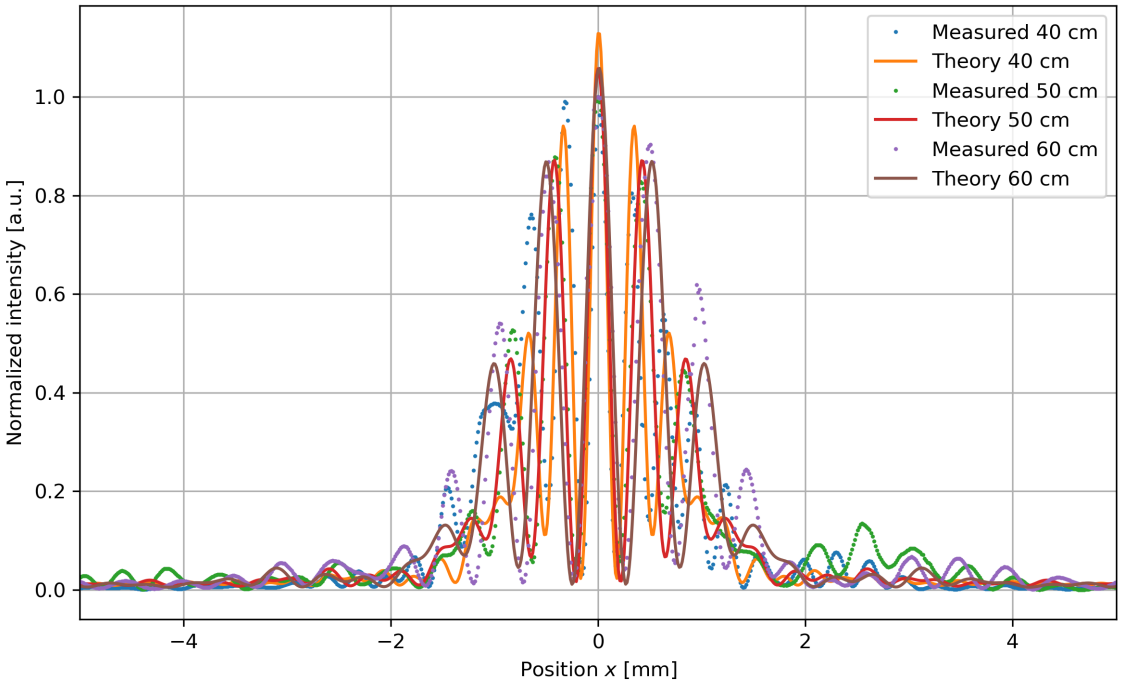


Figure 31: Comparison of the measured double-slit Fresnel diffraction profiles at $z = 40\,\text{cm}$, $50\,\text{cm}$, and $60\,\text{cm}$ with the corresponding theoretical curves.

The residuals for the three Fresnel comparisons are shown in Figures 32, 33, and 34.

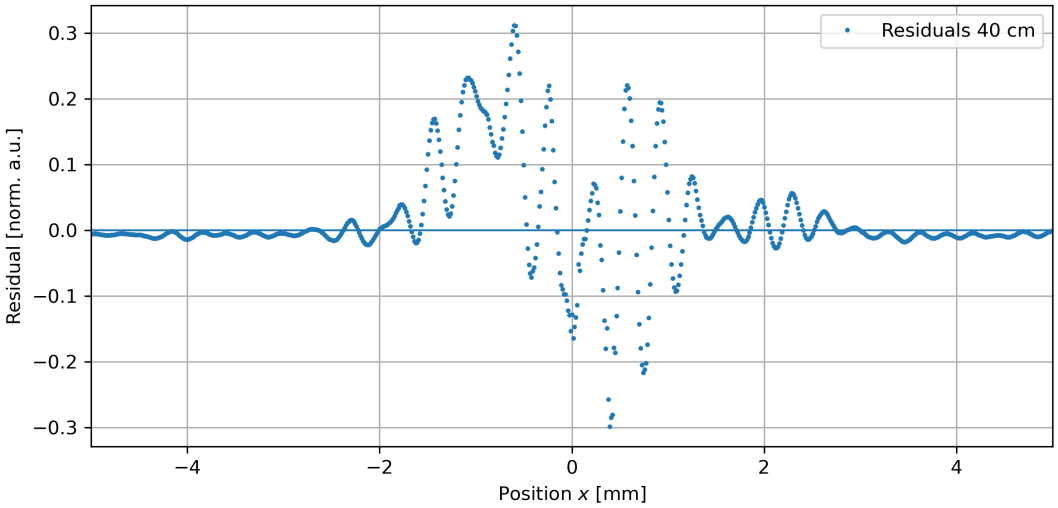


Figure 32: Residuals of the double-slit Fresnel theory comparison at $z = 40\,\text{cm}$.

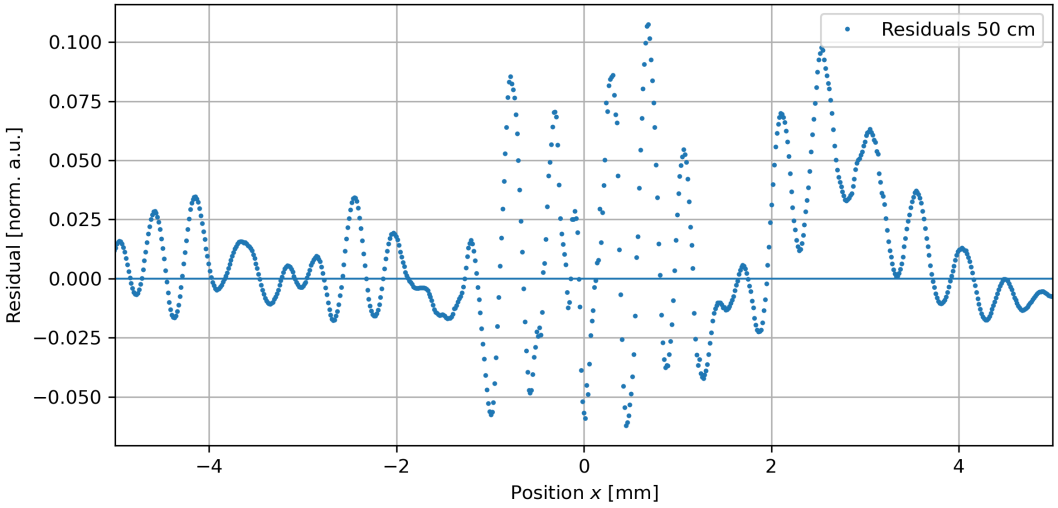


Figure 33: Residuals of the double-slit Fresnel theory comparison at $z = 50\,\mathrm{cm}$.

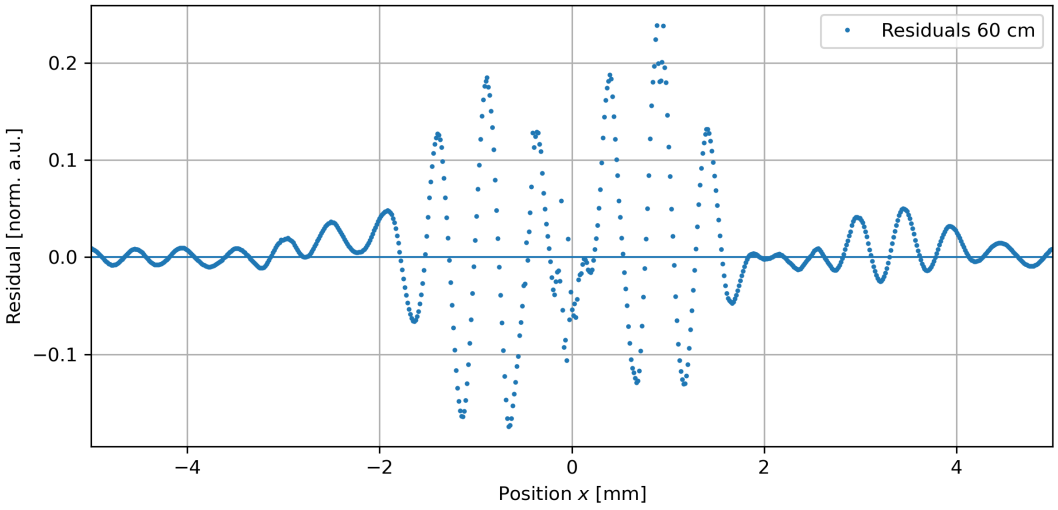


Figure 34: Residuals of the double-slit Fresnel theory comparison at $z = 60\,\mathrm{cm}$.

Figure 35 shows the Fresnel numbers N_b and N_g as functions of propagation distance. Both decrease with increasing distance, as expected from their inverse proportionality to z.

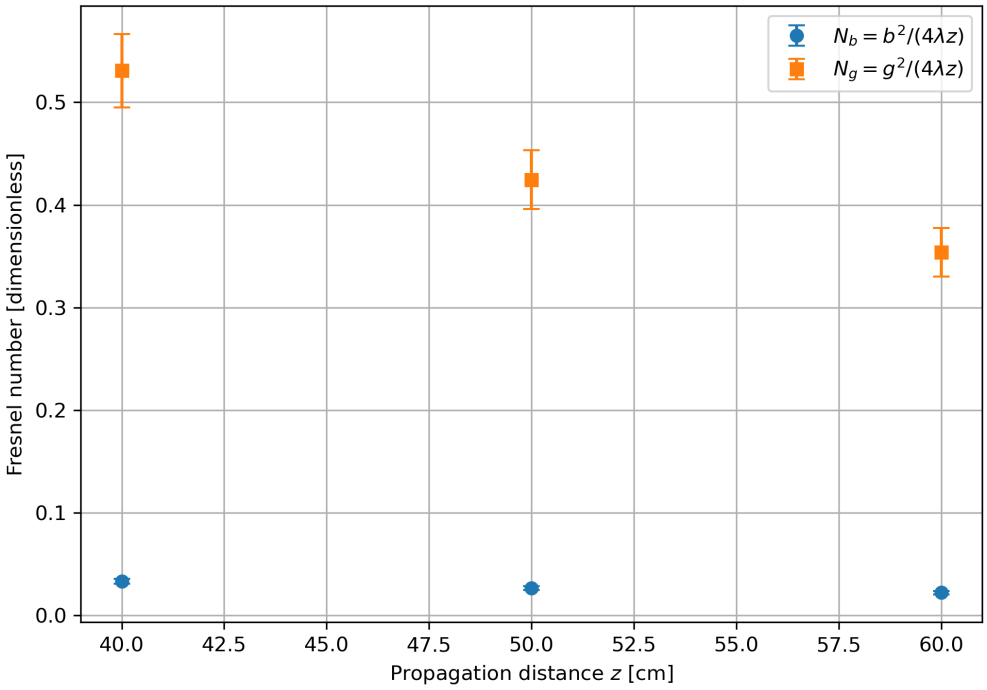


Figure 35: Fresnel numbers $N_b = b^2/(4\lambda z)$ and $N_g = g^2/(4\lambda z)$ as functions of propagation distance.

10.7 Uncertainty Analysis

The uncertainties in this task come from the slit parameters, propagation distances, profile centering, and intensity noise.

The slit parameters used in the theory were taken from Task 2b:

$$b = (0.1843 \pm 0.0062)\,\text{mm},$$

and

$$g = (0.7349 \pm 0.0244)\,\text{mm}.$$

The propagation distance uncertainty was

$$u(z) = 0.5\,\text{cm}.$$

The uncertainty in locating the profile center was

$$u(p_0) = 2 \text{ pixels}.$$

With

$$w_{\text{pix}} = 14\,\mu\text{m},$$

this gives

$$u(x_0) = 28\,\mu\text{m}.$$

The uncertainties in the Fresnel numbers were calculated using Gaussian uncertainty propagation. For

$$N_b = \frac{b^2}{4\lambda z},$$

the uncertainty is

$$u(N_b) = N_b \sqrt{\left(2\frac{u(b)}{b}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2 + \left(\frac{u(z)}{z}\right)^2}.$$

Similarly, for

$$N_g = \frac{g^2}{4\lambda z},$$

the uncertainty is

$$u(N_g) = N_g \sqrt{\left(2\frac{u(g)}{g}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2 + \left(\frac{u(z)}{z}\right)^2}.$$

The wavelength uncertainty was not specified and was therefore neglected. Theory uncertainty bands in the plots were obtained by varying b, g, and z within their uncertainties.

The normalized intensity noise was estimated from the background regions of each profile. The noise estimates were

$$u(I) = 0.001030 \quad \text{for } z = 40\,\text{cm},$$

$$u(I) = 0.002070 \quad \text{for } z = 50\,\text{cm},$$

and

$$u(I) = 0.001607 \quad \text{for } z = 60\,\text{cm}.$$

The RMSE values are larger than these noise estimates, indicating that the residuals are dominated not only by random detector noise, but also by systematic deviations between the ideal model and the measured profiles.

10.8 Interpretation and Discussion

The theoretical Fresnel model describes the measured double-slit profiles reasonably well, especially at $z = 50\,\text{cm}$, where the best agreement was obtained:

$$R^2 = 0.97664, \qquad \text{RMSE} = 0.01920.$$

At $z = 40\,\text{cm}$, the agreement was weaker:

$$R^2 = 0.88251, \qquad \text{RMSE} = 0.04610.$$

At $z = 60\,\text{cm}$, the agreement improved compared with 40 cm, but was still not as good as for 50 cm:

$$R^2 = 0.93756, \qquad \text{RMSE} = 0.03585.$$

The fitted horizontal shifts were small:

$$x_0 = 0.00678\,\text{mm}, \qquad x_0 = 0.00151\,\text{mm}, \qquad x_0 = 0.01027\,\text{mm},$$

for 40 cm, 50 cm, and 60 cm, respectively. These are smaller than the estimated centering uncertainty of 28 μm, so the profiles were well centered.

The Fresnel numbers decrease with increasing propagation distance. For the slit width,

$$N_b = 0.03338$$

at 40 cm, decreasing to

$$N_b = 0.02226$$

at 60 cm. For the slit separation,

$$N_g = 0.53074$$

at 40 cm, decreasing to

$$N_g = 0.35383$$

at 60 cm. Since N_g is not negligible, the finite-distance diffraction structure of the double slit remains important over the measured range.

The deviations between measurement and theory may result from several systematic effects. The double-slit Fresnel model assumes ideal rectangular slits, uniform illumination, perfect alignment, and an exactly known propagation distance. In the actual experiment, the laser beam may not illuminate both slits uniformly, the slit and camera may not have been perfectly parallel, and the exact distance from the slit plane to the CCD sensor may have had additional uncertainty. Background subtraction and detector response may also influence the normalized profile.

Overall, Task 2d satisfies the checklist requirement by showing the acquired Fresnel profiles at three different propagation distances and comparing them quantitatively with the theoretical Fresnel curves. The agreement is generally good, with the best result at $z = 50\,\text{cm}$, and the observed distance dependence is consistent with Fresnel diffraction.

11 Diffraction Grating and Talbot Effect (Task 3)

11.1 Goal of the Analysis

The aim of Task 3 was to analyze Fresnel diffraction from a transmission grating and determine both the grating constant g and the Talbot length t_B. The diffraction profiles were recorded at several propagation distances between the grating and the CCD camera. The measured distances were

$$z = 22\,\text{cm}, 24\,\text{cm}, 26\,\text{cm}, \ldots, 60\,\text{cm}.$$

The manufacturer specification of the grating was

$$40 \text{ lines/cm}.$$

This corresponds to the grating constant

$$g_{\text{man}} = \frac{1}{40}\,\text{cm} = 0.0250\,\text{cm} = 0.250\,\text{mm}.$$

The checklist for this task requires four main steps:

1. calculate the FFT of the grating profiles and represent the result as a function of spatial frequency or wave vector,
2. determine the grating constant from the FFT maxima and compare it with the manufacturer specification,
3. determine the amplitude of the first Fourier-transform peak and plot it as a function of propagation distance,
4. determine the Talbot length and compare it with the theoretical value.

11.2 Relevant Theory

A grating is a periodic aperture with period g. Its transmission function satisfies

$$f(x' + g) = f(x').$$

Therefore, it can be written as a Fourier series,

$$f(x') = \sum_{n=-\infty}^{\infty} c_n \exp\left(-i\frac{2\pi n}{g}x'\right),$$

where c_n are the Fourier coefficients.

In the Fresnel regime, the periodic structure produces self-images at characteristic propagation distances. This phenomenon is known as the Talbot effect. The Talbot length is

$$t_B = \frac{2g^2}{\lambda},$$

where g is the grating constant and λ is the laser wavelength. In this experiment,

$$\lambda = 636\,\text{nm}.$$

Using the manufacturer grating constant,

$$g_{\text{man}} = 0.250\,\text{mm},$$

the theoretical Talbot length is

$$t_{B,\text{man}} = \frac{2g_{\text{man}}^2}{\lambda}.$$

Substitution gives

$$t_{B,\text{man}} = \frac{2(0.250 \times 10^{-3}\,\text{m})^2}{636 \times 10^{-9}\,\text{m}} = 0.1965\,\text{m} = 19.65\,\text{cm}.$$

The Fourier transform of the grating profile contains peaks at spatial frequencies

$$\kappa_n = \frac{n}{g}, \qquad n = 0, \pm1, \pm2, \ldots$$

where κ is the spatial frequency in m^{-1}. Therefore, the first-order Fourier peak occurs at

$$\kappa_1 = \frac{1}{g}.$$

Thus, the grating constant can be determined from

$$g = \frac{1}{\kappa_1}.$$

If the FFT axis is expressed as angular spatial frequency k, then

$$k = 2\pi\kappa,$$

and

$$g = \frac{2\pi}{k_1}.$$

In the Talbot analysis, the amplitude of the first Fourier peak oscillates as a function of propagation distance. The oscillation period of the first Fourier peak amplitude is

$$\Delta z = \frac{t_B}{2}.$$

Therefore, the experimental Talbot length is obtained from

$$t_{B,\mathrm{exp}} = 2\Delta z.$$

11.3 Experimental Parameters and Data Treatment

The grating profiles were recorded with the new CCD line camera, for which

$$w_{\mathrm{pix}} = 14\,\mu\mathrm{m}.$$

Each CSV file contained the pixel number and the measured intensity in arbitrary units. For each profile, the central maximum was used to define the origin of the position axis,

$$x = (p - p_0)w_{\mathrm{pix}}.$$

The measured intensity was corrected by subtracting a background intensity estimated from the edge regions of the CCD profile. The corrected profile was normalized for plotting. For the FFT calculation, the mean value of the normalized profile was subtracted in order to suppress the dominant zero-frequency component.

The FFT was calculated as a function of the spatial frequency axis

$$\kappa = \mathrm{fftfreq}(N, \Delta x),$$

where

$$\Delta x = w_{\mathrm{pix}}.$$

The corresponding angular spatial frequency is

$$k = 2\pi\kappa.$$

The first-order FFT peak was searched for near the expected spatial frequency

$$\kappa_1 = \frac{1}{g_{\mathrm{man}}} = \frac{1}{0.250 \times 10^{-3}\,\mathrm{m}} = 4000\,\mathrm{m}^{-1}.$$

For each measured distance z, the first-order peak position κ_1, the corresponding grating constant g, and the first peak amplitude were extracted.

11.4 Determination of the Grating Constant

For each propagation distance, the first-order FFT peak position was used to calculate the grating constant according to

$$g = \frac{1}{\kappa_1}.$$

The results for all distances are shown in Table 12.

Table 12: First Fourier peak positions and grating constants obtained from the FFT of the grating profiles.

z [cm]	κ_1 [m^{-1}]	k_1 [$\mathrm{rad\,m}^{-1}$]	g [mm]	$u(g)$ [mm]
22	3905.06	24536.16	0.25608	0.00114
24	3862.71	24270.09	0.25886	0.00117
26	4020.77	25263.26	0.24871	0.00108
28	3949.31	24814.23	0.25321	0.00112
30	3940.94	24761.63	0.25375	0.00112
32	3872.96	24334.50	0.25820	0.00116
34	3819.13	23996.32	0.26184	0.00120
36	3485.87	21902.36	0.28687	0.00144
38	3954.08	24844.24	0.25290	0.00112
40	3922.98	24648.78	0.25491	0.00113
42	3844.91	24158.27	0.26008	0.00118
44	3773.69	23710.76	0.26499	0.00123
46	3693.29	23209.01	0.27072	0.00128
48	3953.67	24841.66	0.25293	0.00112
50	3910.75	24571.94	0.25571	0.00114
52	3828.76	24056.81	0.26118	0.00119
54	3741.17	23506.47	0.26730	0.00125
56	3664.31	23023.56	0.27290	0.00130
58	4058.78	25502.07	0.24638	0.00106
60	3858.95	24246.53	0.25914	0.00117

The weighted mean of the grating constant was

$$g_{\mathrm{FFT}} = (0.2587 \pm 0.0021)\,\mathrm{mm}.$$

The manufacturer value was

$$g_{\mathrm{man}} = 0.2500\,\mathrm{mm}.$$

The relative deviation from the manufacturer value was

$$\frac{|g_{\mathrm{FFT}} - g_{\mathrm{man}}|}{g_{\mathrm{man}}} \times 100\% = 3.47\%.$$

Thus, the grating constant determined from the FFT peaks agrees reasonably well with the manufacturer specification.

11.5 First Fourier Peak Amplitude and Talbot Length

The amplitude of the first Fourier peak was extracted for every measured propagation distance. These amplitudes were normalized to the maximum extracted value and fitted with a periodic model,

$$A_1(z) = A_0 + A\cos\left(\frac{2\pi z}{\Delta z} + \phi\right),$$

where Δz is the oscillation period of the first Fourier peak amplitude.

The fitted period was

$$\Delta z = (10.775 \pm 0.516)\,\mathrm{cm}.$$

Since the first Fourier peak amplitude oscillates with half the Talbot length,

$$\Delta z = \frac{t_B}{2},$$

the experimental Talbot length was

$$t_{B,\mathrm{exp}} = 2\Delta z.$$

Therefore,

$$t_{B,\mathrm{exp}} = (21.51 \pm 1.03)\,\mathrm{cm}.$$

The theoretical Talbot length calculated from the manufacturer grating constant was

$$t_{B,\mathrm{man}} = 19.65\,\mathrm{cm}.$$

Using the measured grating constant,

$$g_{\mathrm{FFT}} = (0.2587 \pm 0.0021)\,\mathrm{mm},$$

the theoretical Talbot length becomes

$$t_{B,\mathrm{FFT}} = \frac{2g_{\mathrm{FFT}}^2}{\lambda} = (21.04 \pm 0.34)\,\mathrm{cm}.$$

The relative deviation between the experimental Talbot length and the manufacturer-based theoretical value was

$$\frac{|t_{B,\mathrm{exp}} - t_{B,\mathrm{man}}|}{t_{B,\mathrm{man}}} \times 100\% = 9.65\%.$$

The relative deviation between the experimental Talbot length and the theoretical value calculated from the measured grating constant was

$$\frac{|t_{B,\mathrm{exp}} - t_{B,\mathrm{FFT}}|}{t_{B,\mathrm{FFT}}} \times 100\% = 2.41\%.$$

The normalized difference relative to the measured-g theoretical value was

$$\frac{|t_{B,\mathrm{exp}} - t_{B,\mathrm{FFT}}|}{\sqrt{u(t_{B,\mathrm{exp}})^2 + u(t_{B,\mathrm{FFT}})^2}} = 0.468.$$

Since this value is below one, the experimentally determined Talbot length agrees with the theoretical value calculated from the measured grating constant within one combined standard uncertainty.

11.6 Summary of Main Results

The main numerical results of Task 3 are summarized in Table 13.

Table 13: Summary of the grating constant and Talbot length analysis.

Quantity	Value	Unit
Manufacturer line density	40.0	lines/cm
Manufacturer grating constant g_{man}	0.2500	mm
Measured grating constant g_{FFT}	0.2587	mm
Uncertainty $u(g_{\mathrm{FFT}})$	0.0021	mm
Relative deviation of g from manufacturer	3.47	%
Theoretical Talbot length from g_{man}	19.65	cm
Theoretical Talbot length from g_{FFT}	21.04	cm
Uncertainty of t_B from g_{FFT}	0.34	cm
Fitted first-peak amplitude period Δz	10.78	cm
Uncertainty $u(\Delta z)$	0.52	cm
Experimental Talbot length $t_{B,\mathrm{exp}}$	21.51	cm
Uncertainty $u(t_{B,\mathrm{exp}})$	1.03	cm
Relative deviation from manufacturer-based t_B	9.65	%
Relative deviation from measured-g theoretical t_B	2.41	%
Normalized difference from measured-g theoretical t_B	0.468	–
Amplitude fit RMSE	0.0726	norm. amplitude
Amplitude fit R^2	0.8691	–

11.7 Plots

Figure 36 shows selected measured grating Fresnel profiles at different propagation distances. The profiles change with distance, which is expected in the Fresnel regime.

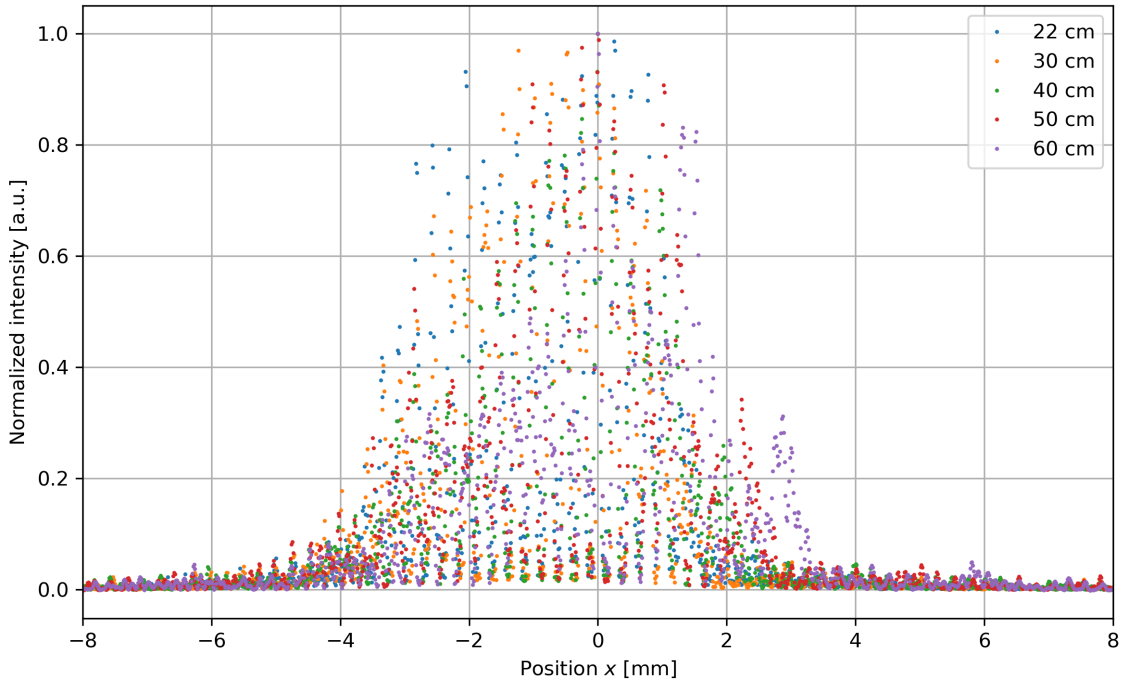


Figure 36: Selected measured Fresnel diffraction profiles of the grating at different propagation distances.

Figure 37 shows the FFT magnitudes of the grating profiles as a function of spatial frequency κ. The expected first-order peak position from the manufacturer grating constant is marked.

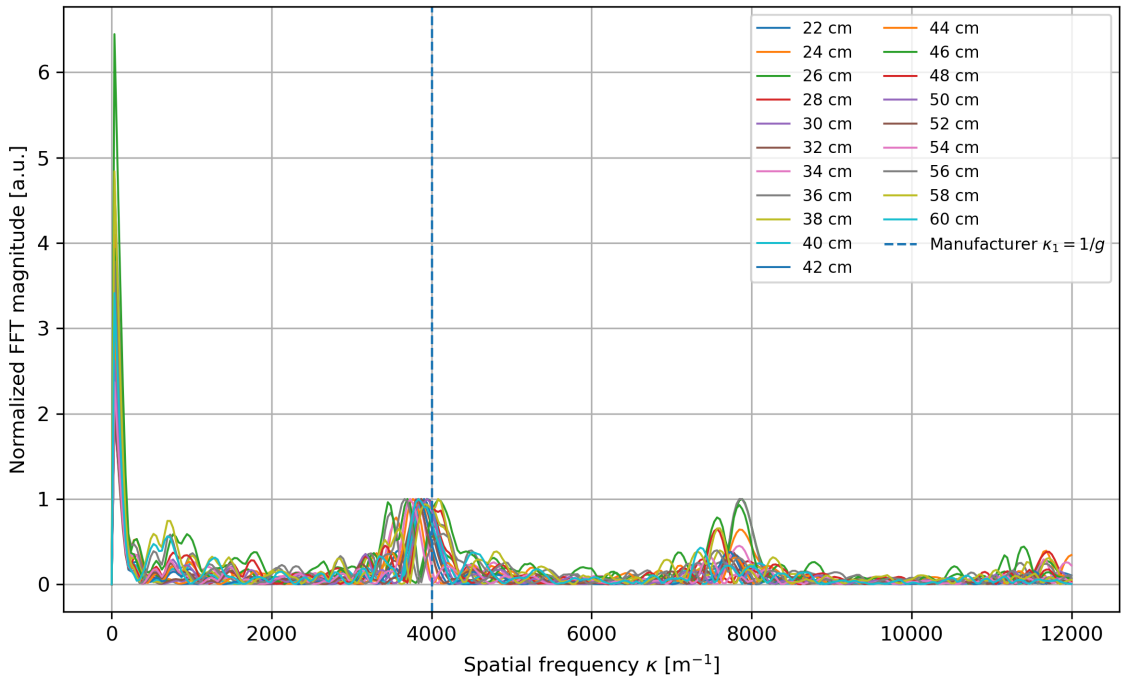


Figure 37: FFT magnitudes of the grating profiles as a function of spatial frequency κ.

Figure 38 shows the same FFT results as a function of angular spatial frequency $k = 2\pi\kappa$.

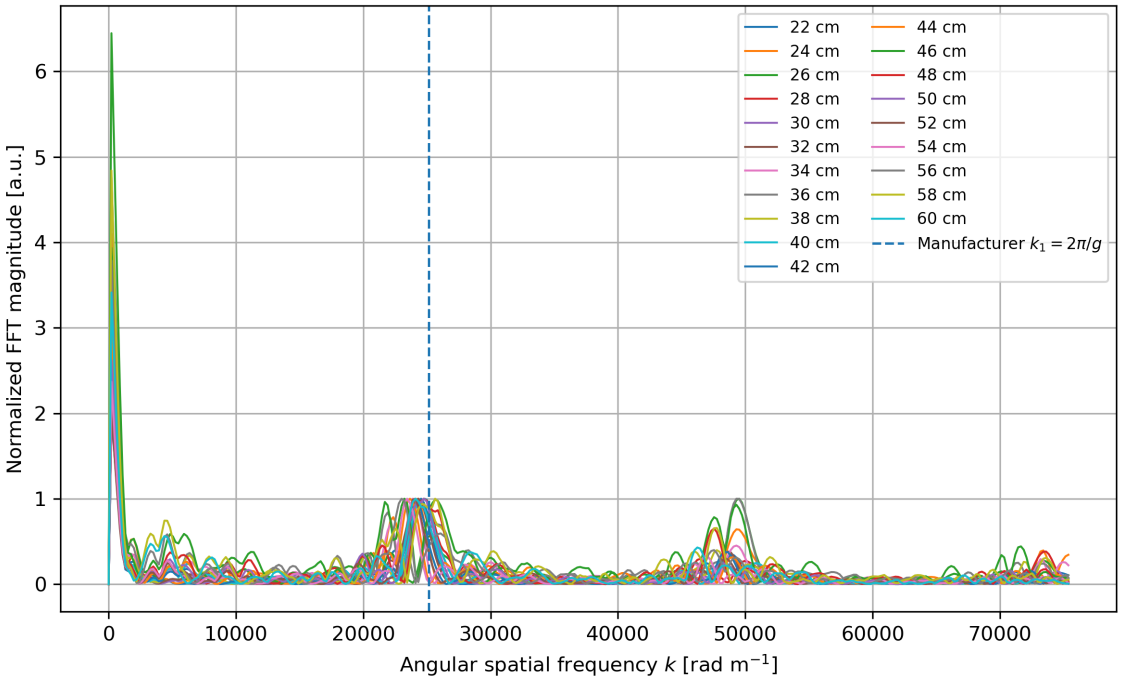


Figure 38: FFT magnitudes of the grating profiles as a function of angular spatial frequency k.

Figure 39 shows the grating constant calculated from the first FFT peak at each distance. The manufacturer value and weighted mean are also shown.

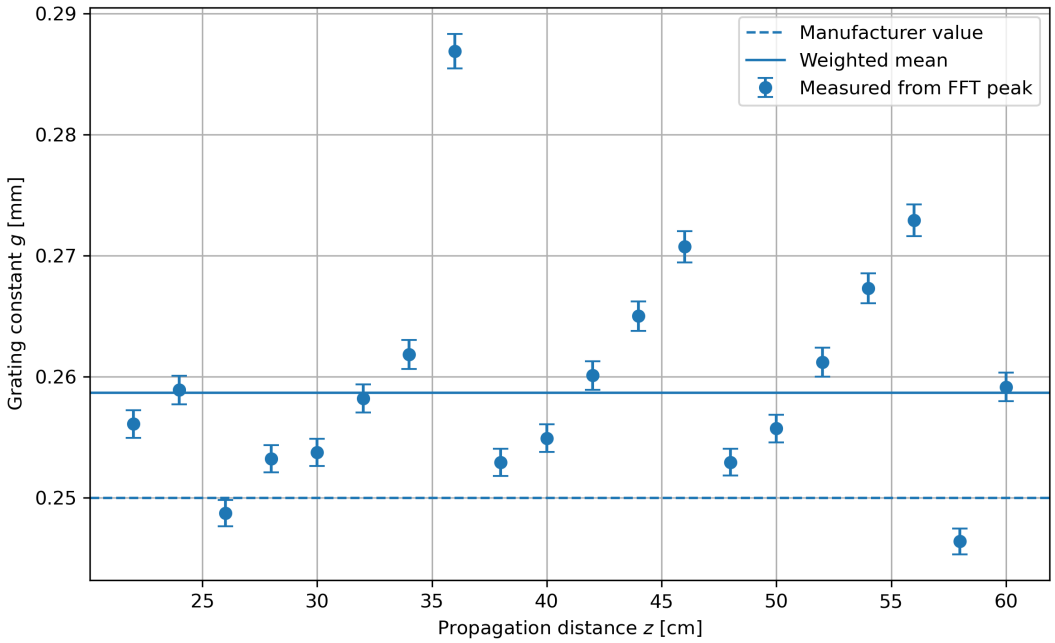


Figure 39: Grating constant determined from the first FFT peak as a function of propagation distance.

Figure 40 shows the normalized first Fourier peak amplitude as a function of propagation distance. The periodic fit was used to determine the amplitude oscillation period and therefore the Talbot length.

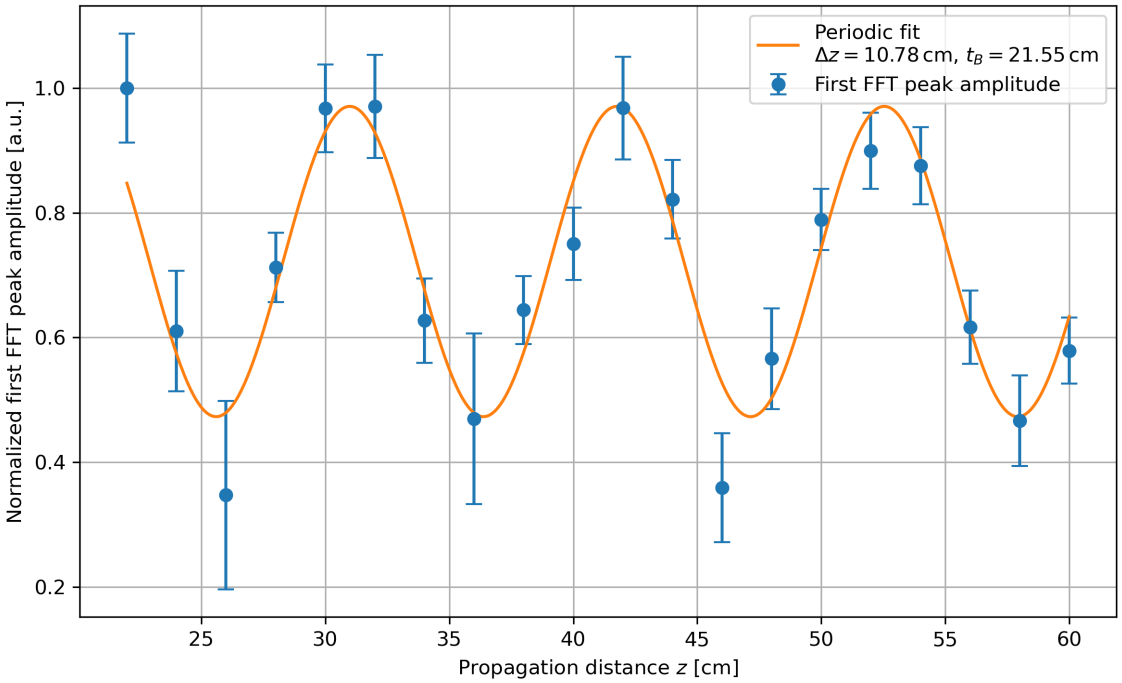


Figure 40: Normalized amplitude of the first Fourier peak as a function of propagation distance. The fitted period corresponds to half the Talbot length.

Figure 41 shows the residuals of the periodic fit used in the Talbot-length analysis.

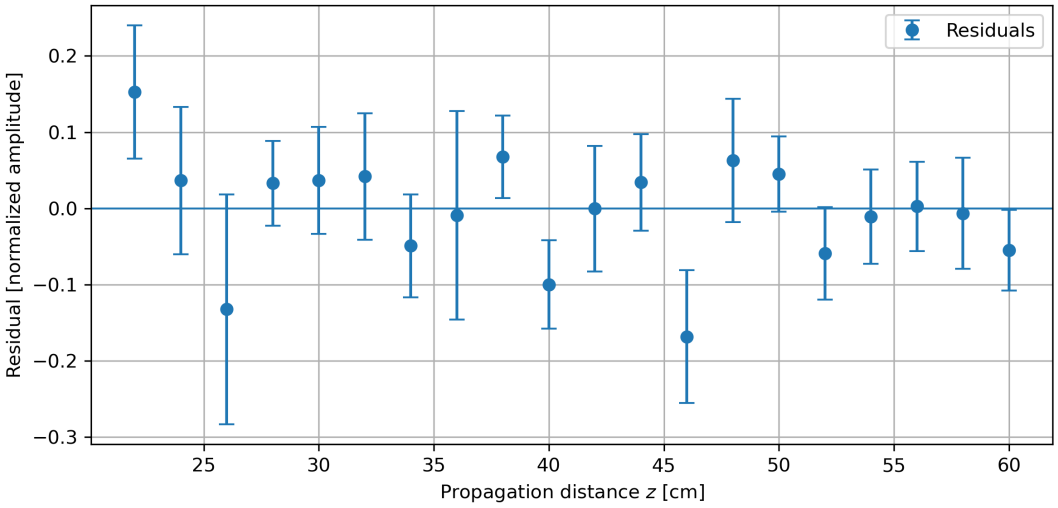


Figure 41: Residuals of the periodic fit to the first Fourier peak amplitude.

11.8 Uncertainty Analysis

The uncertainty of the grating constant was determined from the uncertainty of the first Fourier peak position. The FFT spatial-frequency resolution was

$$\Delta\kappa = \frac{1}{N\Delta x},$$

where $N = 2048$ and $\Delta x = 14\,\mu\mathrm{m}$. The peak-position uncertainty was estimated as

$$u(\kappa_1) = \frac{\Delta\kappa}{2}.$$

Since

$$g = \frac{1}{\kappa_1},$$

the uncertainty in the grating constant is

$$u(g) = \frac{u(\kappa_1)}{\kappa_1^2}.$$

The weighted mean of the grating constants obtained at different distances was used as the final measured grating constant. The uncertainty of this mean included both the weighted statistical uncertainty and the scatter of the individual values.

For the Talbot length calculated from the measured grating constant,

$$t_B = \frac{2g^2}{\lambda},$$

the uncertainty was propagated according to

$$u(t_B) = t_B \sqrt{\left(2\frac{u(g)}{g}\right)^2 + \left(\frac{u(\lambda)}{\lambda}\right)^2}.$$

No wavelength uncertainty was specified, so $u(\lambda)$ was neglected.

The experimental Talbot length was obtained from

$$t_{B,\mathrm{exp}} = 2\Delta z.$$

Thus,

$$u(t_{B,\mathrm{exp}}) = 2u(\Delta z).$$

The fitted uncertainty of the amplitude period was combined with the propagation-distance uncertainty,

$$u(z) = 0.5\,\mathrm{cm},$$

to obtain the final uncertainty of the fitted period. This gave

$$u(t_{B,\mathrm{exp}}) = 1.03\,\mathrm{cm}.$$

11.9 Interpretation and Discussion

The FFT analysis successfully identified a first-order grating peak close to the expected value

$$\kappa_1 = \frac{1}{g_{\mathrm{man}}} = 4000\,\mathrm{m}^{-1}.$$

The measured mean grating constant was

$$g_{\mathrm{FFT}} = (0.2587 \pm 0.0021)\,\mathrm{mm}.$$

This is larger than the manufacturer value of 0.2500 mm by 3.47%. The agreement is reasonable, although the deviation suggests the presence of systematic effects such as finite sampling, uncertainty in identifying the FFT peak, finite illuminated grating size, or distortions due to background subtraction and windowing.

The experimentally determined Talbot length was

$$t_{B,\mathrm{exp}} = (21.51 \pm 1.03)\,\mathrm{cm}.$$

Compared with the theoretical value based on the manufacturer grating constant,

$$t_{B,\mathrm{man}} = 19.65\,\mathrm{cm},$$

the relative deviation was 9.65%. However, when the theoretical Talbot length was calculated using the experimentally measured grating constant,

$$t_{B,\mathrm{FFT}} = (21.04 \pm 0.34)\,\mathrm{cm},$$

the relative deviation decreased to 2.41%. The normalized difference was only

$$0.468,$$

which means that the measured Talbot length agrees with the measured-g theoretical prediction within one combined standard uncertainty.

The fit of the first Fourier peak amplitude had

$$R^2 = 0.8691,$$

which indicates a clear periodic trend but also noticeable scatter. Possible causes include uncertainty in extracting the peak amplitude, variations in laser intensity, imperfect grating alignment, finite CCD size, background subtraction, and the fact that the experimental profiles are not ideal infinite periodic grating images. The amplitude extraction is also sensitive to the finite FFT resolution and to small changes in peak shape with distance.

Overall, Task 3 satisfies the checklist requirements. The FFTs of the grating profiles were calculated and plotted with spatial-frequency units, the grating constant was determined from the first-order FFT peak and compared with the manufacturer value, the first Fourier peak amplitude was plotted as a function of propagation distance, and the Talbot length was determined and compared with theoretical predictions.

12 Discussion

12.1 Overview

The experiment investigated Fraunhofer and Fresnel diffraction from a single slit, a double slit, and a diffraction grating. The main goal was to determine the geometrical parameters of the diffracting objects using several independent methods and to compare the measured diffraction profiles with theoretical predictions.

In the Fraunhofer regime, the single-slit and double-slit profiles were analyzed using minima positions, nonlinear fitting, and Fourier-transform methods. In the Fresnel regime, measured near-field profiles were compared with theoretical Fresnel diffraction curves. For the grating, the FFT of the measured Fresnel profiles was used to determine the grating constant and the Talbot length.

Overall, the experimentally determined parameters were generally close to the manufacturer values. The best agreement was obtained for quantities determined from peak or fringe spacings, such as the single-slit width from minima, the double-slit separation from minima, and the grating constant from FFT peak positions. Larger discrepancies were observed when a parameter had to be extracted from the detailed shape or width of an FFT autocorrelation feature, especially for the slit width in Tasks 1b and 2c.

12.2 Single-Slit Fraunhofer Diffraction

In Task 1a, the single-slit width was determined from the positions of the Fraunhofer minima. The measured distances $2x_n$ between the $\pm n$-th order minima showed a clear linear dependence on diffraction order n, as expected from

$$2x_n = \frac{2f\lambda}{b} n.$$

The fit gave

$$b_{1a} = (0.3955 \pm 0.0141)\,\mathrm{mm}.$$

This agrees well with the manufacturer value

$$b_{\mathrm{man}} = 0.4000\,\mathrm{mm},$$

with a relative deviation of only

$$1.12\%.$$

The high fit quality,

$$R^2 = 0.99999,$$

shows that the Fraunhofer minima followed the expected theoretical relation very closely. The RMSE was comparable to the estimated uncertainty of the measured minima distances, so the assumed pixel uncertainty of 2 pixels was reasonable.

Task 1b used the FFT of the single-slit Fraunhofer intensity profile. The FFT method gave

$$b_{\mathrm{FFT}} = (0.4738 \pm 0.0161)\,\mathrm{mm}.$$

This differs from the Task 1a result by

$$19.78\%,$$

and the normalized difference was

3.65.

This means that the FFT-derived slit width did not agree with the minima-derived value within combined uncertainty. The discrepancy is likely due to the sensitivity of the FFT method to the finite CCD range, background subtraction, smoothing, and the difficulty of identifying the first zero of the autocorrelation feature. The FFT of the intensity profile should ideally produce the autocorrelation of the slit function, but the experimental intensity profile is finite, noisy, and not perfectly ideal. Therefore, the apparent first zero can be shifted from the expected value.

For the single-slit width, the minima method was therefore more reliable than the FFT method in this experiment.

12.3 Single-Slit Fresnel Diffraction

In Task 1c, single-slit Fresnel profiles were measured at

$$z = 27\,\text{cm}, \quad 31\,\text{cm}, \quad 35\,\text{cm}.$$

The theoretical Fresnel curves were calculated using the slit width from Task 1a. The agreement between measured and theoretical profiles was very good. The coefficients of determination were

$$R^2 = 0.998925, \quad 0.998975, \quad 0.999417,$$

for 27 cm, 31 cm, and 35 cm, respectively.

The Fresnel number decreased from

$$N_b = 0.2277$$

at 27 cm to

$$N_b = 0.1757$$

at 35 cm, consistent with

$$N_b = \frac{b^2}{4\lambda z}.$$

The measured profiles therefore showed the expected distance dependence of Fresnel diffraction. The small residuals indicate that the one-dimensional Fresnel model was suitable for the single-slit data. Remaining deviations may be due to small alignment errors, uncertainty in the slit-camera distance, nonuniform laser illumination, and imperfect background subtraction.

12.4 Double-Slit Fraunhofer Diffraction

In Task 2a, the double-slit interference minima were used to determine the slit separation g. The measured distances $2x_n$ increased linearly with $n + 1/2$, as predicted by

$$2x_n = \frac{2f\lambda}{g}\left(n + \frac{1}{2}\right).$$

The result was

$$g_{2a} = (0.7592 \pm 0.0264)\,\text{mm}.$$

This agrees with the manufacturer value

$$g_{\text{man}} = 0.7500\,\text{mm},$$

with a relative deviation of

$$1.22\%.$$

The fit quality was high,

$$R^2 = 0.99860,$$

although the RMSE was slightly larger than the estimated uncertainty in the minima positions. This suggests small systematic effects, such as imperfect centering of the diffraction pattern or uncertainty in the exact position of the minima.

In Task 2b, the full double-slit Fraunhofer profile was fitted using

$$I(k_x) = A\text{sinc}^2\left(\frac{b(k_x - k_{x,0})}{2}\right)\cos^2\left(\frac{g(k_x - k_{x,0})}{2}\right) + C.$$

The fitted slit parameters were

$$b_{2b} = (0.1843 \pm 0.0062)\,\text{mm},$$

and

$$g_{2b} = (0.7349 \pm 0.0244)\,\text{mm}.$$

The fitted slit separation agreed with both the manufacturer value and the Task 2a value within uncertainty. The normalized difference between g_{2b} and g_{2a} was

$$0.674,$$

which indicates good agreement.

The fitted slit width was lower than the manufacturer value of

$$0.2000\,\text{mm}$$

by

$$7.84\%.$$

This discrepancy is larger than for the slit separation. This is expected because the slit separation is determined mainly by the fringe spacing, which is easy to measure accurately, while the slit width is determined by the broader diffraction envelope, which is more sensitive to background correction, normalization, finite detector range, and imperfections in the slit illumination.

12.5 Double-Slit FFT Analysis

In Task 2c, the FFT of the double-slit Fraunhofer intensity profile was used to determine both b and g. The autocorrelation interpretation predicts that the central feature is related to the slit width and the side peaks are related to the slit separation.

The FFT results were

$$b_{2c} = (0.2313 \pm 0.0084)\,\text{mm},$$

and

$$g_{2c} = (0.7398 \pm 0.0248)\,\text{mm}.$$

The FFT-derived slit separation agreed very well with the values from Tasks 2a and 2b. The normalized difference was

$$0.534$$

relative to Task 2a and

$$0.141$$

relative to Task 2b. This confirms that the slit separation is robustly encoded in the separation of the autocorrelation side peaks.

However, the FFT-derived slit width differed significantly from the Task 2b result. The normalized difference between b_{2c} and b_{2b} was

$$4.51.$$

This shows that the slit width extracted from the FFT central feature was not reliable for this dataset. As in Task 1b, the width of the autocorrelation feature is sensitive to windowing, finite data length, smoothing, background subtraction, and the exact criterion used to define the feature edge. Therefore, the FFT method gave a reliable slit separation but a less reliable slit width.

12.6 Double-Slit Fresnel Diffraction

In Task 2d, double-slit Fresnel profiles were recorded at

$$z = 40\,\text{cm}, \quad 50\,\text{cm}, \quad 60\,\text{cm}.$$

The theoretical curves were calculated using the fitted parameters from Task 2b:

$$b = (0.1843 \pm 0.0062)\,\text{mm},$$

and

$$g = (0.7349 \pm 0.0244)\,\text{mm}.$$

The best agreement was obtained at

$$z = 50\,\text{cm},$$

with

$$R^2 = 0.97664$$

and

$$\text{RMSE} = 0.01920.$$

The agreement was weaker at 40 cm,

$$R^2 = 0.88251,$$

and moderate at 60 cm,

$$R^2 = 0.93756.$$

The double-slit Fresnel comparison was therefore less accurate than the single-slit Fresnel comparison. This is reasonable because the double-slit profile is more sensitive to alignment, relative illumination of the two slits, and the exact slit separation. If the two slits are not illuminated equally, or if the camera is slightly tilted or displaced, the interference structure can deviate from the ideal one-dimensional model.

The calculated Fresnel numbers showed that

$$N_g$$

was significantly larger than

$$N_b,$$

because $g > b$. This means that finite-distance effects related to the slit separation remain important in the measured distance range.

12.7 Grating and Talbot Effect

In Task 3, grating diffraction profiles were recorded from

$$z = 22\,\text{cm}$$

to

$$z = 60\,\text{cm}.$$

The manufacturer specification was

$$40\,\text{lines/cm},$$

corresponding to

$$g_{\text{man}} = 0.2500\,\text{mm}.$$

From the first-order FFT peaks, the measured grating constant was

$$g_{\text{FFT}} = (0.2587 \pm 0.0021)\,\text{mm}.$$

This differs from the manufacturer value by

$$3.47\%.$$

The agreement is reasonable, although the measured value is systematically larger. This may be due to finite FFT resolution, uncertainty in peak extraction, finite illuminated grating width, or small calibration errors in the pixel-to-position conversion.

The theoretical Talbot length based on the manufacturer value was

$$t_{B,\mathrm{man}} = 19.65\,\mathrm{cm}.$$

Using the measured grating constant instead gave

$$t_{B,\mathrm{FFT}} = (21.04 \pm 0.34)\,\mathrm{cm}.$$

The experimental Talbot length obtained from the oscillation of the first Fourier peak amplitude was

$$t_{B,\mathrm{exp}} = (21.51 \pm 1.03)\,\mathrm{cm}.$$

This differs from the manufacturer-based theoretical value by

$$9.65\%,$$

but differs from the measured-g theoretical value by only

$$2.41\%.$$

The normalized difference between $t_{B,\mathrm{exp}}$ and $t_{B,\mathrm{FFT}}$ was

$$0.468,$$

so the experimental Talbot length agrees with the theory based on the measured grating constant within one combined standard uncertainty.

The fit of the first Fourier peak amplitude had

$$R^2 = 0.8691.$$

This indicates a clear periodic trend, but with noticeable scatter. The scatter is expected because peak amplitudes are sensitive to background, finite detector range, laser intensity fluctuations, grating alignment, and noise in the FFT.

12.8 Comparison of Methods

A general trend observed throughout the experiment is that distances or periods determined from peak positions were more reliable than quantities determined from peak amplitudes or feature widths.

For example, the following results agreed well with manufacturer or theoretical values:

$$b_{1a} = (0.3955 \pm 0.0141)\,\mathrm{mm},$$

$$g_{2a} = (0.7592 \pm 0.0264)\,\mathrm{mm},$$

$$g_{2b} = (0.7349 \pm 0.0244)\,\mathrm{mm},$$

$$g_{2c} = (0.7398 \pm 0.0248)\,\mathrm{mm},$$

and

$$g_{\mathrm{FFT}} = (0.2587 \pm 0.0021)\,\mathrm{mm}.$$

In contrast, the FFT-derived slit widths showed larger discrepancies:

$$b_{1b} = (0.4738 \pm 0.0161)\,\mathrm{mm},$$

and

$$b_{2c} = (0.2313 \pm 0.0084)\,\mathrm{mm}.$$

This suggests that FFT peak positions are robust, while FFT feature widths are more sensitive to systematic effects. The position of a side autocorrelation peak gives a clear distance such as g, whereas the edge or zero of a central autocorrelation feature can be distorted by finite-window effects and smoothing.

12.9 Main Sources of Uncertainty and Systematic Error

The most important uncertainty sources were uncertainty in the focal length f, uncertainty in locating minima or profile centers, finite CCD pixel width, uncertainty in propagation distance z, background subtraction, finite FFT resolution, laser intensity fluctuations, imperfect alignment of the slit, grating, lens, and CCD camera, nonuniform illumination of the aperture, finite CCD size and finite data-window effects.

The focal length uncertainty,

$$u(f) = 10\,\mathrm{mm},$$

was a dominant scale uncertainty in the Fraunhofer analyses. Since the extracted aperture dimensions depend directly on f, this uncertainty propagated directly into b and g.

For Fresnel measurements, the distance uncertainty,

$$u(z) = 0.5\,\mathrm{cm},$$

affected the theoretical curves and Fresnel numbers. In addition, the exact position of the CCD sensor plane may not coincide exactly with the external camera reference position, which could introduce a systematic uncertainty in z.

For FFT analyses, finite-window effects were especially important. The FFT assumes a finite sampled signal, while the ideal diffraction pattern extends infinitely. Truncation of the diffraction profile can broaden or shift features in the Fourier transform. This effect likely contributed to the discrepancies in the FFT-derived slit widths.

Alignment was also critical. Since the CCD camera is a line detector, small vertical or horizontal misalignments can change the measured profile. This is especially important for the double slit and grating, where the intensity pattern contains finer structure than in the single-slit case.

12.10 Overall Evaluation

The experiment successfully demonstrated the relationship between diffraction and Fourier analysis. The Fraunhofer measurements showed that aperture dimensions can be extracted from minima positions and from fitting theoretical intensity functions. The Fresnel measurements showed the expected dependence of diffraction profiles on propagation distance. The grating measurements demonstrated the Talbot effect and gave an experimental Talbot length consistent with theory when the measured grating constant was used.

The most reliable quantitative results were obtained from linear fits of minima positions and from FFT peak positions. The less reliable results were those based on FFT feature widths and peak amplitudes. This distinction is important because it shows that the FFT method is powerful, but its accuracy depends strongly on how the relevant feature is extracted.

Despite some discrepancies, especially in the FFT-derived slit widths, the overall results support the theoretical descriptions of Fraunhofer diffraction, Fresnel diffraction, and the Talbot effect.

13 Conclusion

In this experiment, Fraunhofer and Fresnel diffraction from a single slit, a double slit, and a diffraction grating were investigated using a CCD line camera. The measured diffraction profiles were analyzed using minima positions, nonlinear fitting, Fourier transforms, and Fresnel diffraction theory.

For the single slit, the width determined from the Fraunhofer minima was

$$b_{1a} = (0.3955 \pm 0.0141)\,\mathrm{mm},$$

which agrees well with the manufacturer value of 0.400 mm. The FFT method gave

$$b_{1b} = (0.4738 \pm 0.0161)\,\mathrm{mm},$$

which showed a larger deviation, indicating that the FFT zero-position method was more sensitive to finite-window effects and background correction. The single-slit Fresnel profiles at 27 cm, 31 cm, and 35 cm agreed very well with the theoretical Fresnel curves.

For the double slit, the slit separation obtained from the interference minima was

$$g_{2a} = (0.7592 \pm 0.0264)\,\mathrm{mm},$$

which agrees with the manufacturer value of 0.750 mm. The full Fraunhofer fit gave

$$b_{2b} = (0.1843 \pm 0.0062)\,\text{mm},$$

and

$$g_{2b} = (0.7349 \pm 0.0244)\,\text{mm}.$$

The FFT analysis gave

$$b_{2c} = (0.2313 \pm 0.0084)\,\text{mm},$$

and

$$g_{2c} = (0.7398 \pm 0.0248)\,\text{mm}.$$

The slit separation values from all three methods agreed within uncertainty, while the slit width from the FFT showed a larger discrepancy. The double-slit Fresnel profiles at 40 cm, 50 cm, and 60 cm showed reasonable agreement with theory, with the best agreement at 50 cm.

For the grating, the manufacturer specification of 40 lines/cm corresponds to

$$g_{\text{man}} = 0.250\,\text{mm}.$$

From the FFT peak positions, the measured grating constant was

$$g_{\text{FFT}} = (0.2587 \pm 0.0021)\,\text{mm}.$$

The experimental Talbot length determined from the oscillation of the first Fourier peak amplitude was

$$t_{B,\text{exp}} = (21.51 \pm 1.03)\,\text{cm}.$$

This agrees within uncertainty with the theoretical value calculated from the measured grating constant,

$$t_{B,\text{FFT}} = (21.04 \pm 0.34)\,\text{cm}.$$

Overall, the experiment confirmed the expected theoretical relationships for Fraunhofer diffraction, Fresnel diffraction, and the Talbot effect. The most reliable results were obtained from minima positions and FFT peak positions, while quantities extracted from FFT feature widths were more sensitive to systematic effects such as finite detector range, background subtraction, and smoothing.

14 References

M. Ziese, O17e Diffraction, University of Leipzig, May 2026.